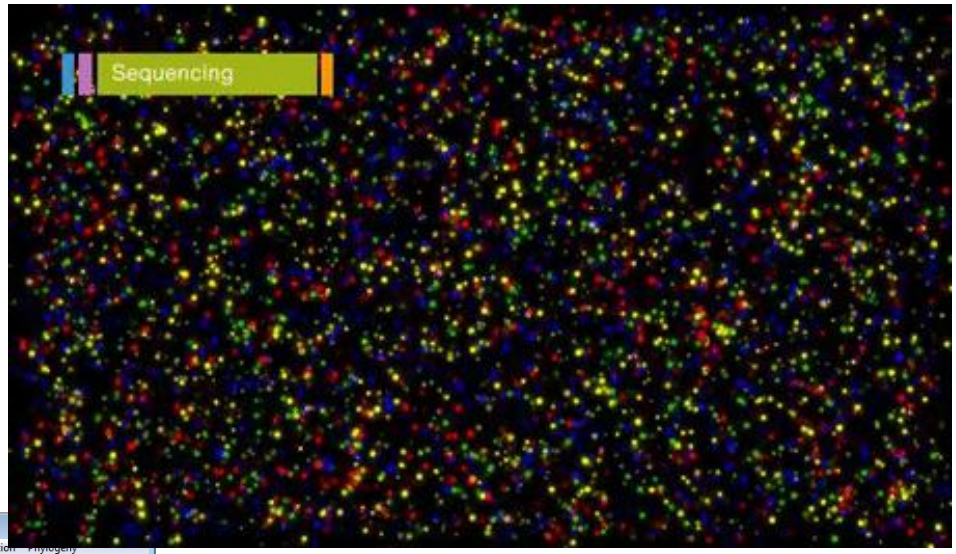
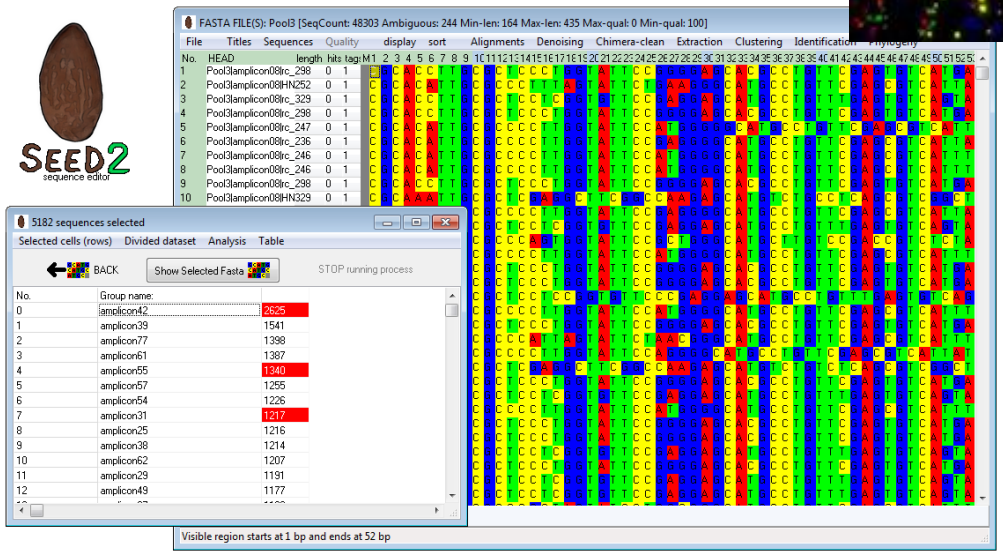
Bioinformatics





Tomáš Větrovský
vetrovsky@biomed.cas.cz
Laboratory of Environmental Microbiology
Institute of Microbiology of the Academy of Sciences of the Czech Republic

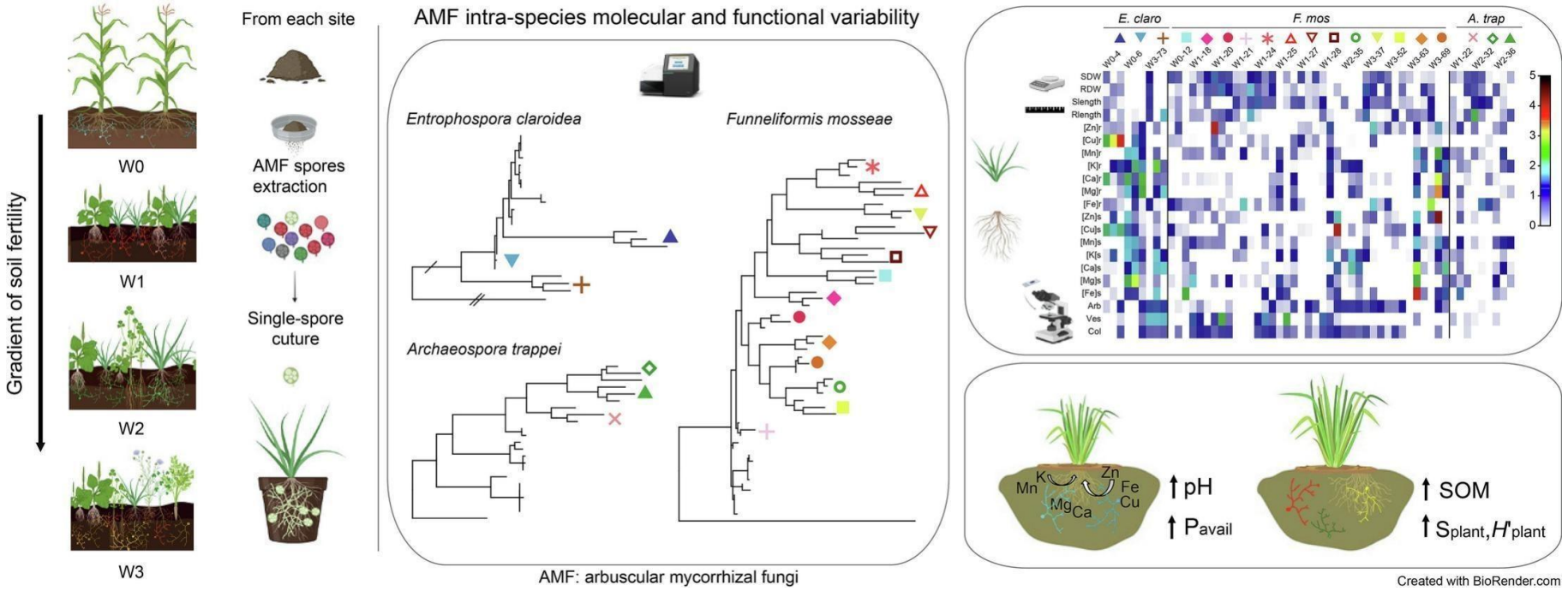
Structure of the day

1. PART 1 (**9:00-10:30**):
   Introduction to amplicon processing
2. PART 2 (**10:30-12:00 + 12:30-14**):
   Amplicon processing using pipeline SEED2
3. PART 3 (**14:00-15:00**):
   Introduction to basic statistical analysis

Fungal amplicon data usage

Taxonomic composition

Identifying which fungi are present (from phylum down to species level, where possible)



Fungal amplicon data usage

Study community composition and fungal ecology

Diverse ecological roles

- free-living vs. symbiotic
- lower ecological versatility among fungal species compare to bacteria

Parasitic



Saprotrophic



Mycorrhizal



Fungal amplicon data usage

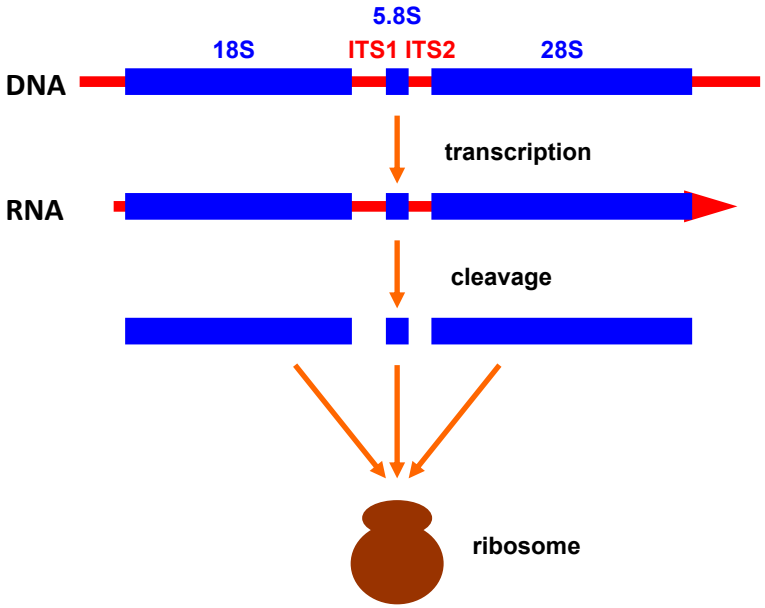
Biogeography of fungi

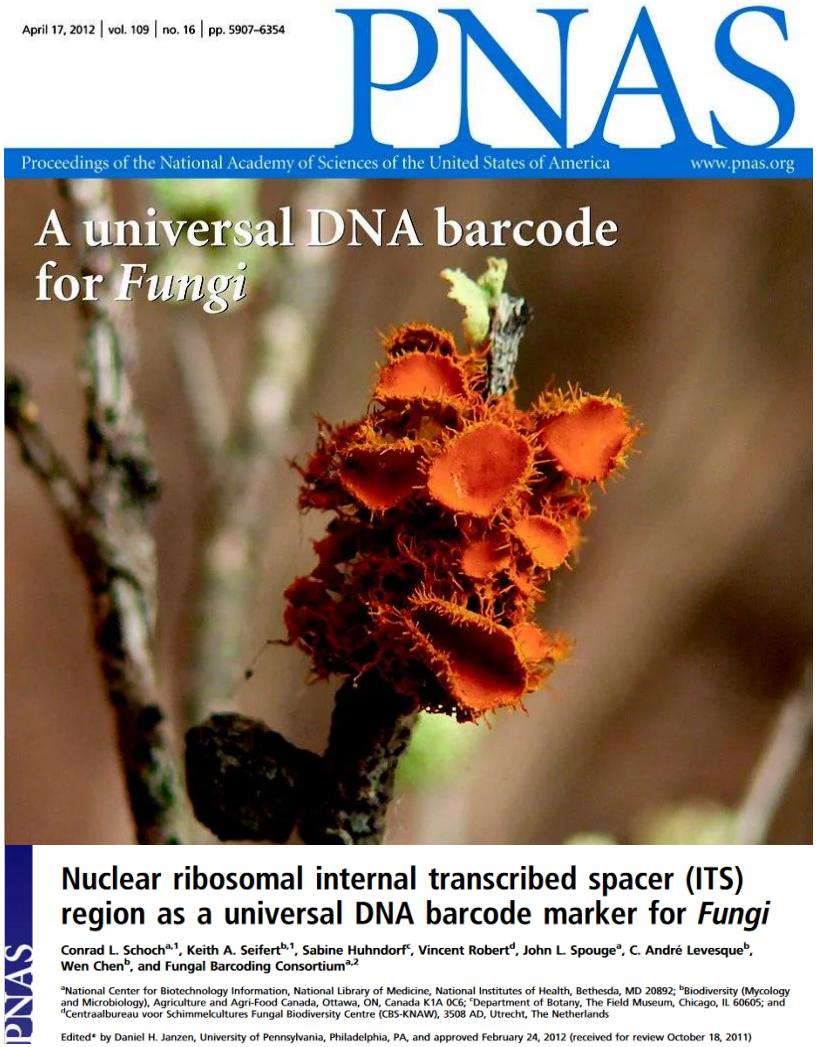
- Many fungi are affected by plants as their hosts, symbionts, or habitat
- Fungi are a unique group comprising unicellular organisms as well as extremely large individuals
- Distribution of (many) fungi through spores is virtually unlimited
- Fungi are often endemic



Fungal markers used for taxonomical identification

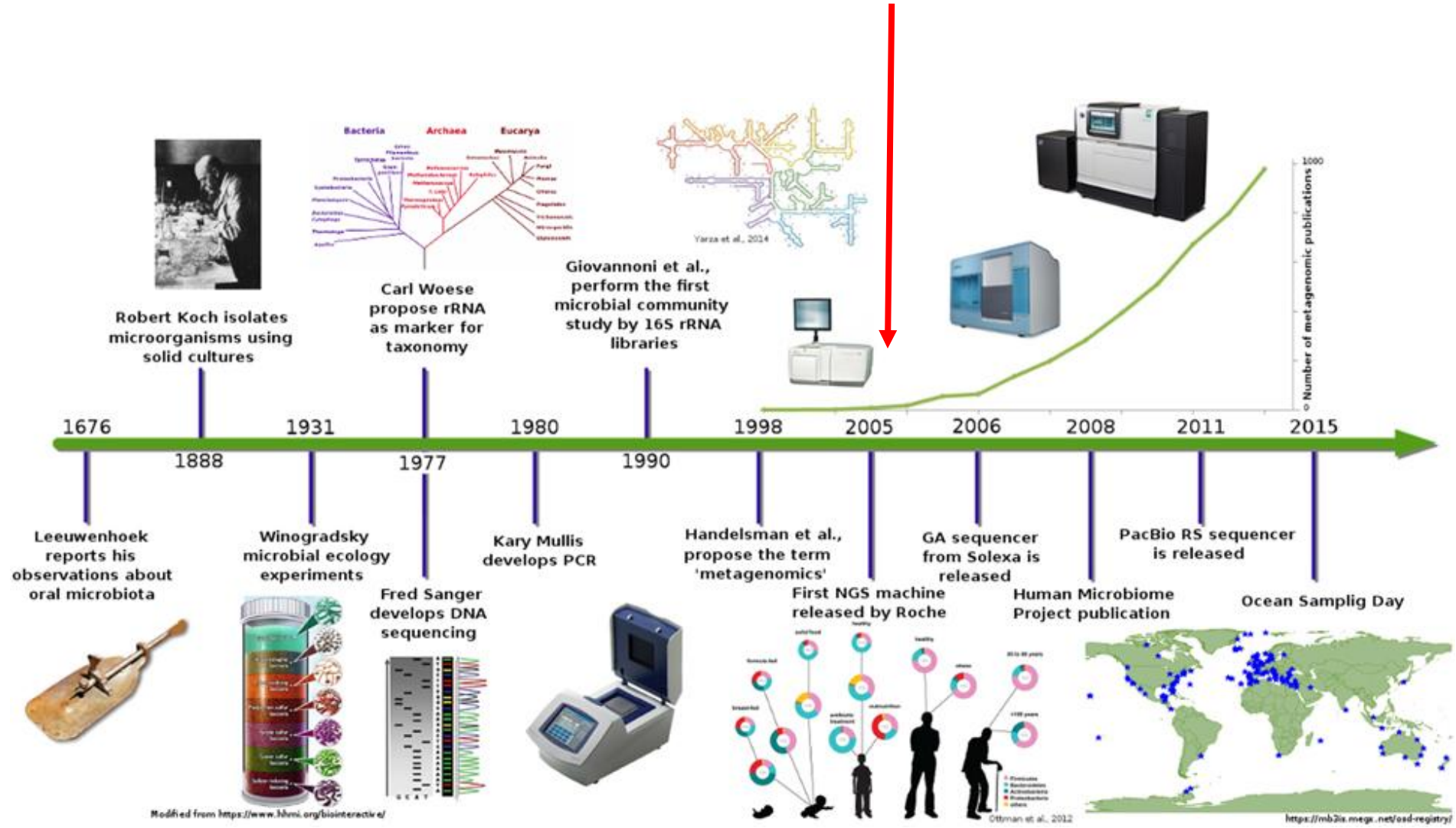
- many molecular markers - ITS, LSU, SSU, RPB2, β-tubulin, TEF1-α but…
- ITS region is considered the universal barcode of Fungi



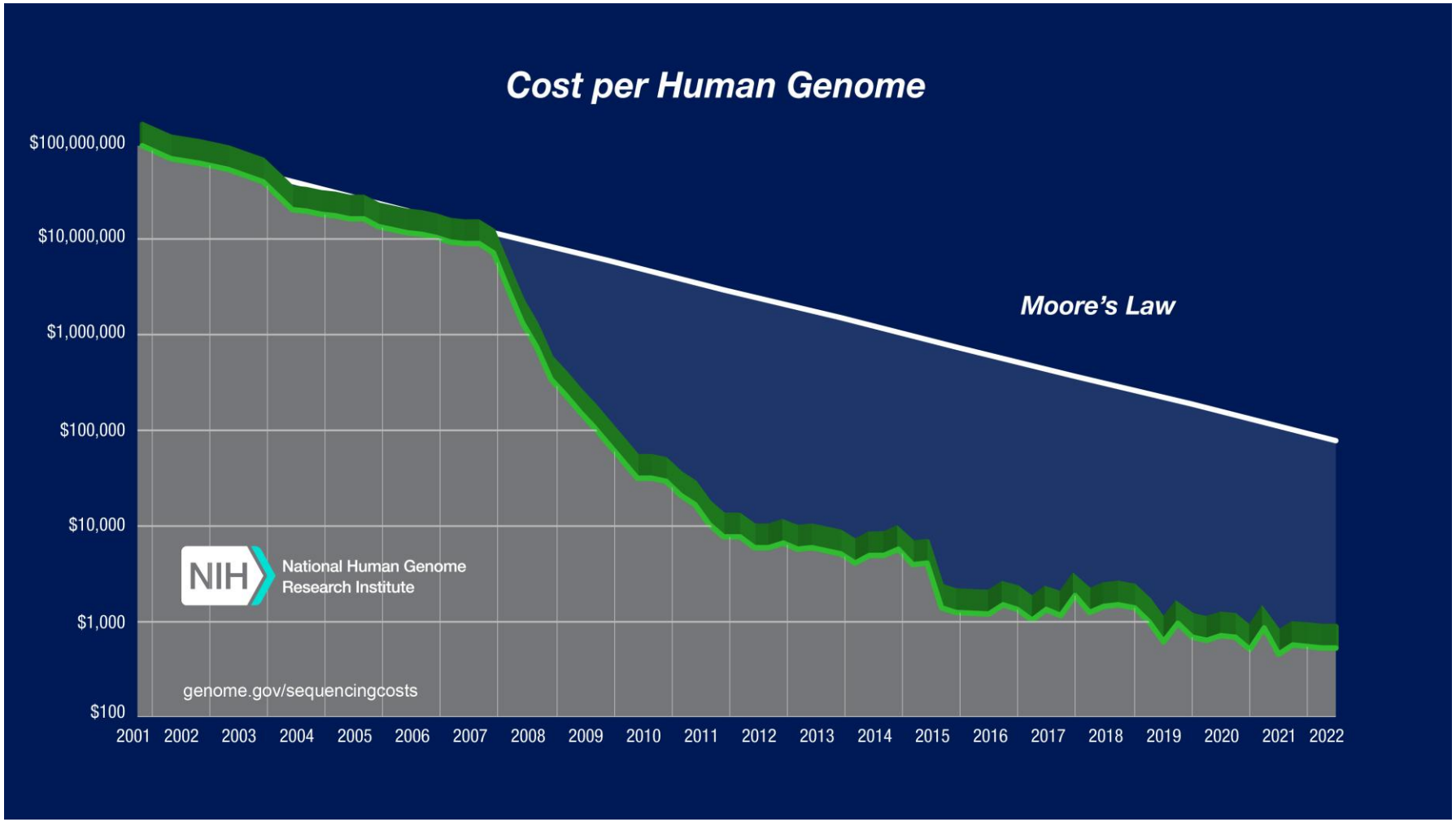


Genomics and DNA sequencing

high throughput sequencing (HTS) techniques were introduced by 454 Pyrosequencing

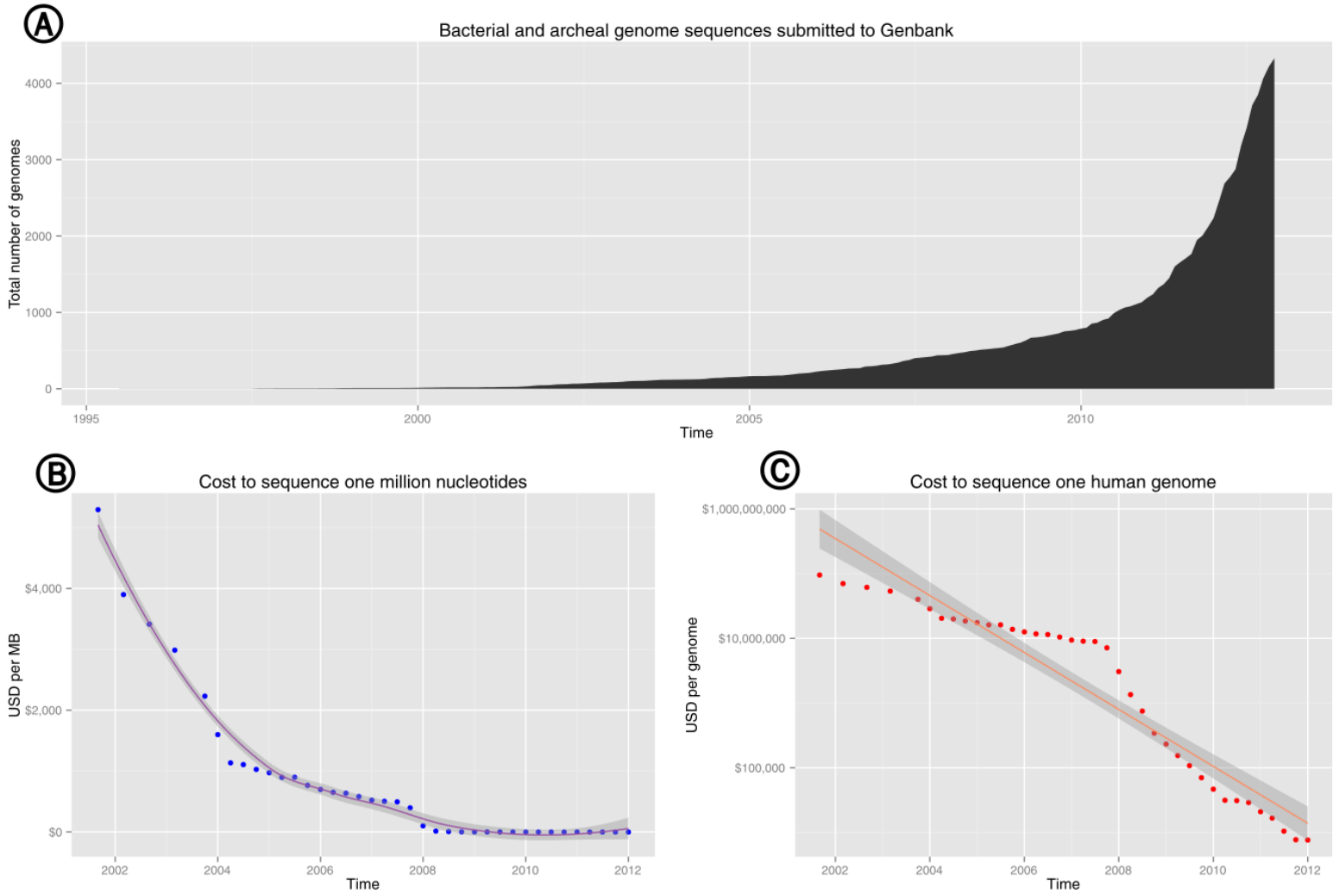


Fall of sequencing costs



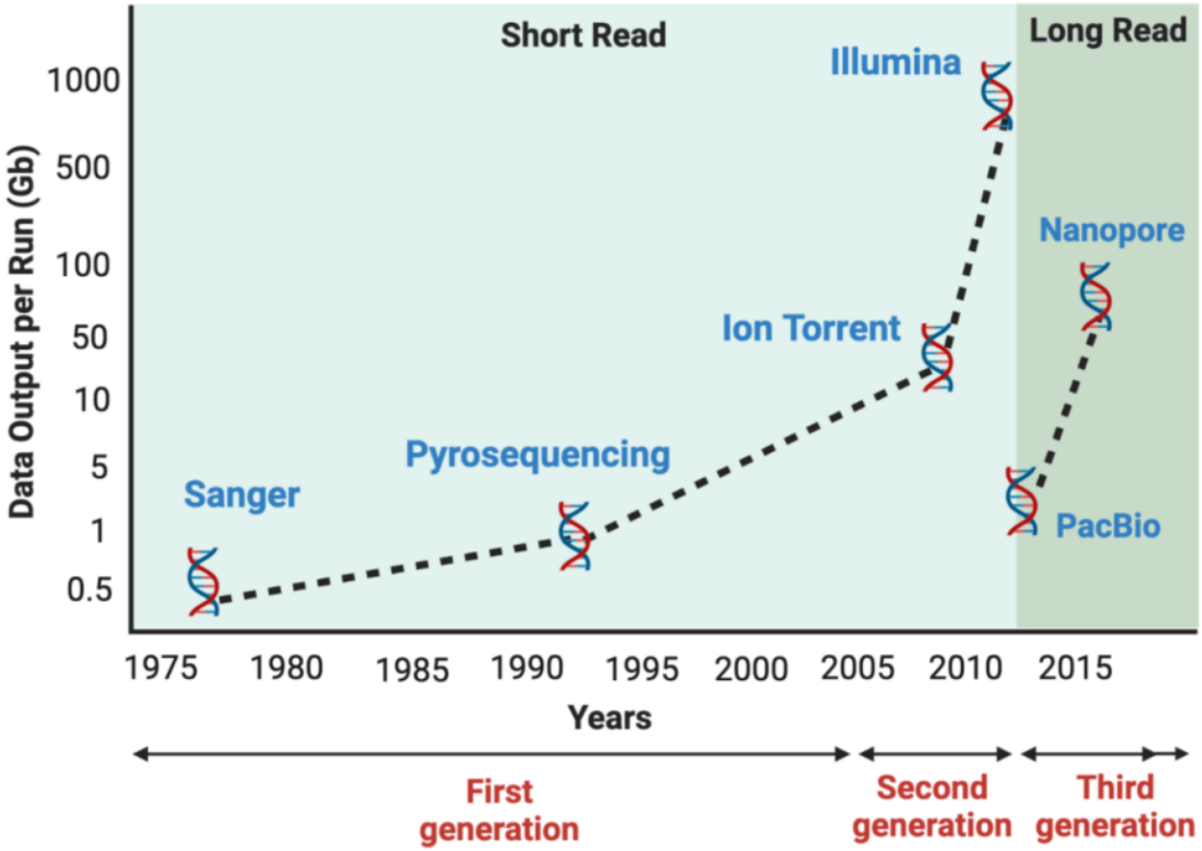
$1,000 genome

High throughput sequencing technologies have become essential in studies on genomics

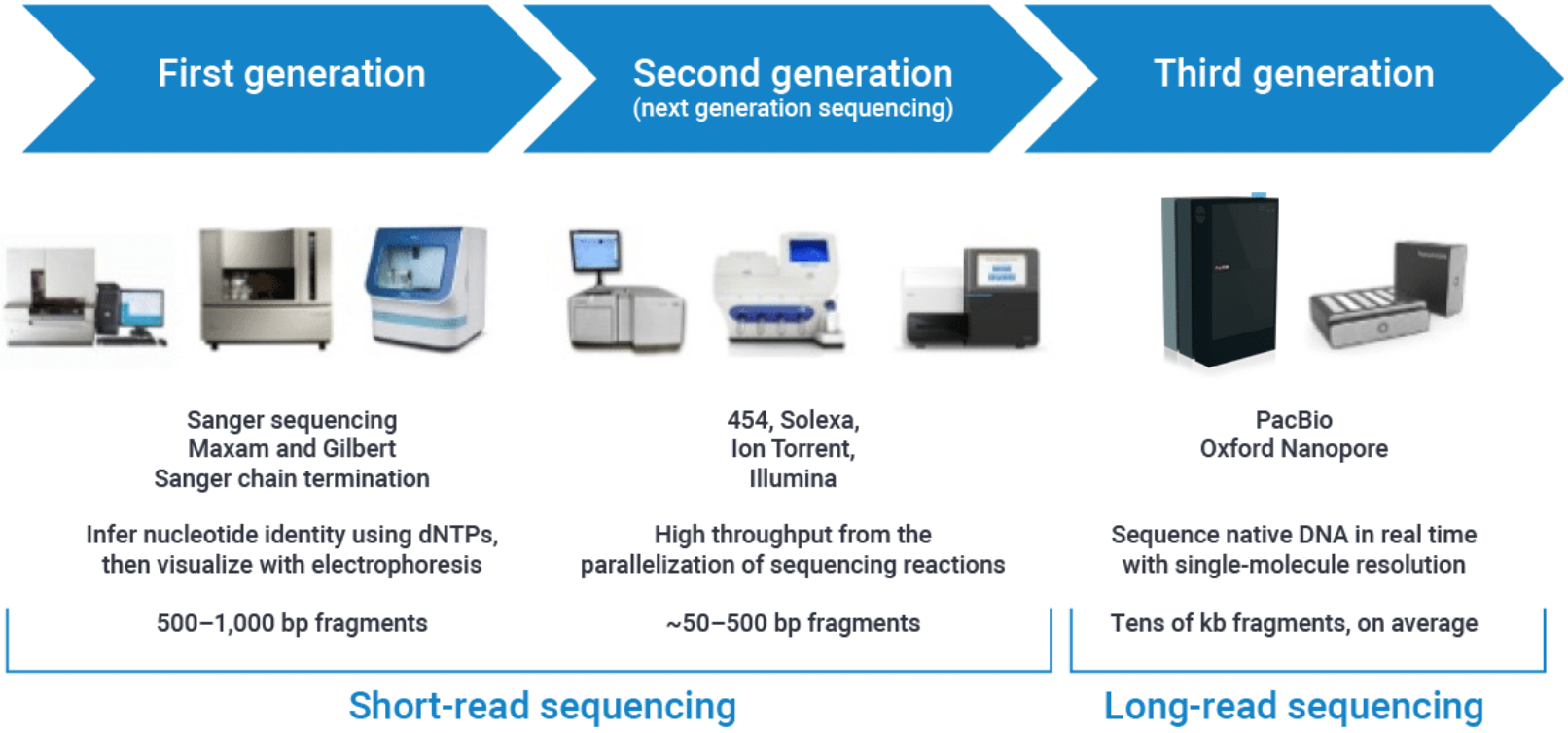


The number of genome projects has increased as technological improvements continue to lower the cost of sequencing. **(A)** Exponential growth of genome sequence databases since 1995. **(B)** The cost in US Dollars (USD) to sequence one million bases. **(C)** The cost in USD to sequence a 3,000 Mb (human-sized) genome on a log-transformed scale.

Evolution of sequencing technologies



The development of sequencing technologies over the past four decades can be categorized into three generations. The first generation was represented by **Sanger sequencing**, providing the foundation for DNA sequencing. The second generation introduced **massively parallel sequencing** with platforms such as Illumina and Ion Torrent, enabling high-throughput sequencing. The current third generation includes PacBio and Nanopore, offering **long-read and single-molecule sequencing** capabilities.



2nd generation sequencing

- exponentially increased sequence throughput and accuracy
- allows simultaneous sequencing of millions of different DNA fragments (cDNA, RNA) of approx. 30-1000 bp length (depending on the chosen platform and sequencing kit)
- fragment amplification (emulsion PCR, bridge amplification) - higher signal during nucleotide incorporation during sequencing, allowing detection

3nd generation sequencing

- it does not use amplification to increase the signal (should be higher accuracy)
- produces long reads
- good sequencing of GC rich areas
- Epigenetics
- PacBio and Nanopore (MinION)

Third generation



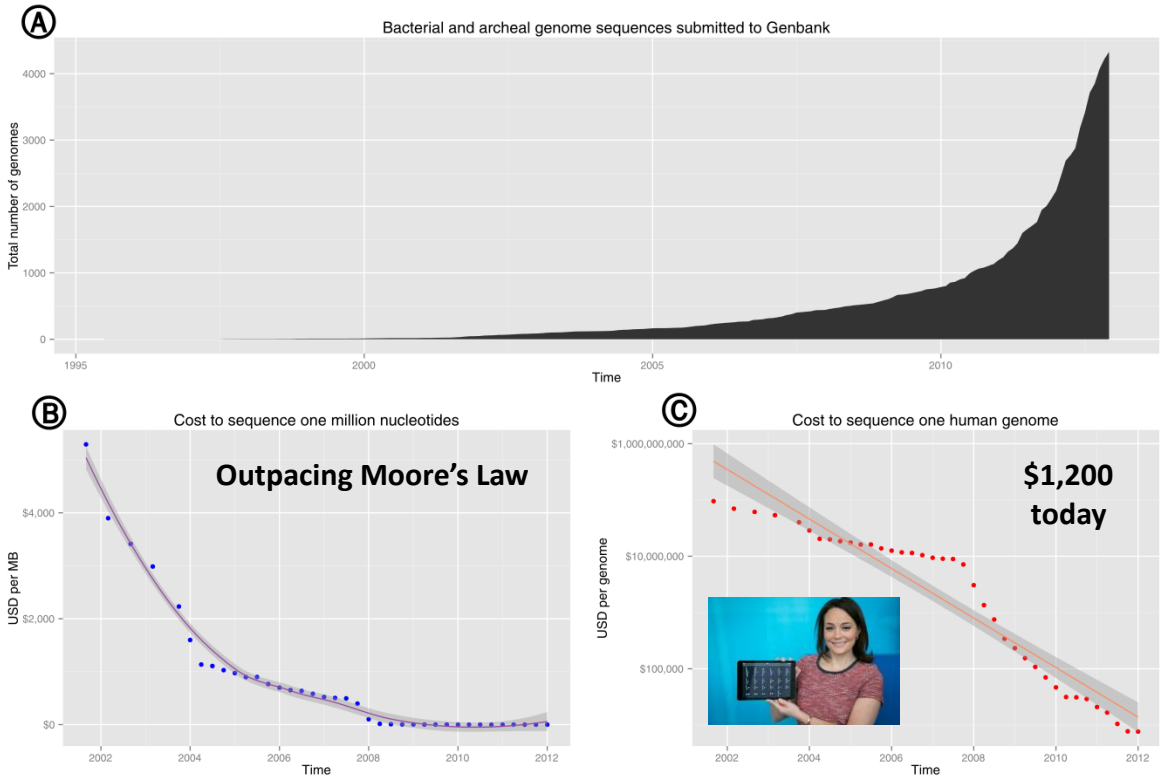
PacBio
Oxford Nanopore

Sequence native DNA in real time
with single-molecule resolution

Tens of kb fragments, on average

Long-read sequencing

High throughput sequencing technologies have become essential in studies on genomics



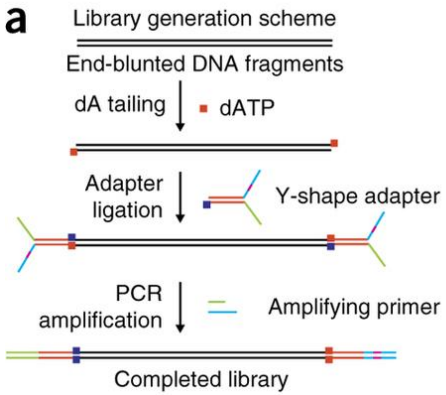
The number of genome projects has increased as technological improvements continue to lower the cost of sequencing. **(A)** Exponential growth of genome sequence databases since 1995. **(B)** The cost in US Dollars (USD) to sequence one million bases. **(C)** The cost in USD to sequence a 3,000 Mb (human-sized) genome on a log-transformed scale.

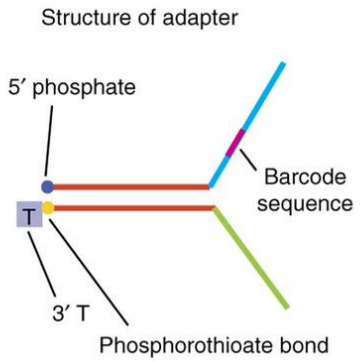
Next (Second) Generation Sequencing - short fragments, high throughput, low price…

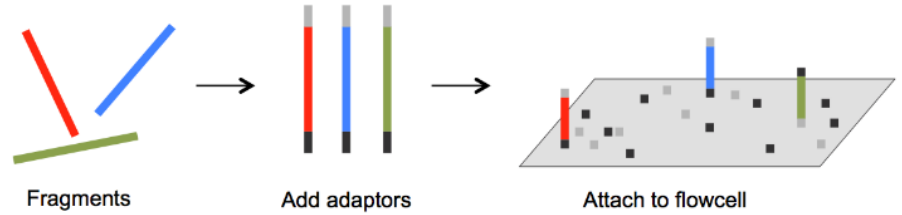


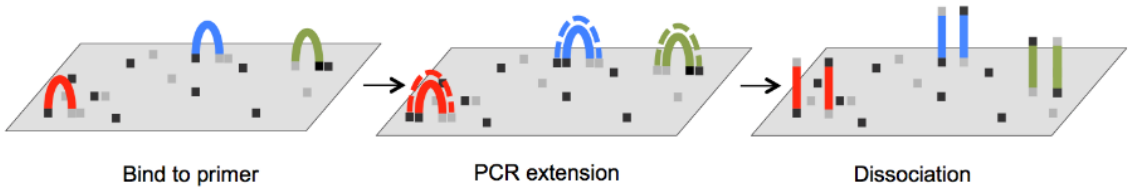
Illumina sequencing

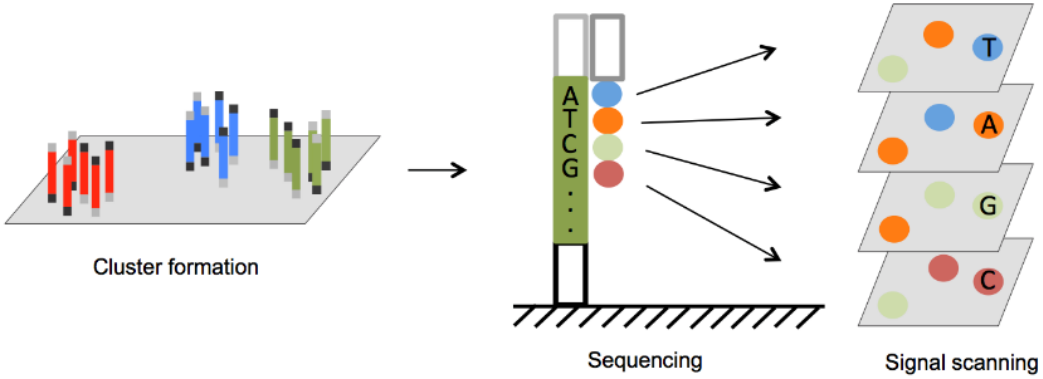
- reversible dye-terminators
- identification of single bases introduced into DNA strands
- DNA attaches to the flow cell via complementary sequences.
- strand bends over and attaches to a second oligo forming a bridge
- polymerase synthesizes the reverse strand
- two strands release and straighten. Each forms a new bridge (bridge amplification)
- result is a cluster of DNA forward and reverse strands clones.



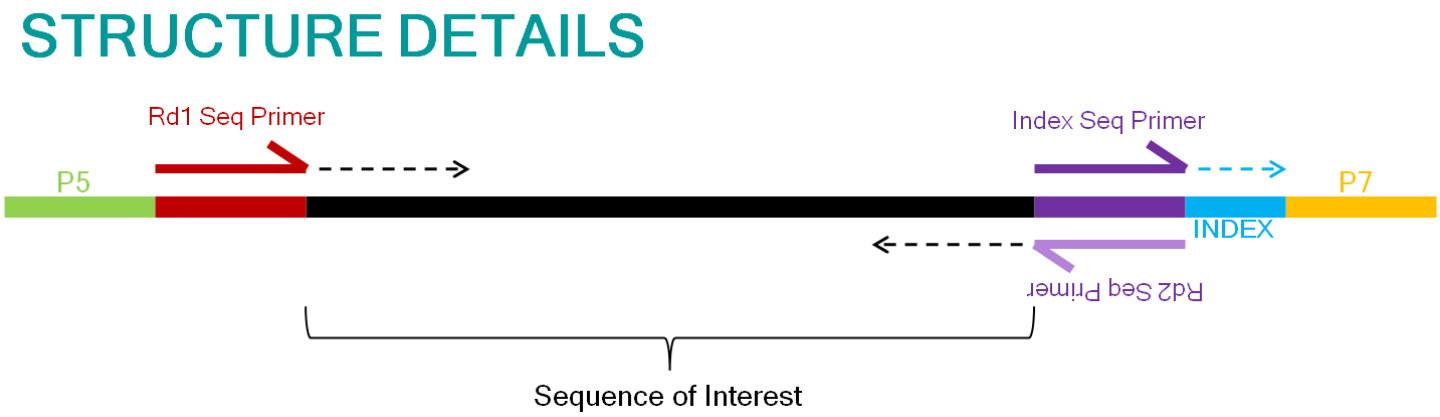


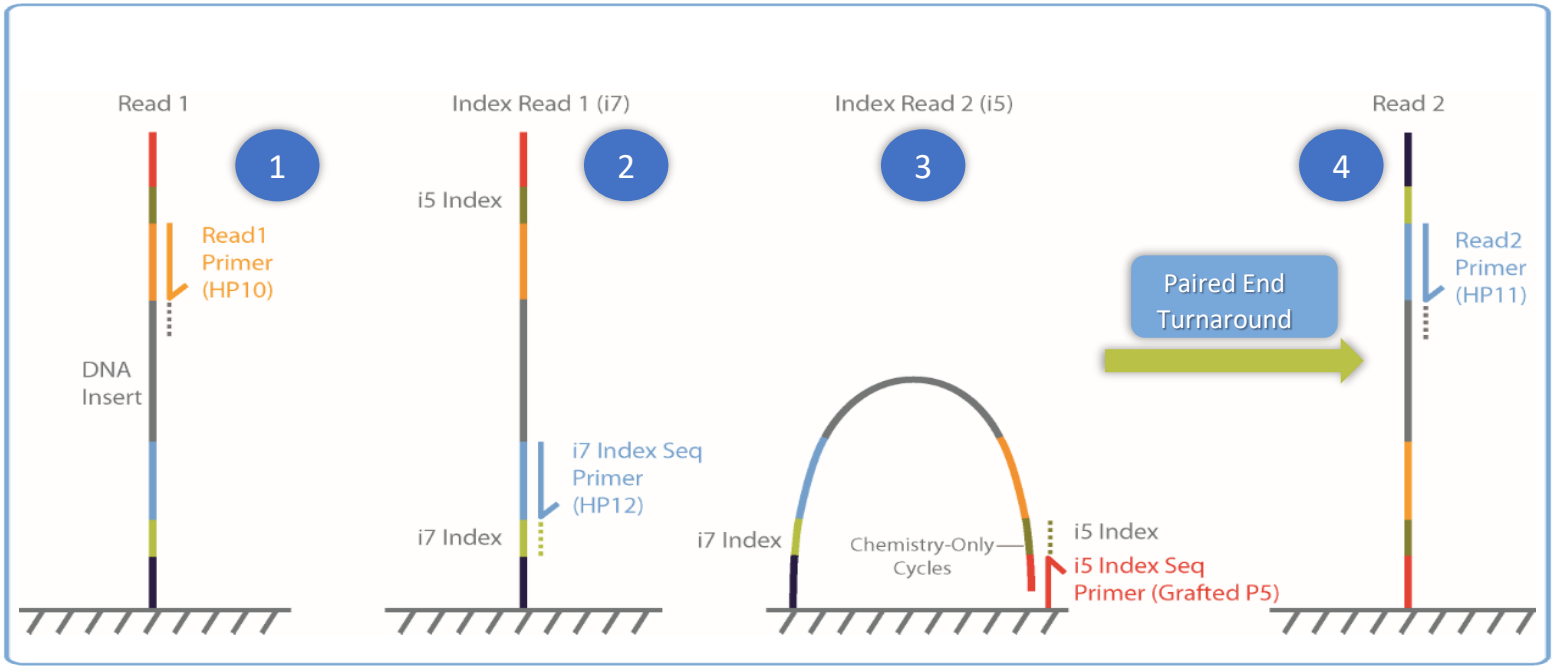






Illumina Sequencing

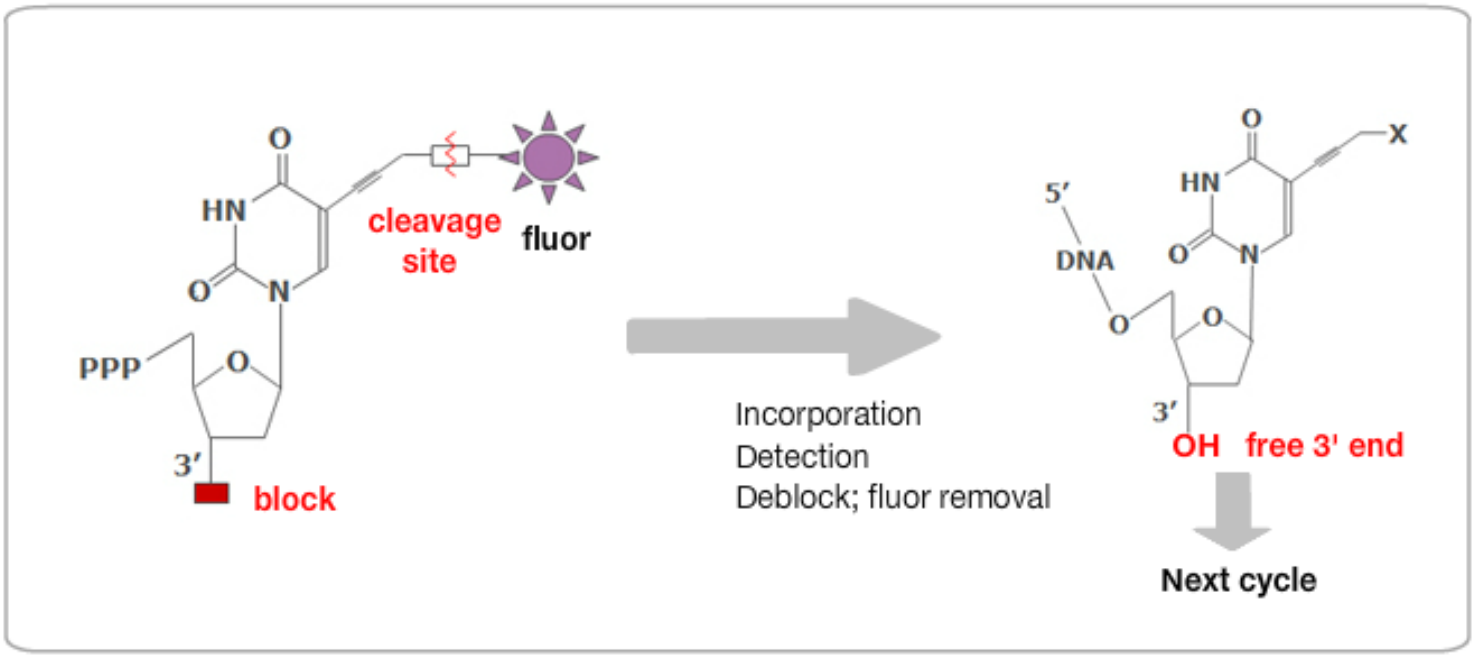


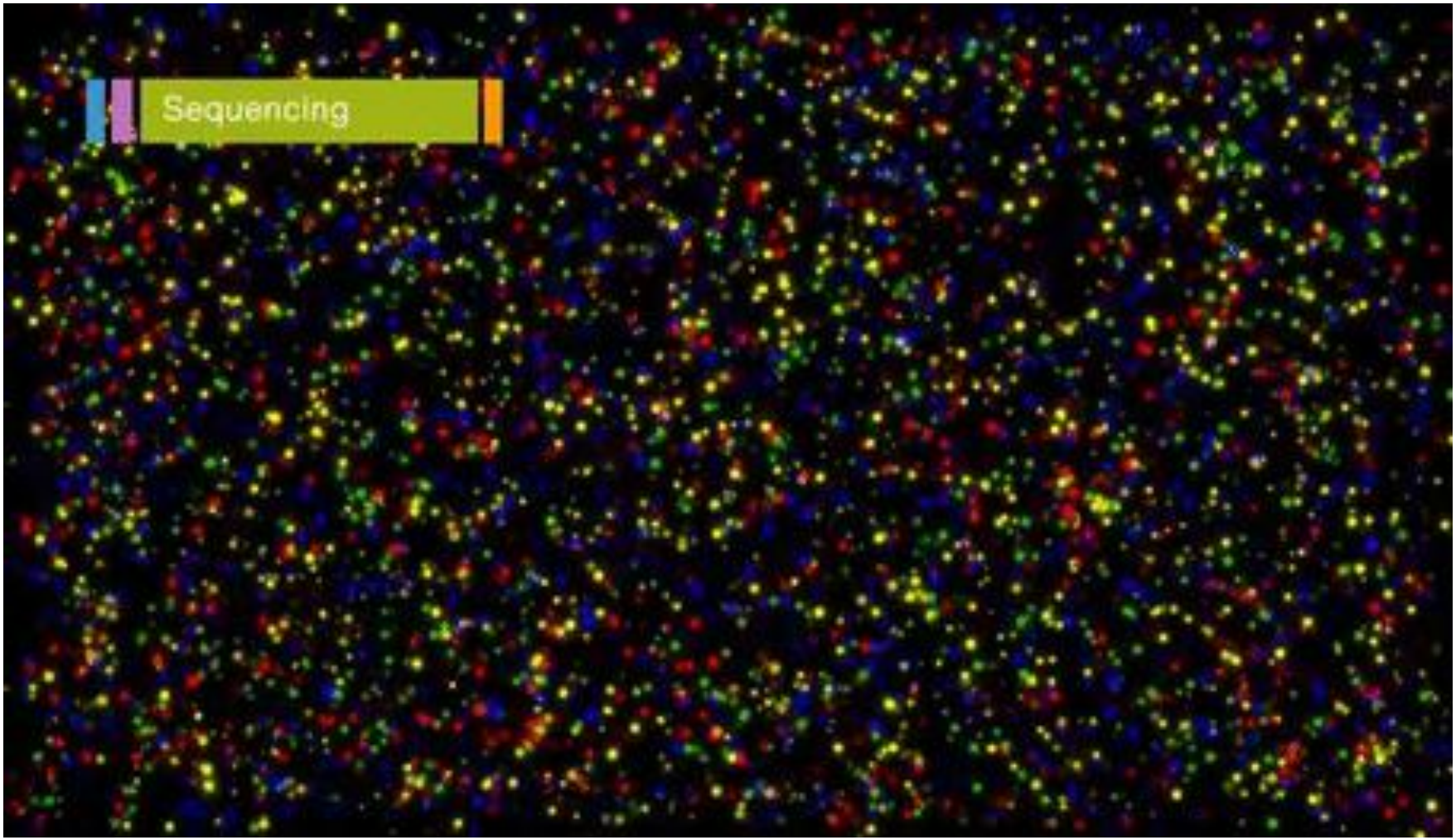


Detection: fluorescently labelled bases with reversible terminator

The MiSeq sequences the DNA clusters using Illumina's Sequencing By Sythesis (SBS) Chemistry which relies on Reversible Terminator Chemistry (RTC).

- All 4 labeled nucleotides in 1 reaction
- Higher accuracy




Sequencing

Illumina



https://www.youtube.com/watch?v=fCd6B5HRaZ8

3rd gen sequencing – Single molecule sequencing

- initially very low accuracy: single molecule sequencing leads to ↓ signal to noise ratio
- still lower throughput than 2nd generation
- main technologies are:
  - PacBio CLR/HiFi – sequencing by template synthesis
  - Oxford Nanopore Sequencing (ONT) – sequencing by 'flow'

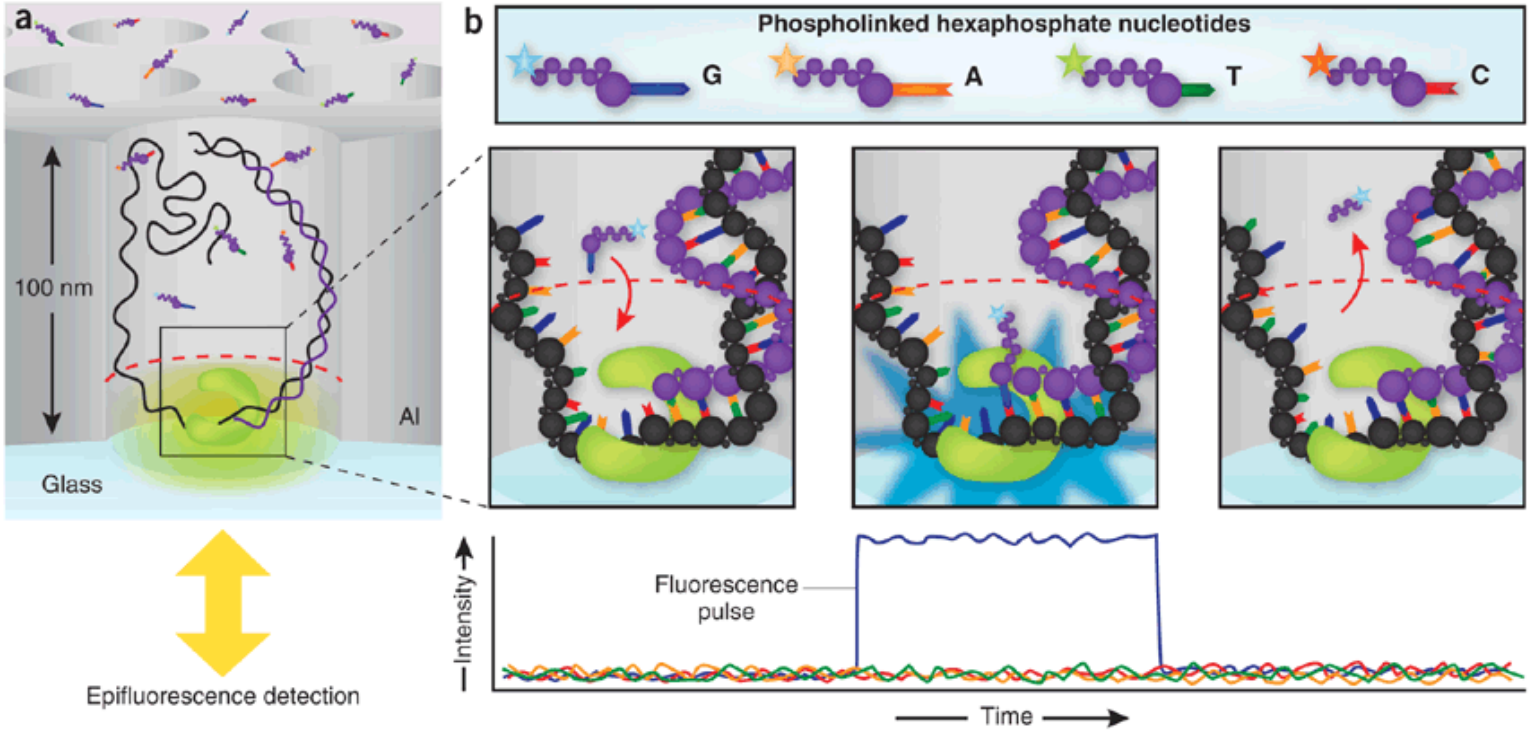




3rd gen sequencing – Single molecule sequencing

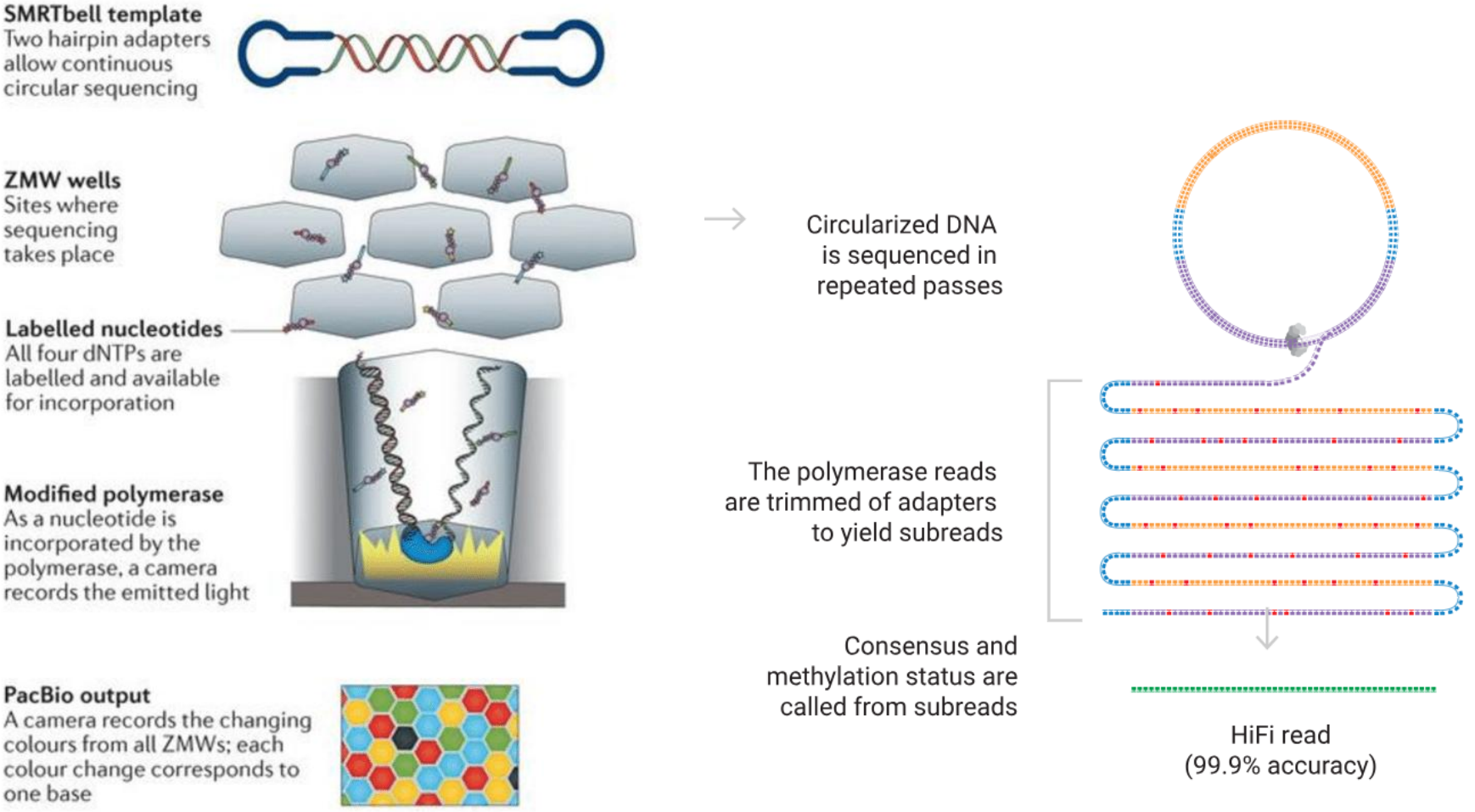
Single molecule real time sequencing (**SMRT**) is a parallelized single molecule real time DNA sequencing method. **DNA polymerase enzyme is affixed at the bottom of a zero-mode waveguide** (ZMW). The ZMW is a structure that creates an illuminated observation volume that is small enough to observe only a single nucleotide of DNA being incorporated by DNA polymerase. **Each of the four DNA bases is attached to one of four different fluorescent dyes**. When a nucleotide is incorporated by the DNA polymerase, the fluorescent tag is cleaved off and diffuses out of the observation area of the ZMW where its fluorescence is no longer observable.





PacBio – HiFi long reads

- initially accuracy was ~80%
- sequences up to 25kb
- same template is sequenced several times
- longer reads, less passes, lower accuracy

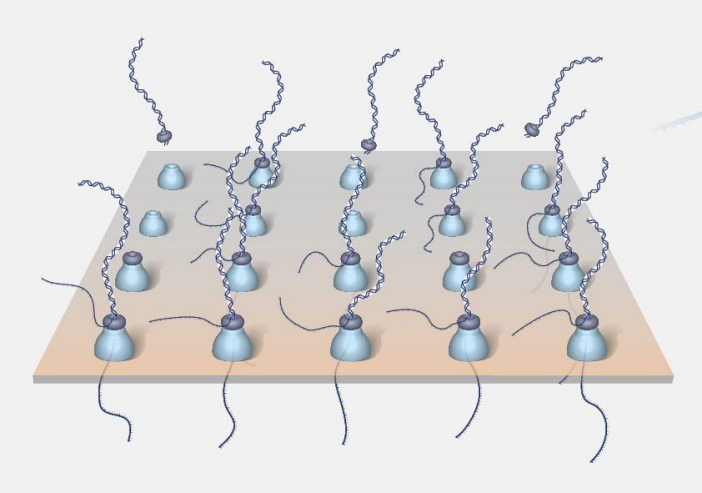


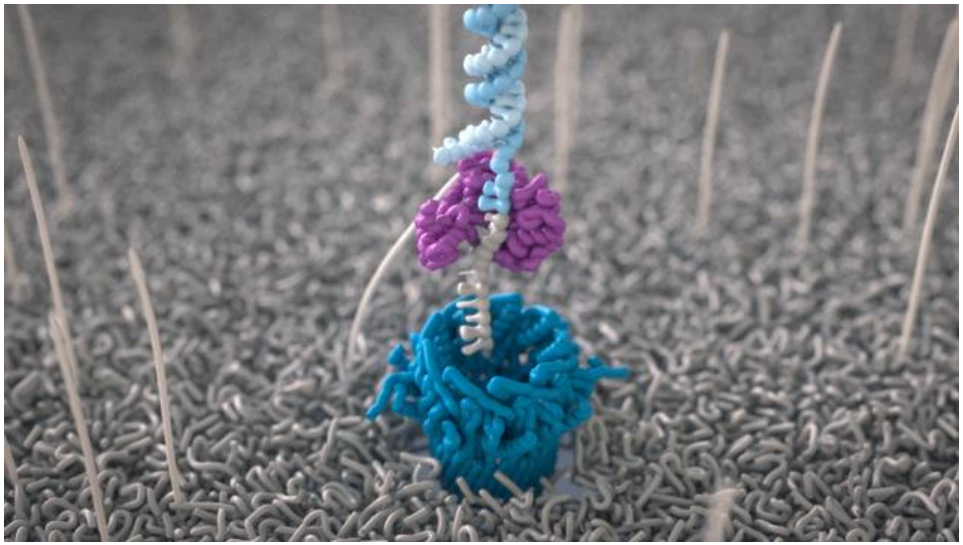
PacBio



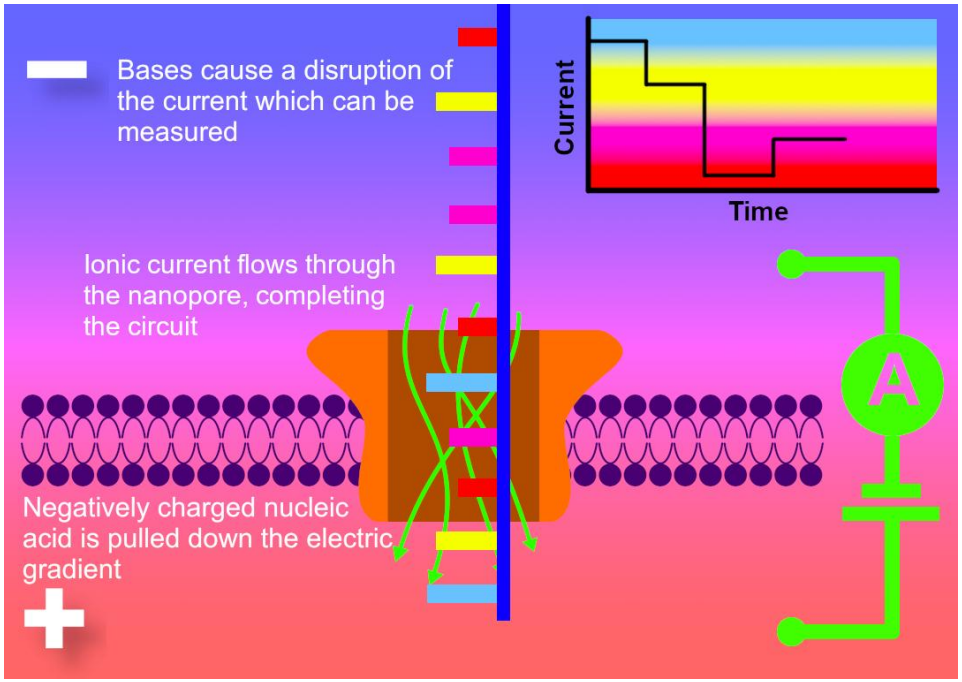
https://www.youtube.com/watch?v=_lD8JyAbwEo

Oxford Nanopore sequencing





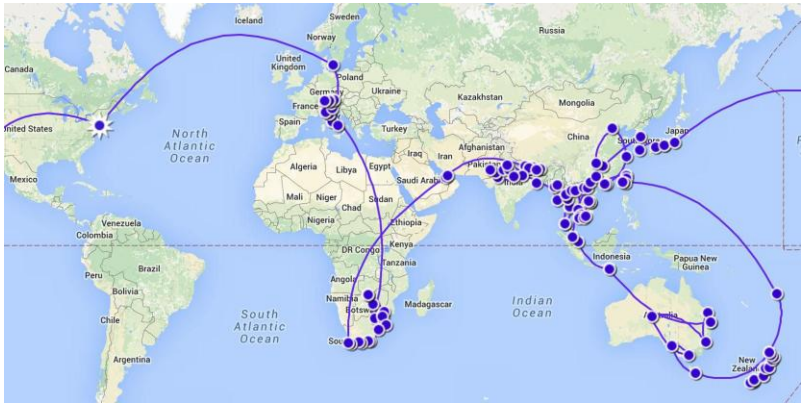
- Sequences native DNA and RNA molecules
- No template amplification; no immobilization
- Single strand moves through pore by potential difference: same principle of electrophoresis



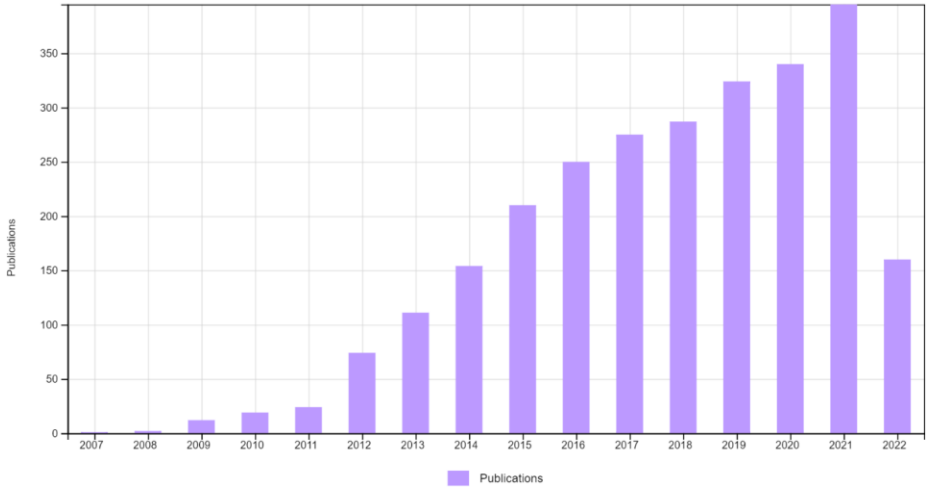
Fungal amplicon data

How to study global fungal distribution?

Travel around the world …



… or based on published data



Fungal amplicon data - resources

unite community

https://unite.ut.ee/

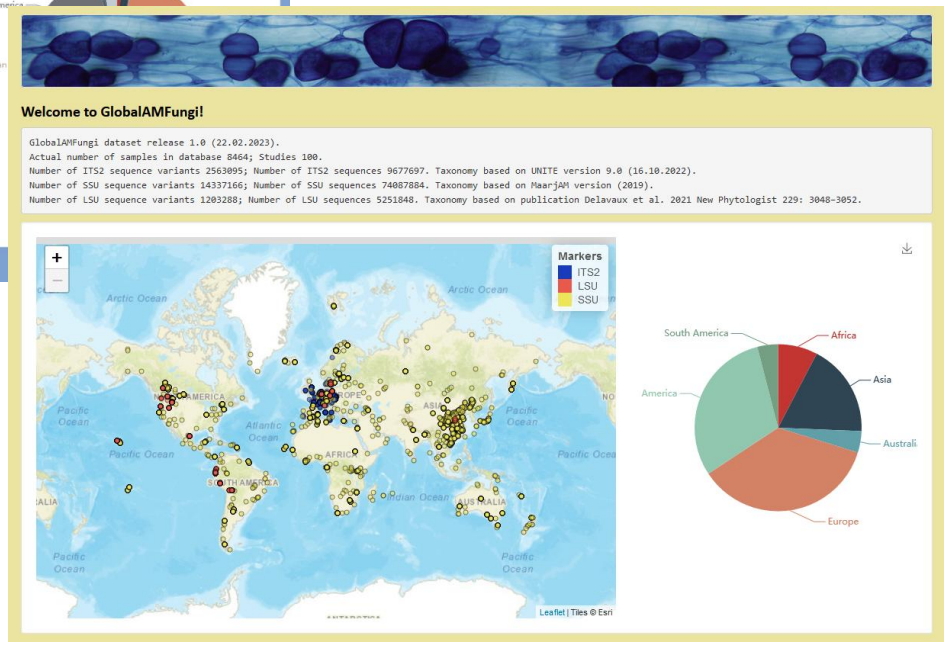


GlobalFungi

https://globalfungi.com/

GlobalAMFungi

https://globalamfungi.com/



GlobalFungi (Nov 2023: Release 5)

846 studií -> 84 972 vzorků

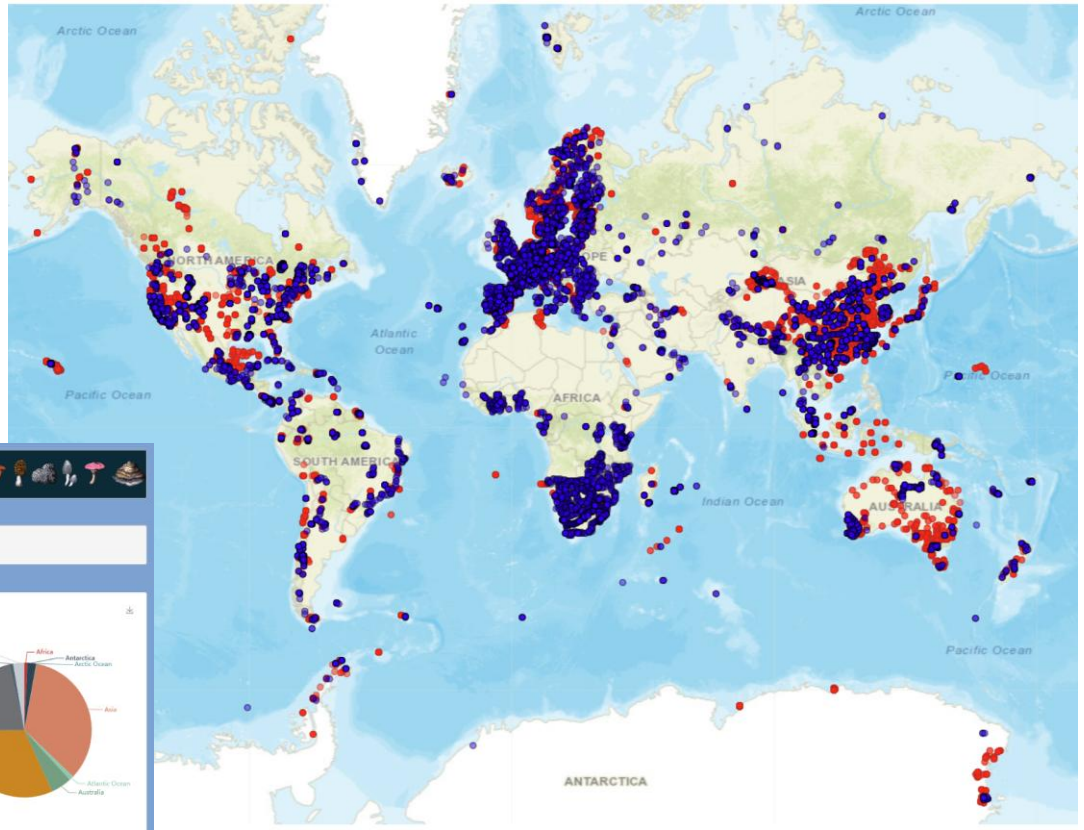
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1 233 mil. ITS1 sekvencí
3 474 mil. ITS2 sekvencí

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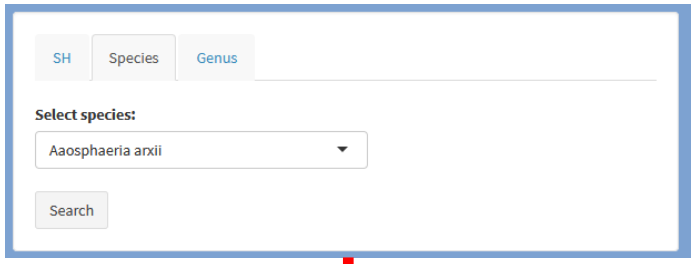
593 mil. ITS variant
Taxonomie - UNITE 10 (4.4.2024)

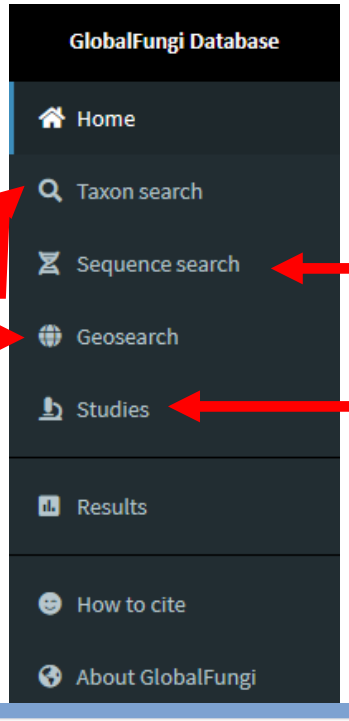


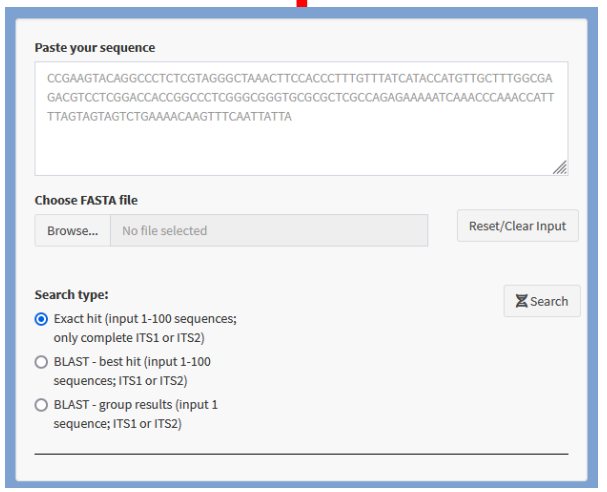


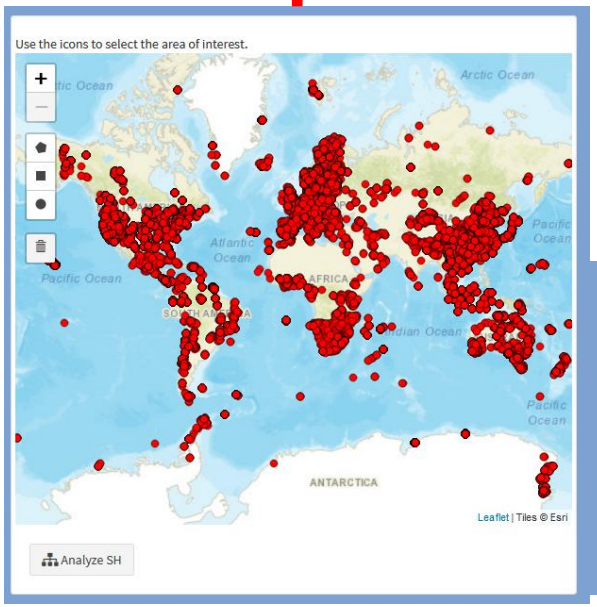
Vzorky z předchozí verze (červeně); Nové vzorky (modře)

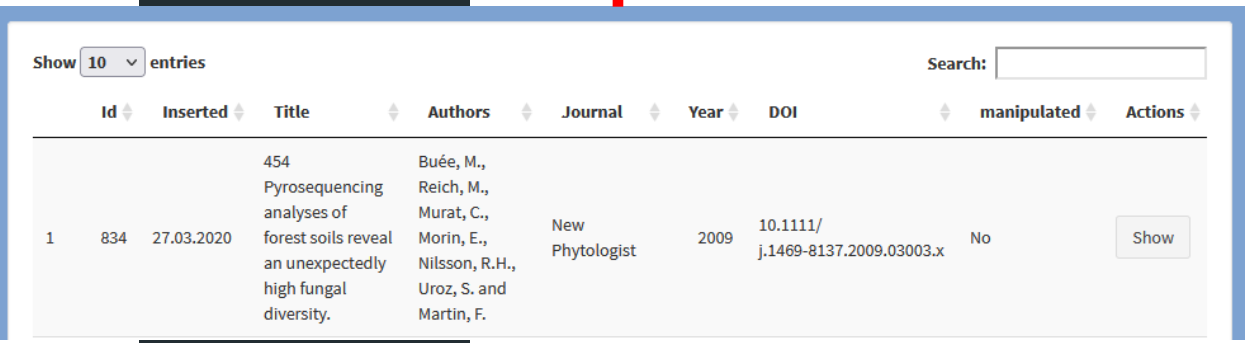
Main features







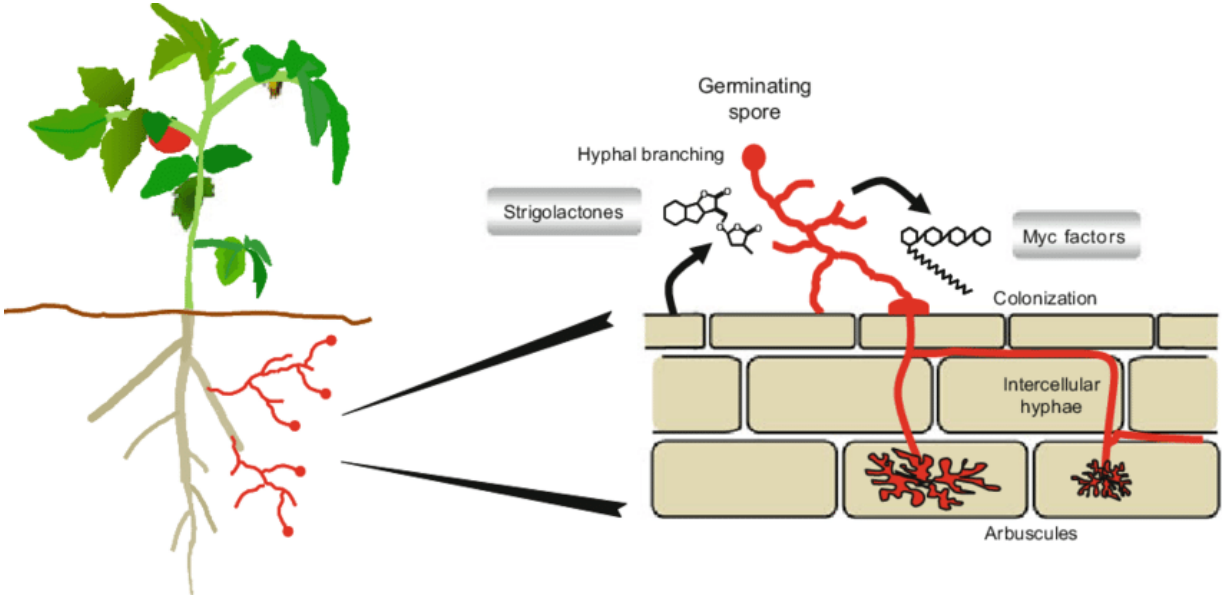


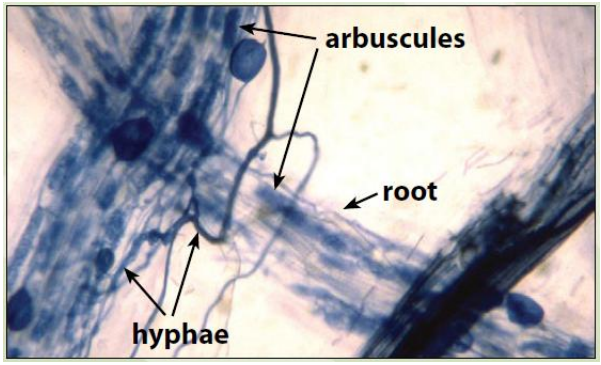


GlobalAMFungi

Arbuscular mycorrhizal (AM) fungi

- crucial mutualistic symbionts of the majority of plant species (> 70%)
- essential roles in plant nutrient uptake and stress mitigation (great importance for agriculture)
- limited knowledge of the biogeography of AM fungi and the environmental factors that influence it
- there are currently 332 described





GlobalAMFungi

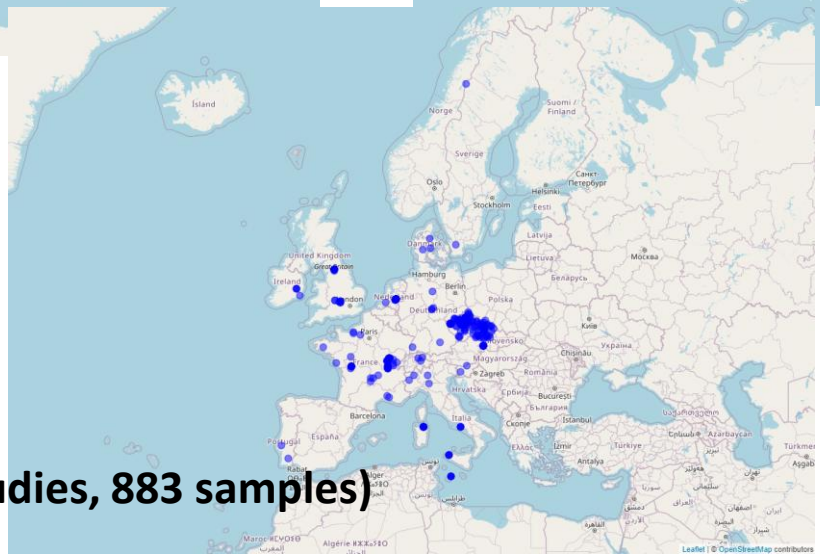
SSU (75 studies, 6912 samples)



LSU (17 studies, 669 samples)

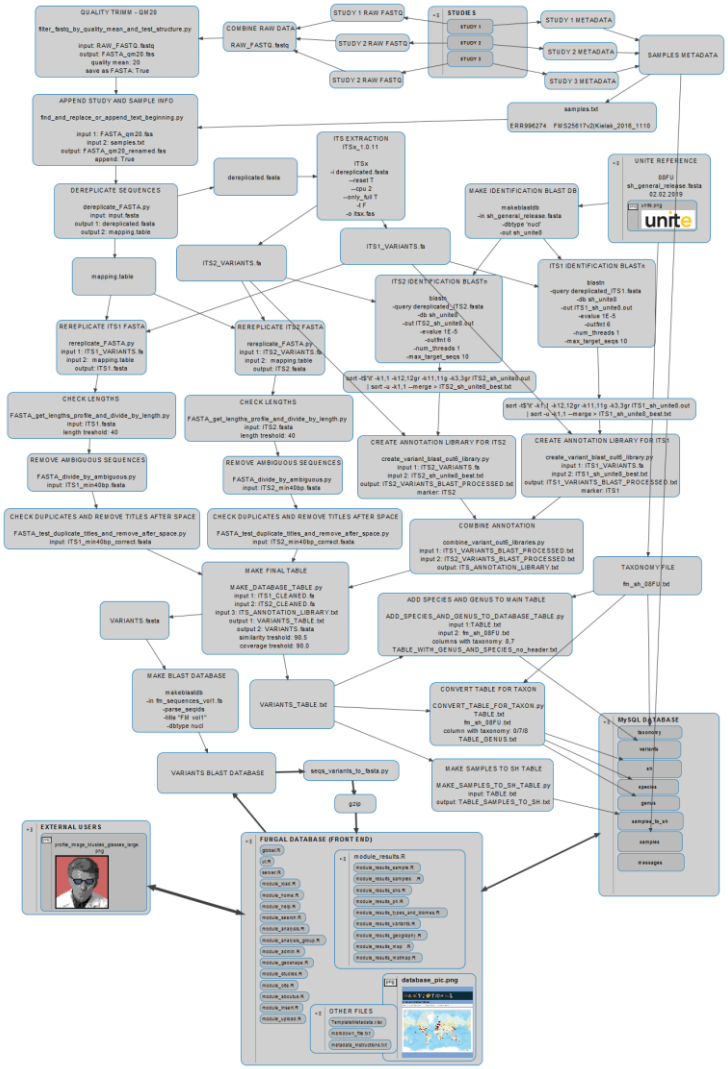


ITS2 (8 studies, 883 samples)

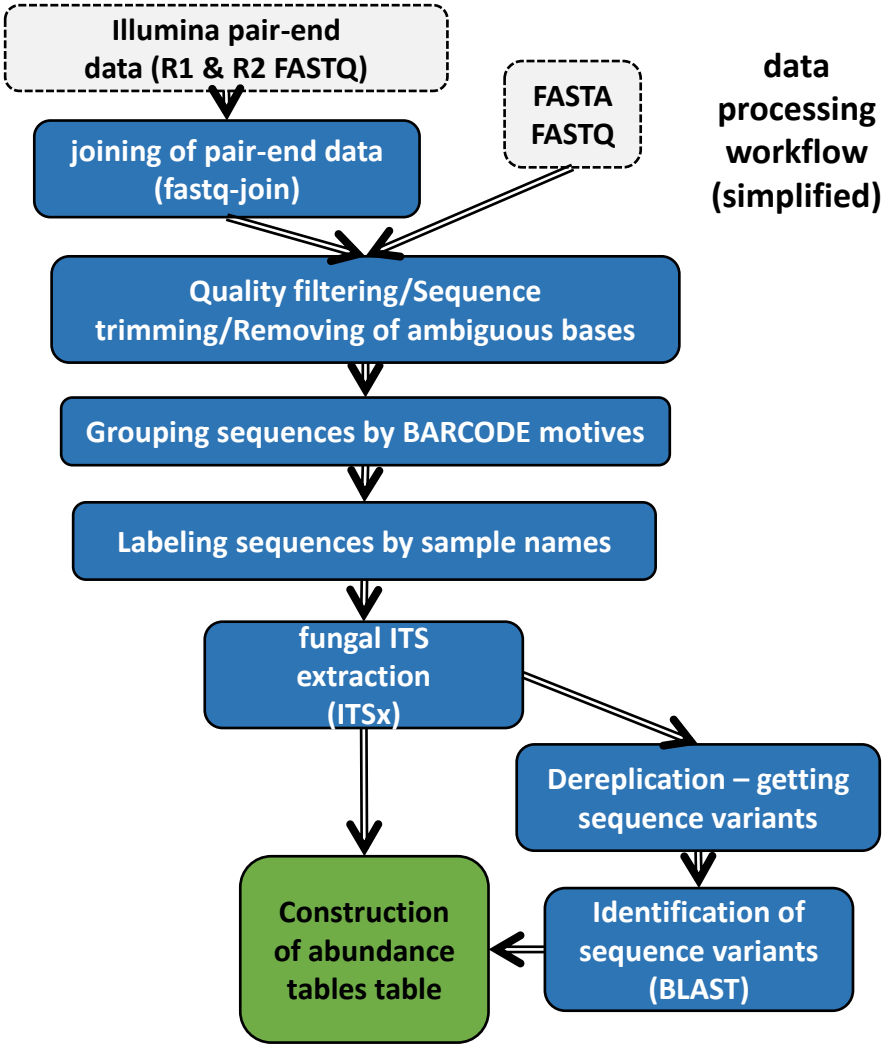


Bioinformatics behind it…

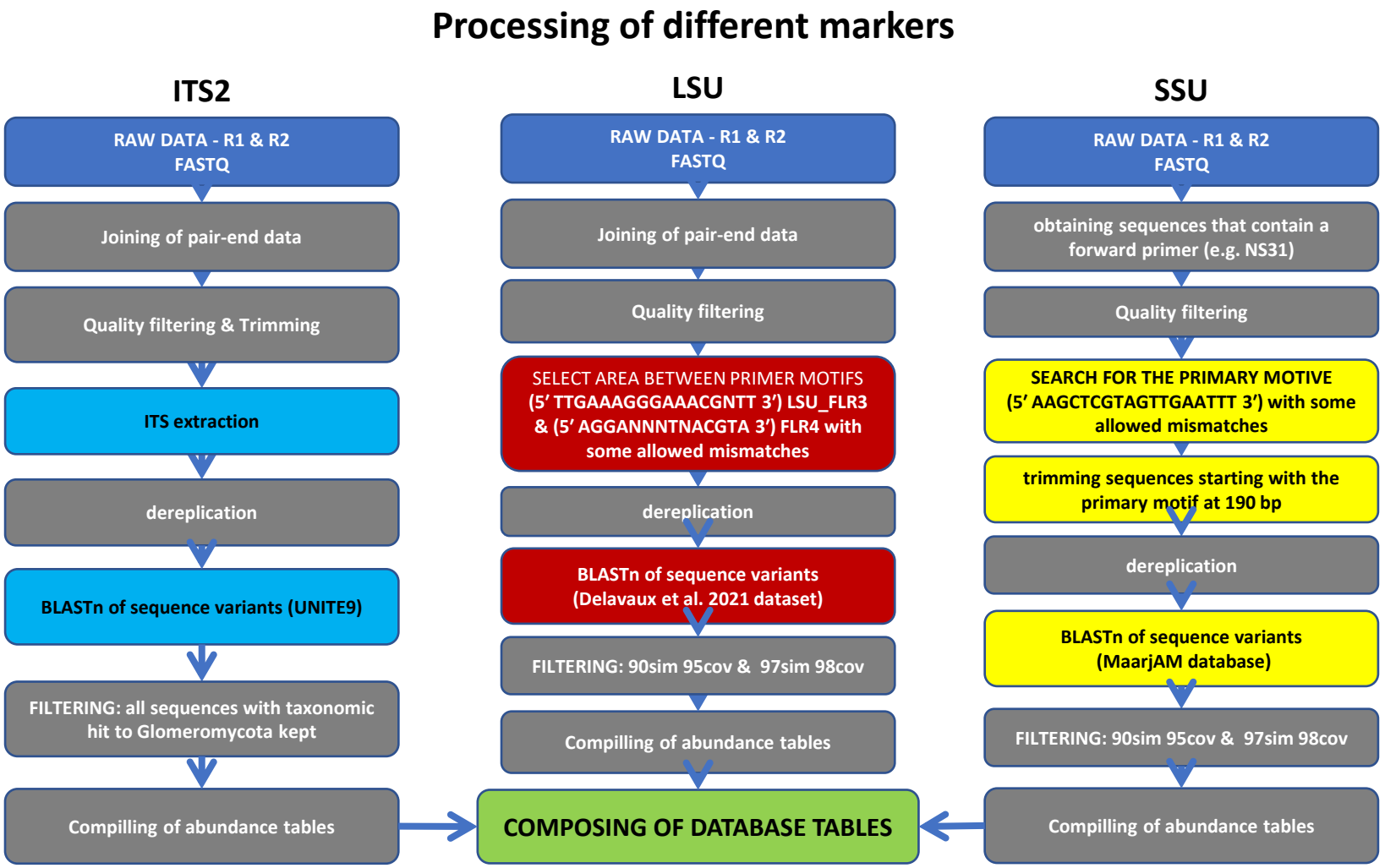
data processing workflow (complete)



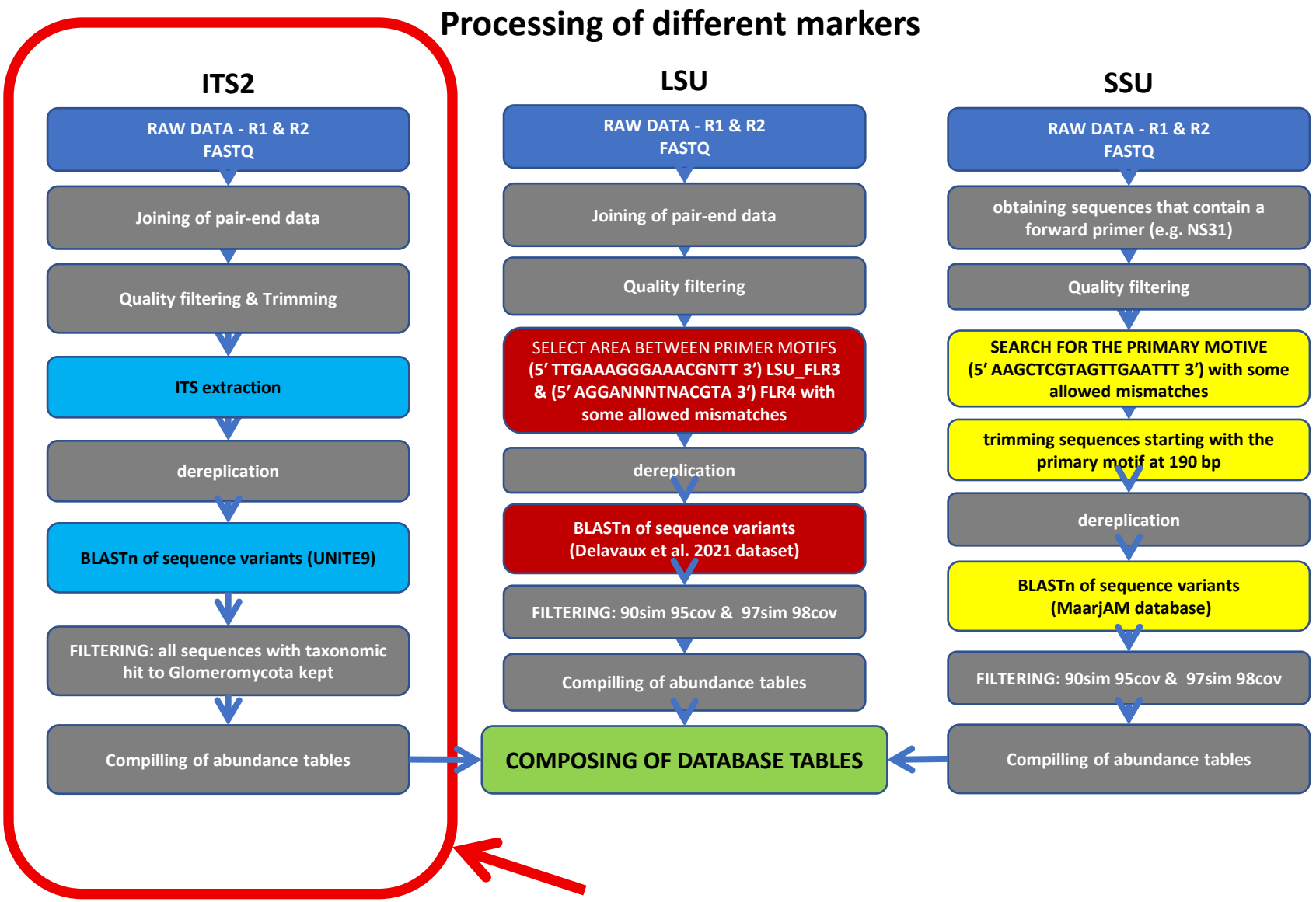
data processing workflow (simplified)



Bioinformatics behind it…

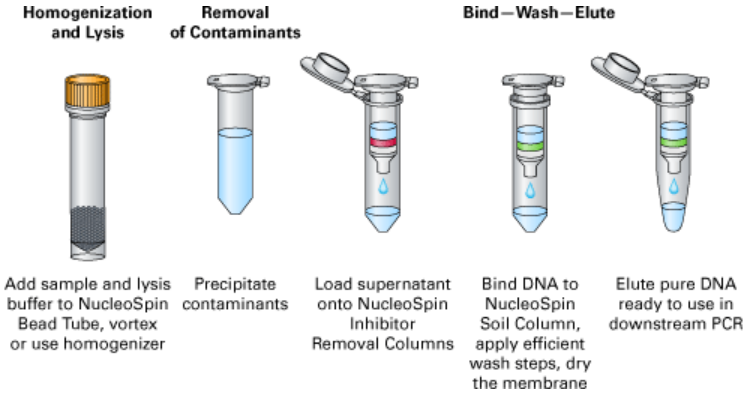


Bioinformatics behind it…

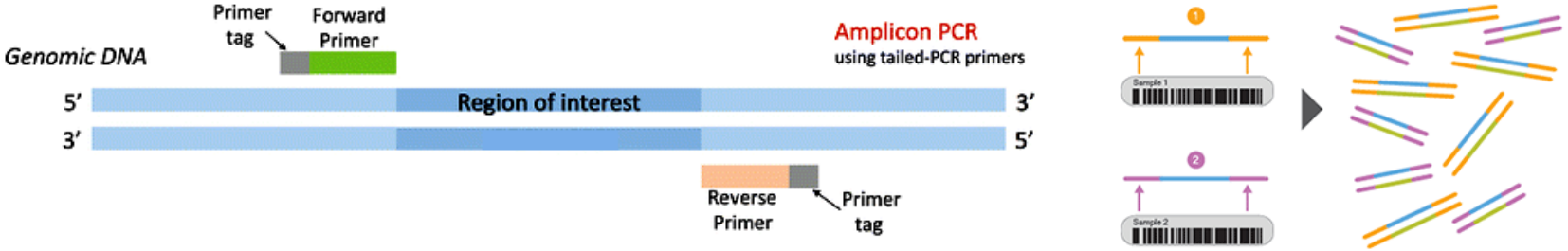


Illumina amplicon sequence library preparation

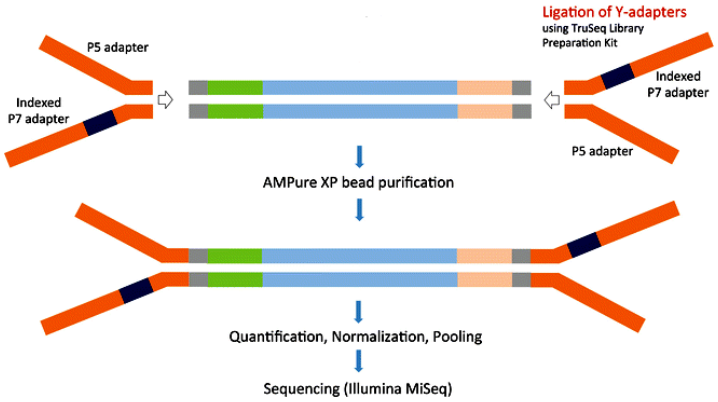
1. DNA isolation



2. PCR with barcoded primers (multiplexing)



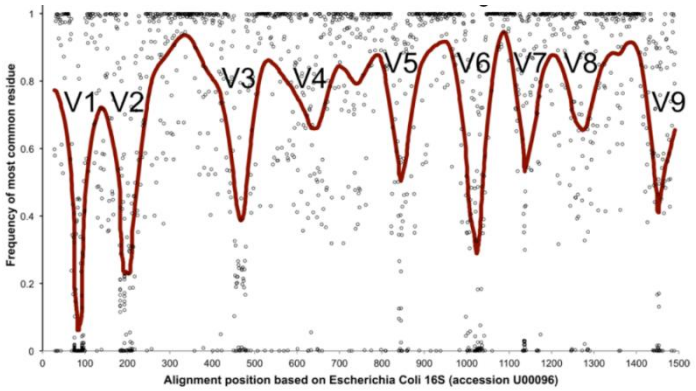
3. Ligation of sequencing adapters – to attach short oligonucleotides (60bp) to your DNA fragments, these oligonucleotides are used to attach to the sequencing flow cell and they are also used as barcode of library



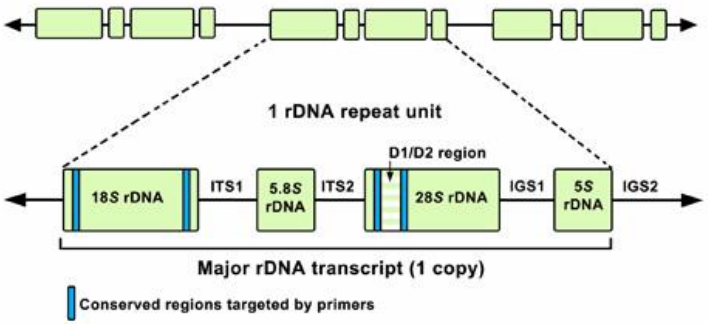
4. Quantification of the library by qPCR – to quantify of the exact amount of ligated fragments

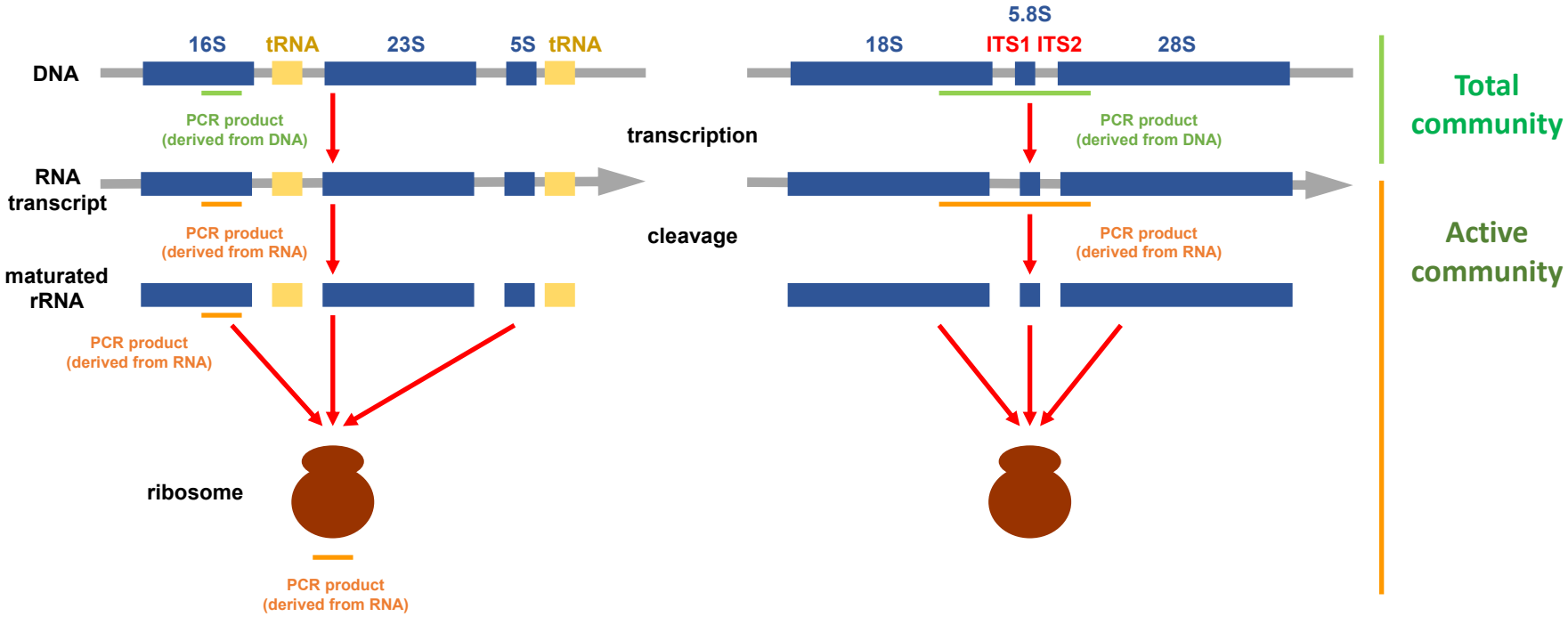
Most used marker genes

Bacterial 16S ribosomal RNA gene



Fungal internal transcribed spacer (ITS)



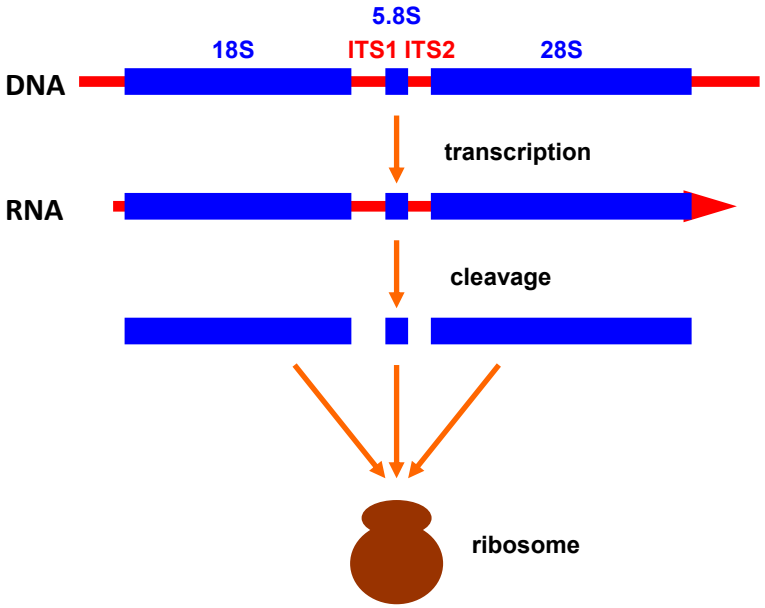


1-15 copies of rDNA per genome

X0–X00? copies of ITS per genome

Fungal markers used for taxonomical identification

ITS region is considered the universal barcode of Fungi



>bfe0c1ab088e7e908fc275fe7b95db91
AATTCTCAACCATGTCTTGAATTAGGCATGGCTTGGACTTGGGGGTTGCTGGCACACGTTGTC
AAGCTCTCCTGAAATGCATTAGCGATGTTCAGCAAGCCTGAACGTGCACGGCCTTTTCGACGT
GATAACGATCGTCGTGGGCTGGAGCGTAGGGTGAGCAGTGAATCGCTTCTAATCTATACCTC
AGTTACTAGTCGGTCGTGAGACTGACGAACGAGCGAGGCTTGATCTTGGACCCCAAGCCTTA
GTTACTAGTCGTTCGTGAGGCCGACGAACGCGCAGGGTTAGGACCATTTGAAC

ITS identification based Species Hypothesis concept

unite community

https://unite.ut.ee/

Boletus edulis (SH1644963.08FU)

Kingdom	Phylum	Class	Order	Family	Genus	Species
Fungi	Basidiomycota	Agaricomycetes	Boletales	Boletaceae	Boletus	Boletus edulis

PNAS

Nuclear ribosomal internal transcribed spacer (ITS) region as a universal DNA barcode marker for *Fungi*

Conrad L. Schoch[a,1], Keith A. Seifert[b,1], Sabine Huhndorf[c], Vincent Robert[d], John L. Spouge[a], C. André Levesque[b], Wen Chen[b], and Fungal Barcoding Consortium[a,2]

[a]National Center for Biotechnology Information, National Library of Medicine, National Institutes of Health, Bethesda, MD 20892; [b]Biodiversity (Mycology and Microbiology), Agriculture and Agri-Food Canada, Ottawa, ON, Canada K1A 0C6; [c]Department of Botany, The Field Museum, Chicago, IL 60605; and [d]Centraalbureau voor Schimmelcultures Fungal Biodiversity Centre (CBS-KNAW), 3508 AD, Utrecht, The Netherlands

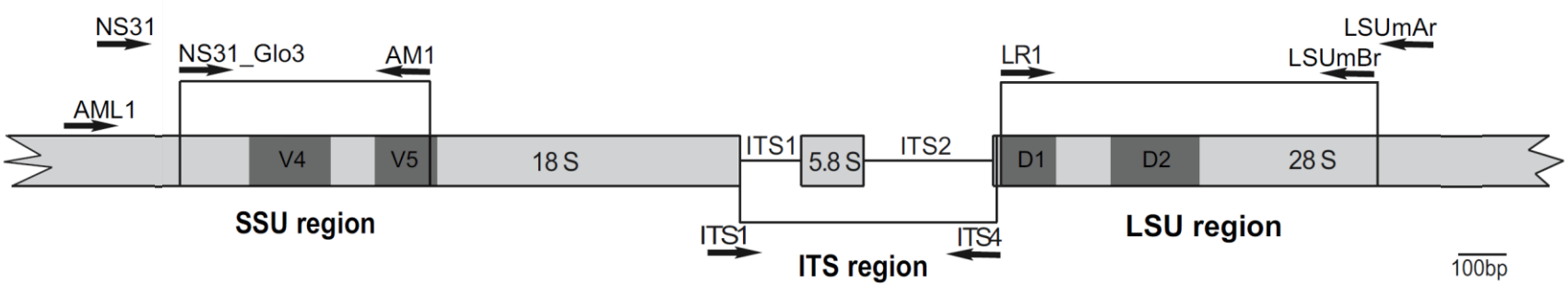
Edited* by Daniel H. Janzen, University of Pennsylvania, Philadelphia, PA, and approved February 24, 2012 (received for review October 18, 2011)

Koljalg et al. 2013 Molecular Ecology

Identification of arbuscular mycorrhizal fungi (AMF)

AMF are underrepresented in studies using general fungal primers (GlobalFungi)

- Three different barcoding regions – SSU, ITS, LSU
- SSU region is mostly used in studies focused on community ecology of AMF
- ITS and LSU regions are largely used by AMF taxonomists



AMF specific primers

Using long reads to cover multiple markers

SSU, ITS, LSU

X

Long reads (PacBio)

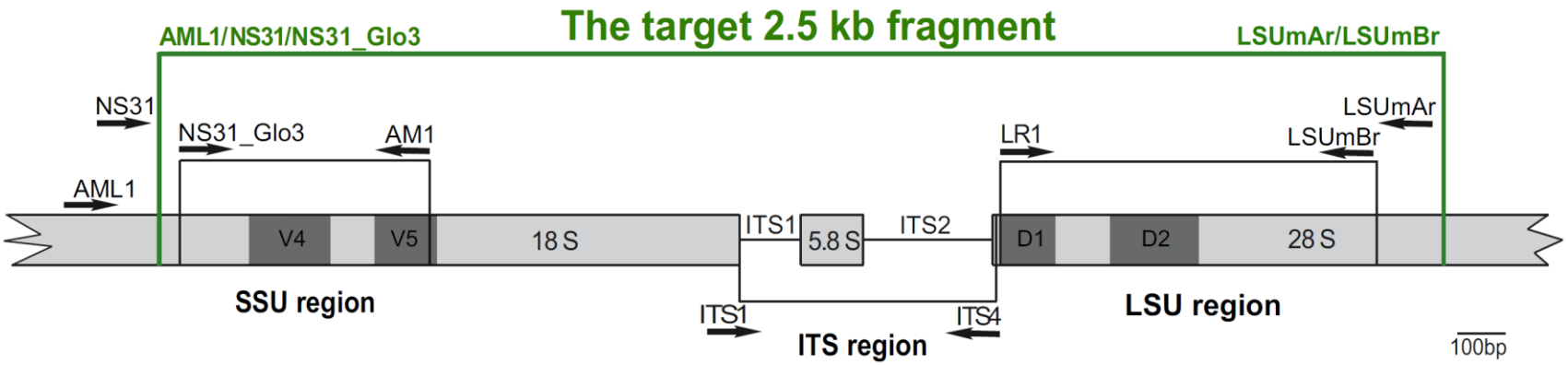
New Phytologist

Research

PacBio sequencing of Glomeromycota rDNA: a novel amplicon covering all widely used ribosomal barcoding regions and its applicability in taxonomy and ecology of arbuscular mycorrhizal fungi

Zuzana Kolaříková[1], Renata Slavíková[1], Claudia Krüger[1], Manuela Krüger[1] and Petr Kohout[1,2,3]

[1]Institute of Botany of the Czech Academy of Sciences, Průhonice CZ-252 43, Czech Republic; [2]Faculty of Science, Charles University in Prague, Prague CZ-128 44, Czech Republic; [3]Institute of Microbiology of the Czech Academy of Sciences, Prague CZ-142 20, Czech Republic



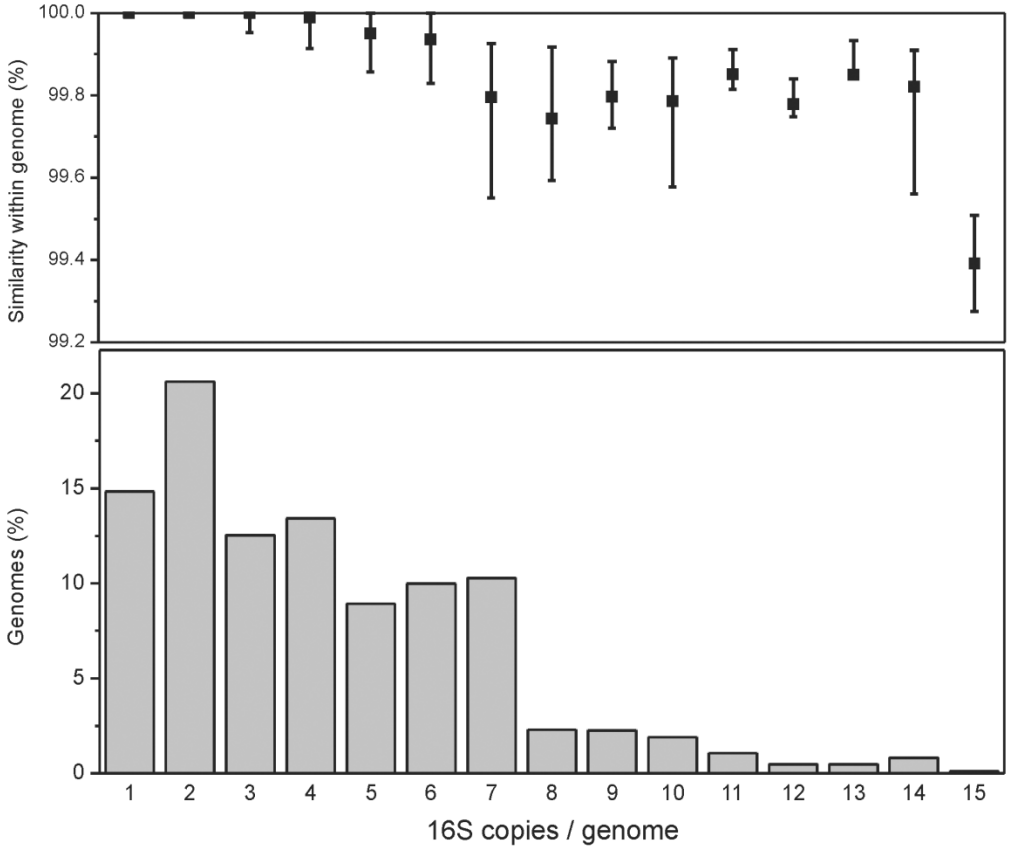
Kolaříková et al. 2021 New Phytologist

16S rDNA gene vs. alternative (low-copy) markers

Pros: highly populated reference databases

Cons: muticopy nature of bacterial 16S rDNA gene

- possibility of high intragenomic variability - diversity over estimation (number of OTUs)
- relative abundance estimation is skew -> normalisation by 16S copy number of closest taxon



16S rRNA within-genome similarity and copy numbers in bacterial genomes.

Upper panel: the similarity of genomes with various copy numbers: the values indicated represent the first, the second and the third quartile.

Lower panel: distribution of 16S rRNA copy numbers per genome in 1,690 sequenced bacterial genomes.

T. Větrovský & P. Baldrian - PloS one, 2013

Sequencing platforms for amplicon sequencing (most used)



454 Pyrosequencing

Not supported anymore (most studies 2009-2012)

Errors in homopolymeric regions

long reads (up to 700 bp)



illumina

Illumina

The most used

Error rate less than 1%

pair-end data (Illumina)



ion torrent
by life technologies

IonTorrent

Very chaep sequencing

Lot of errors

Medium read size 200-600 bp



PACIFIC BIOSCIENCES

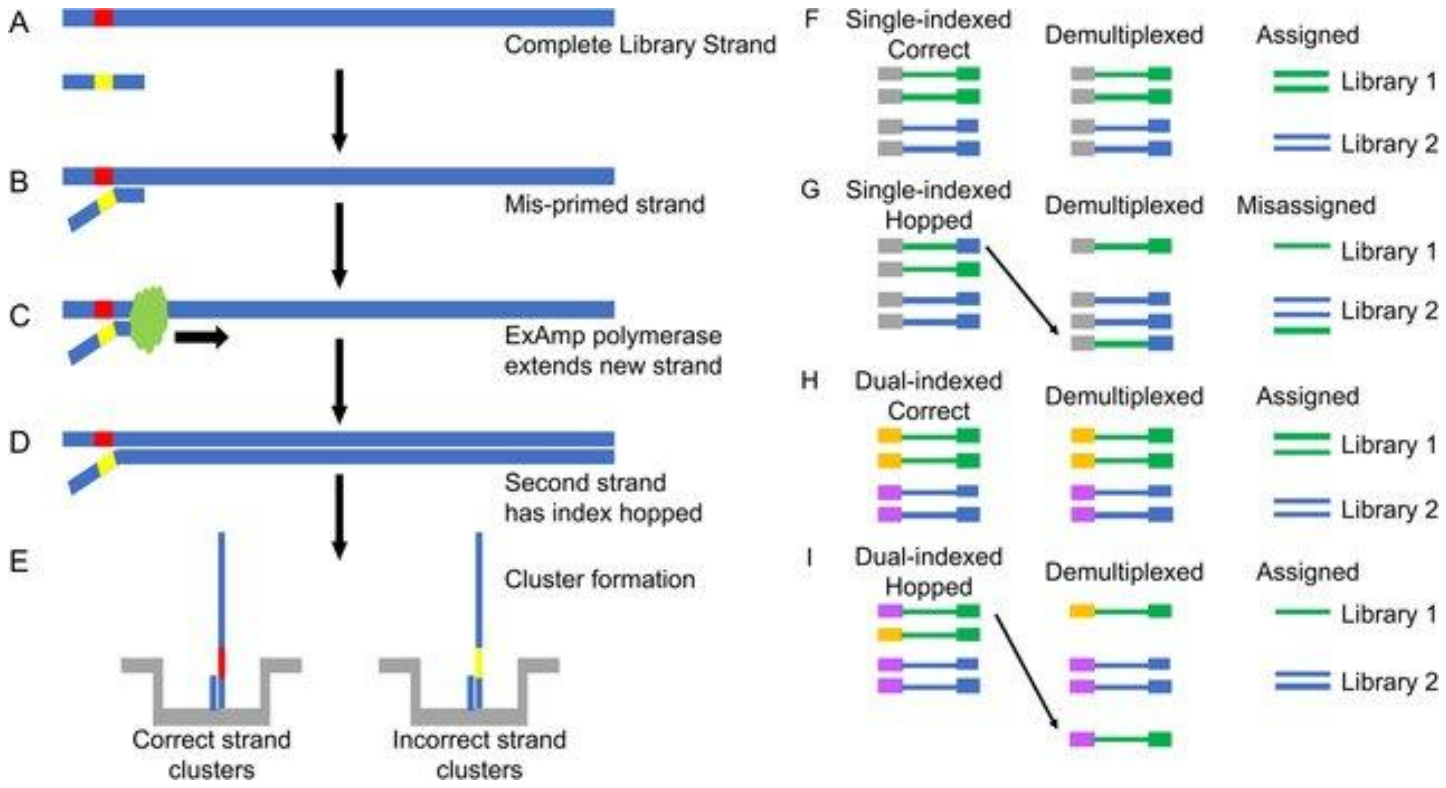
PacBio

Still rare

Repeated sequncing of the same region

extra long reads (up to 10.000 bp)

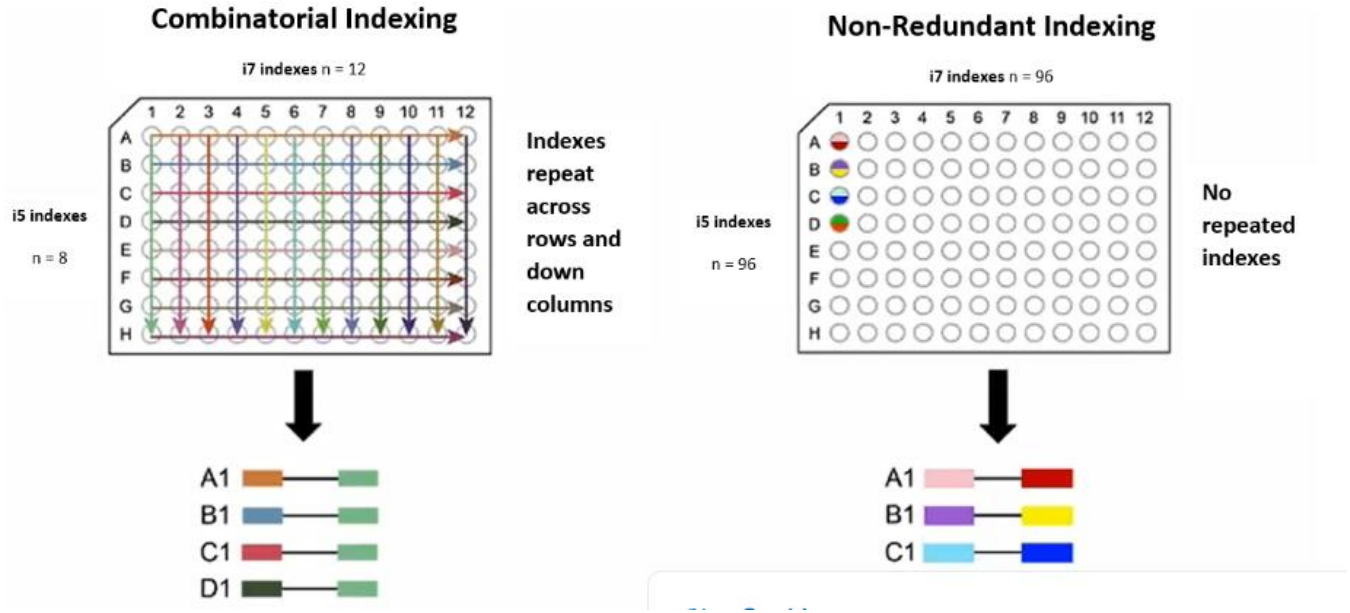
Index hoping



Index hopping occurs due to the **physical incorporation of a sample index from one library into a sample molecule from another library**. The end result is an incorrect read assignment between samples.

Dual Indexes

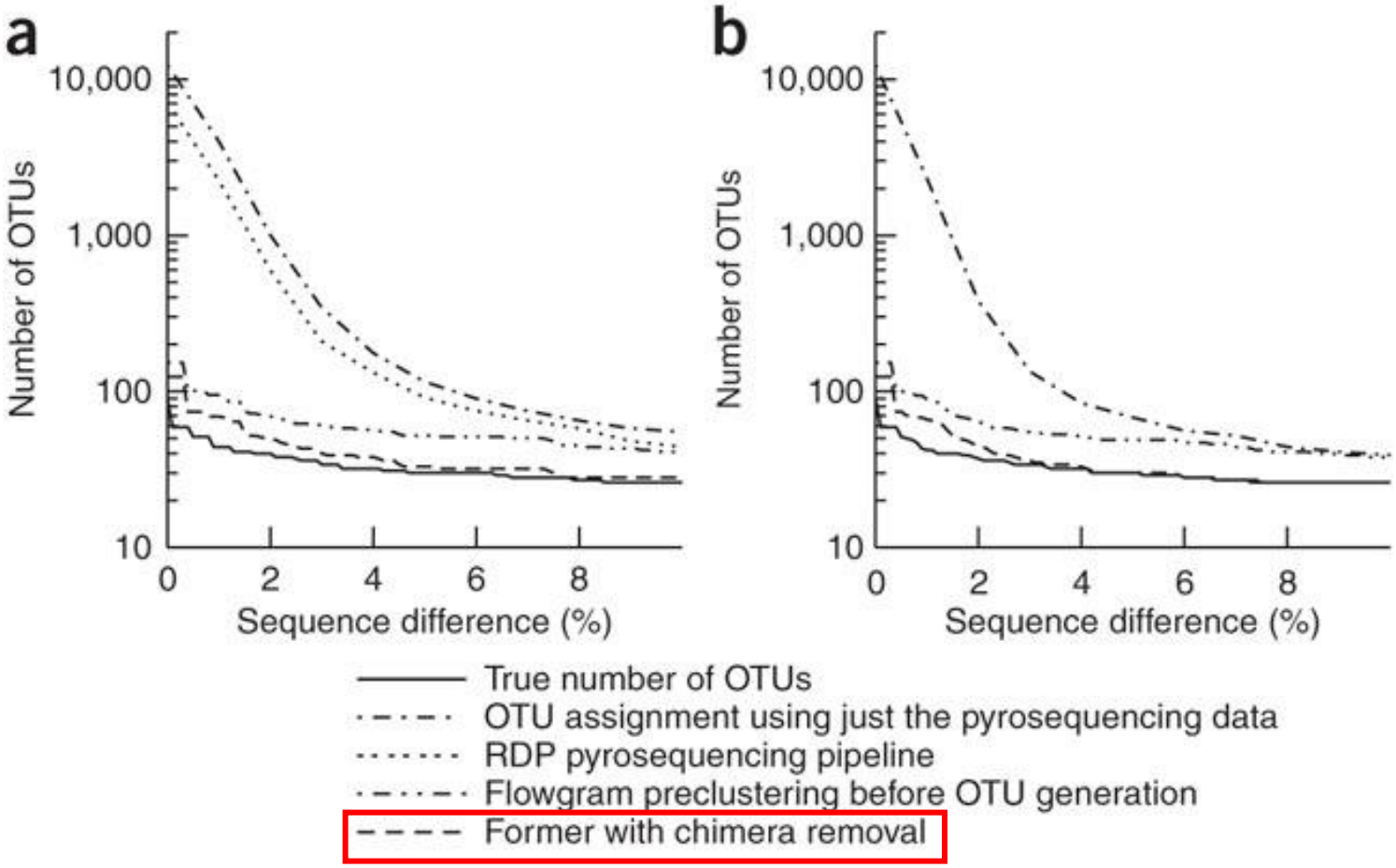
UDI (Unique Dual Indexes)



https://knowledge.illumina.com/library-preparation/general/library-preparation-general-reference_material-list/000002344

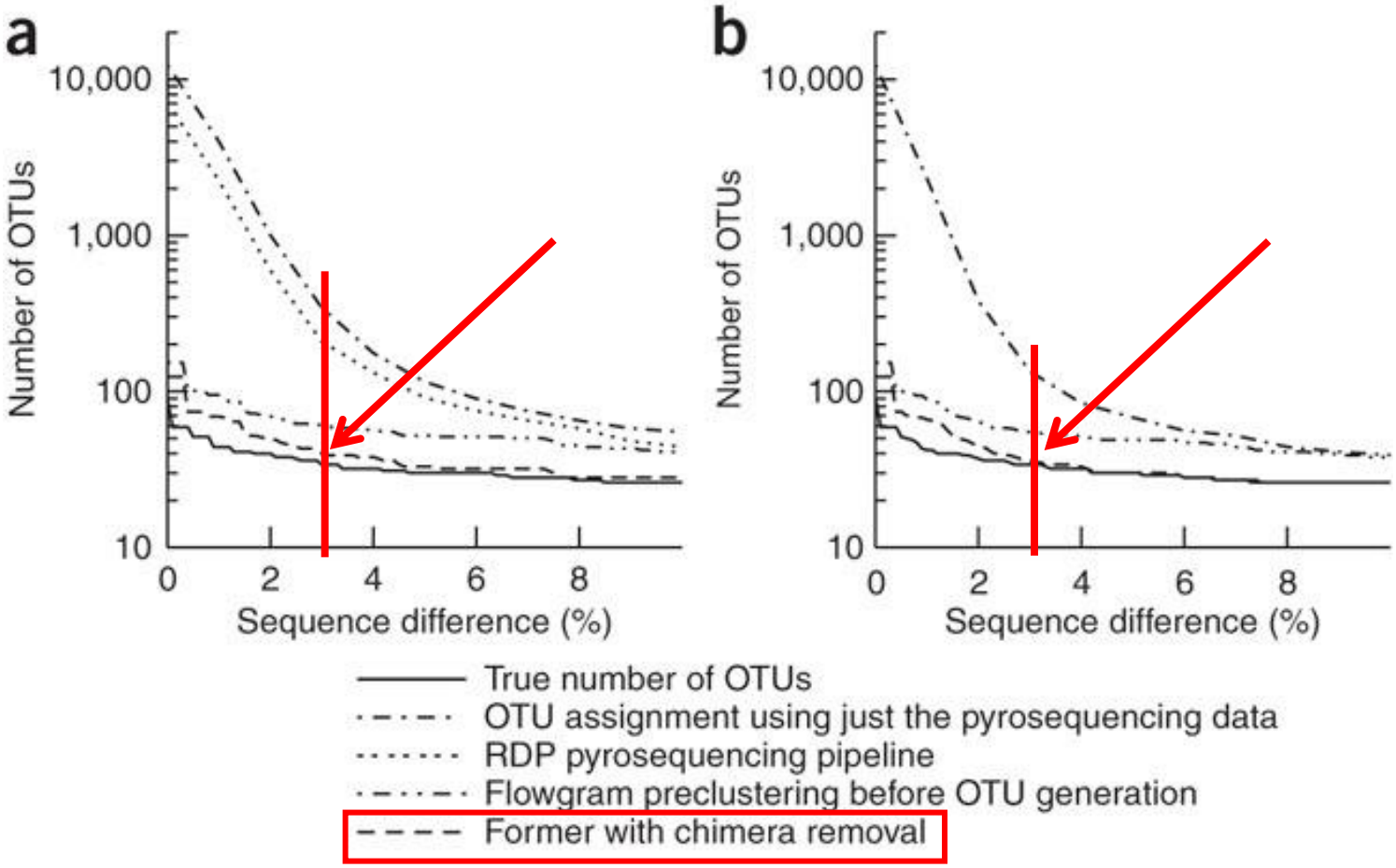
there is a **different index on each side of the sequenced strand**, allowing you to filter sequences where index hopping has occurred

Searching for true number of taxa (OTUs) in the data



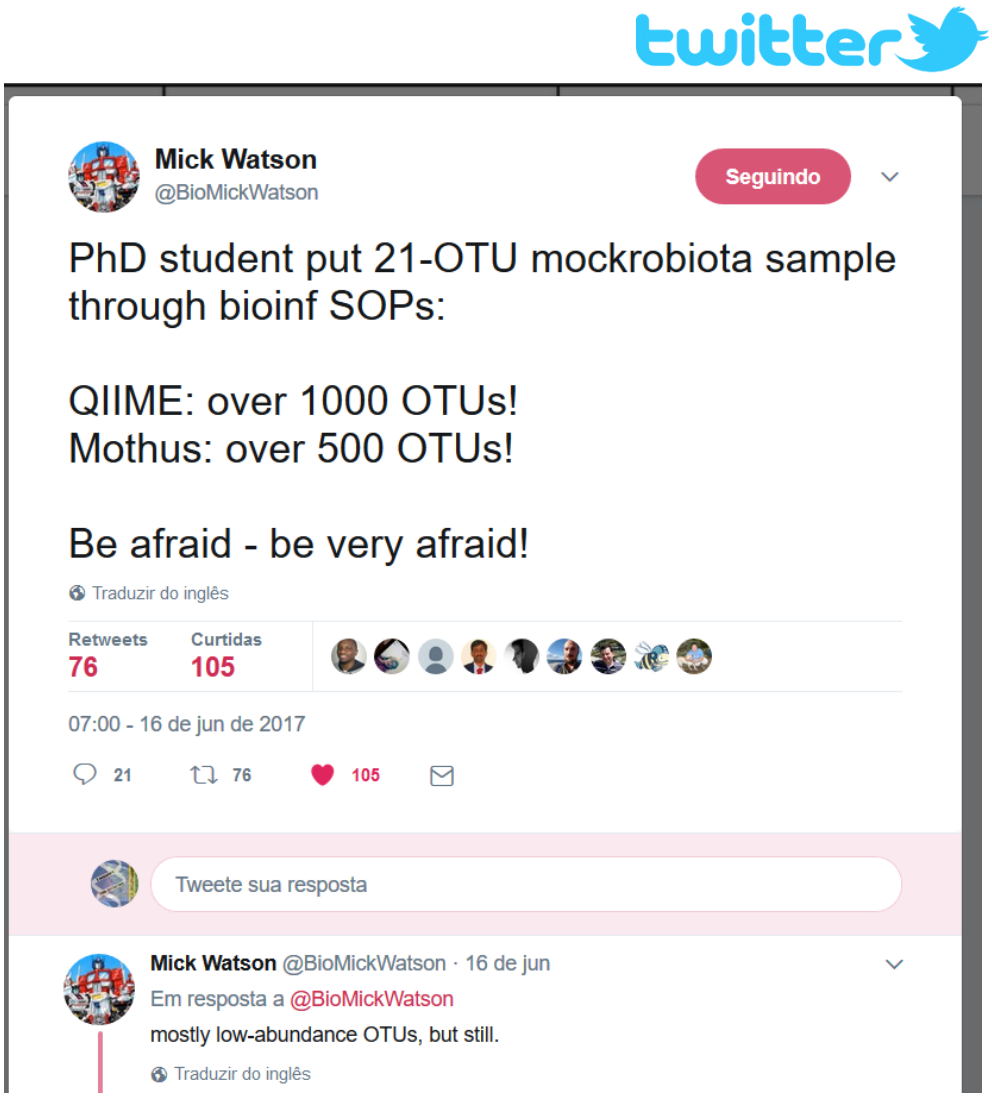
Quince, Christopher, et al. "Accurate determination of microbial diversity from 454 pyrosequencing data." *Nature methods* 6.9 (2009): 639.

Searching for true number of taxa (OTUs) in the data

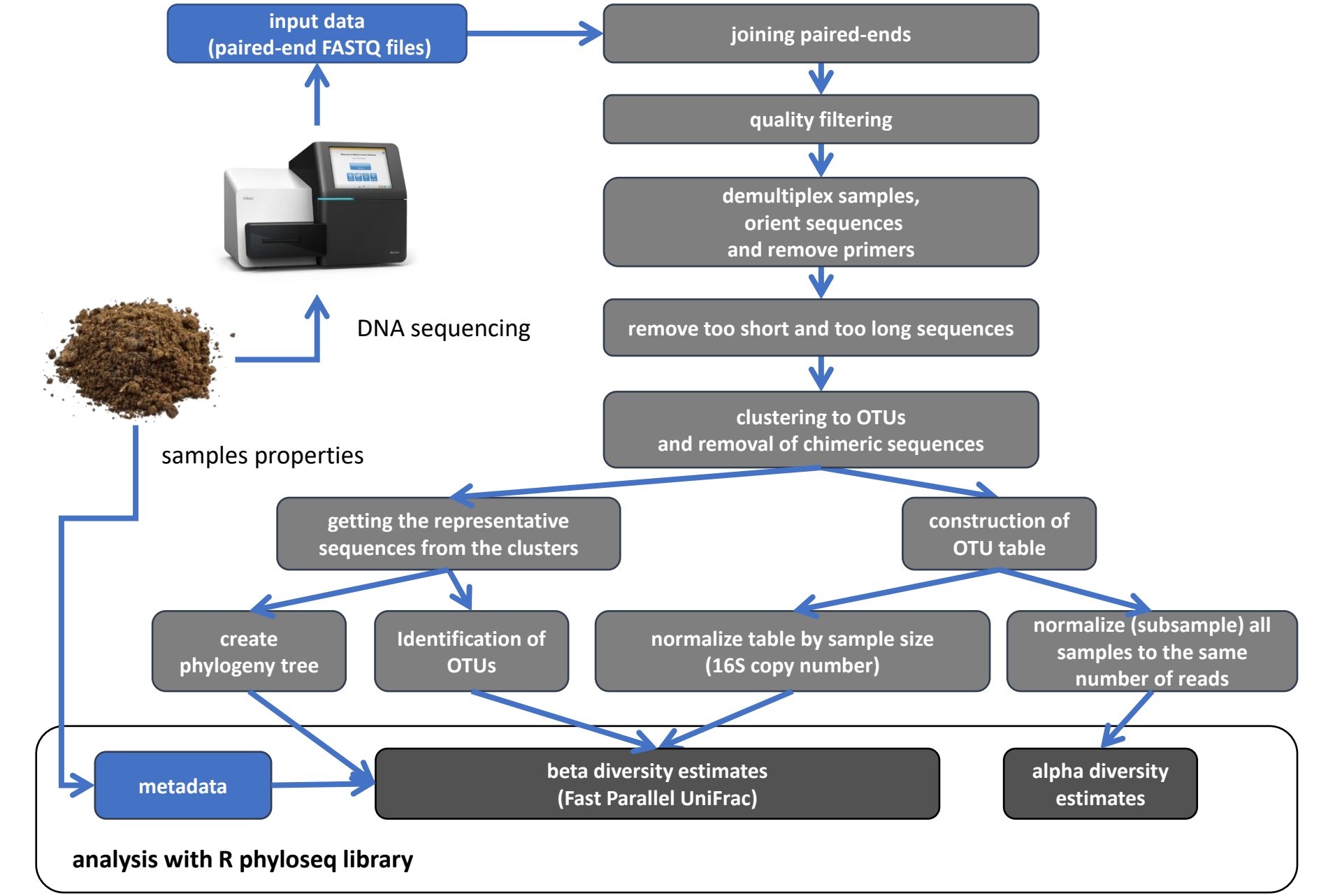


Quince, Christopher, et al. "Accurate determination of microbial diversity from 454 pyrosequencing data." *Nature methods* 6.9 (2009): 639.

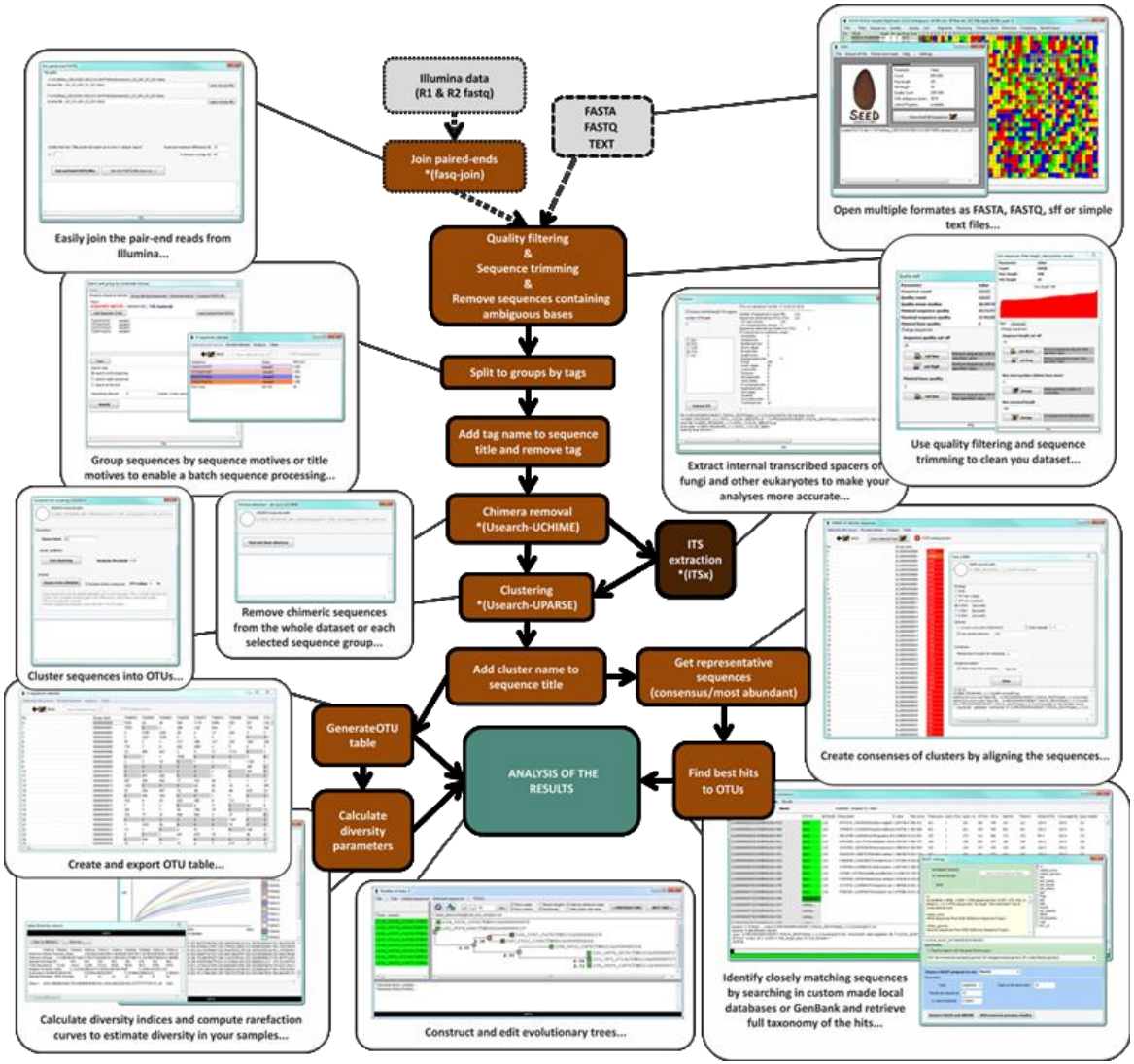
This is still a big concern to all microbial community analysis pipelines!



Amplicons pipeline workflow



GUI based alternative for Windows (http://www.biomed.cas.cz/mbu/lbwrf/seed/)



SEED 2: a user-friendly platform for amplicon high-throughput sequencing data analyses

- editing of sequences and their titles
- sorting
- quality trimming
- pair-end joining
- grouping of sequences based on sequence motifs or sequence titles
- batch processing of sequence groups
- denoising
- chimera removal
- ITS extraction
- sequence alignments and clustering
- OTU table construction
- construction of consensus sequences
- creation of local databases for BLAST
- searching either local databases or the whole NCBI
- retrieval of taxonomical classification from the NCBI
- calculation of diversity parameters
- many more...

Větrovský, Baldrian & Morais (2018) SEED 2: a user-friendly platform for amplicon high-throughput sequencing data analyses. Bioinformatics, bty071, 2018

SEED is alternative to

https://qiime2.org/

QIIME 2™ is a next-generation microbiome bioinformatics platform that is extensible, free, open source, and community developed.

- command line, Unix (Linux) based

https://mothur.org/

mothur - one of the most widely used tools for analyzing 16S rRNA gene sequence data



- command line, multiplatform

RAW DATA - R1 & R2 FASTQ

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PROCESSING OF THE RESULTS

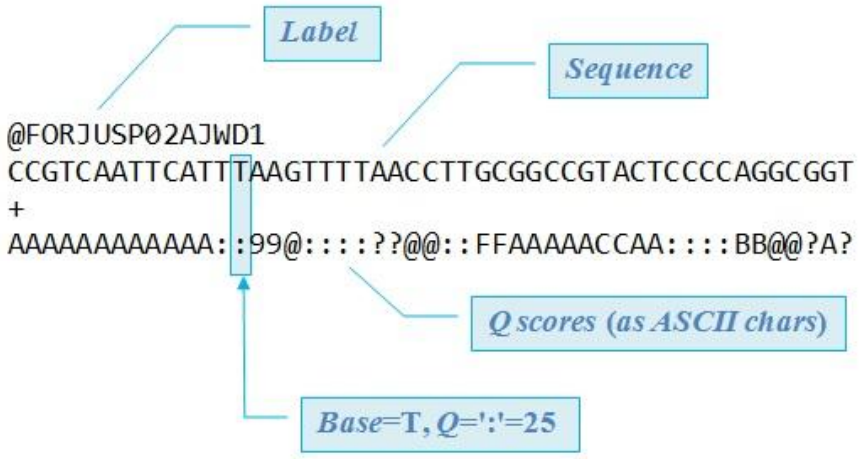
RAW DATA

BAC_R1.fastq - first sequence

@M03794:8:000000000-AJCUU:1:2114:9990:17907 1:N:0:7
AACAGCCGGACTACTGGGGTTTCTAATCCTGTTTGCTCCCCACGCTTTCGTGCCTCAGTGTCAATGACCGTGTAGCAAGCTGCCTTCGCAATTGGT
GTTCTATGTCATATCTAAGCATTTCACCGCTACATGACATATTCCGCTTACCTCCACGATATTCAAGACTAATAGTATCAATGGCAGTTCCCAAGTT
AAGCTCGGGGATTTCACCACGGACTTACTAGCCCACCTACGCACCCTTTAAACCCAGT
+
BCCCCFCCCCCCGGGGGGGFGGHHHCHHHHHHHHHHHHHHGG2FGGGGGHGHGGHHHFHHHHHHHHHHHHHGGHHHHHHHHHHHHHHHH
HGGGGHHHHGHGHHHHHHHHHFHHHHHHHHGHHHHHHGGGGGHHHGHHHHHHHHHGGGGHHHHHHHGCGFDHHGGHHHHHHHHGG
>GGGHHG/GHGHHHHHHFHFHGHHGHHG?DGGGGGGGGGGGGACGGGGGGGGGFGGFAAFF;@ADFFFFFFB/FFFFF;

BAC_R2.fastq - first sequence

@M03794:8:000000000-AJCUU:1:2114:9990:17907 2:N:0:7
ACGAAGTGTGCCAGCAGCCGCGGTAATACGGAGGGTGCAAGCGTTATCCGGATTCACTGGGTTTAAAGGGTGCGTAGGTGGGCTAGTAAGT
CCGTGGTGAAATCCCCGAGCTTAACTTGGGAACTGCCATTGATACTATTAGTCTTGAATATCGTGGAGGTAAGCGGAATATGTCATGTAGCGG
TGAAATGCTTAGATATGACATAGAACACCAATTGCGAAGGCAGCTTGCTACACGGTCATTGACACTG
+
BBBBBBBFFBFDGFFFGGGCGGCGGGHHHGDEDGGGFGGHHHGGGGGFDEE?EEGBGFHHHHGFFGGFFGGFEFGDFDGFFEGGHHHGFEGF
HFHEEFFFHHHHFHGGGFGFHHHFHHHHHHBGHHFBCHFHBGGGHHHHGHHHHHHGFFHHHGHGD<GGAGHHHGGG@CFHHFCGHH:C
CG?AAAGFEFEFGGGBFFFFFFGFFBBFGFFBDE?BBBB.@9-9..A.B/:AFFFEF.@;AAF//99FFF/



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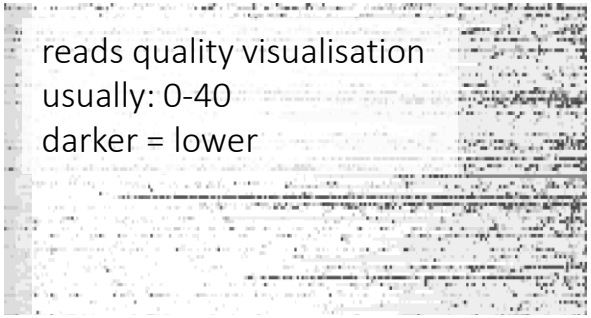
Identification of OTUs

Construction of OTU table

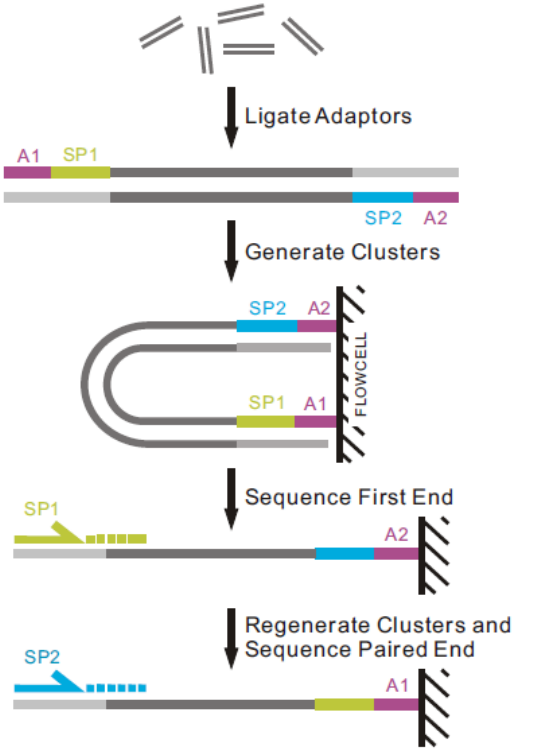
Estimation of diversity indices

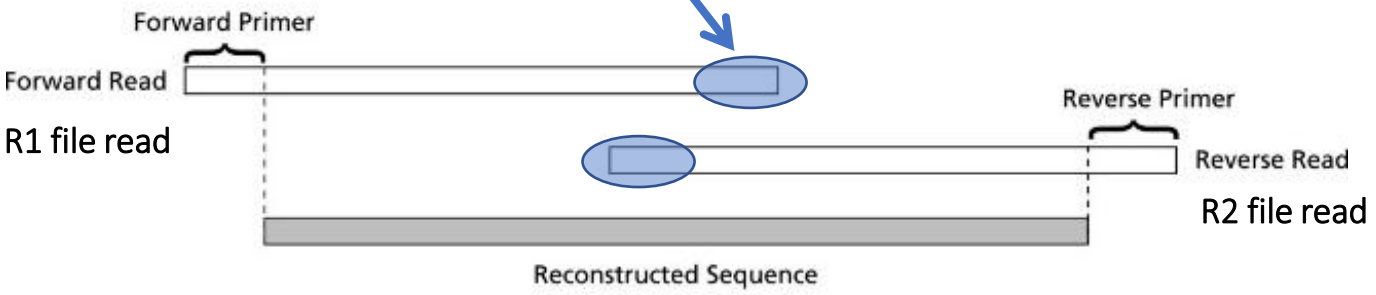
PROCESSING OF THE RESULTS

Joining of pair-end data (Illumina)



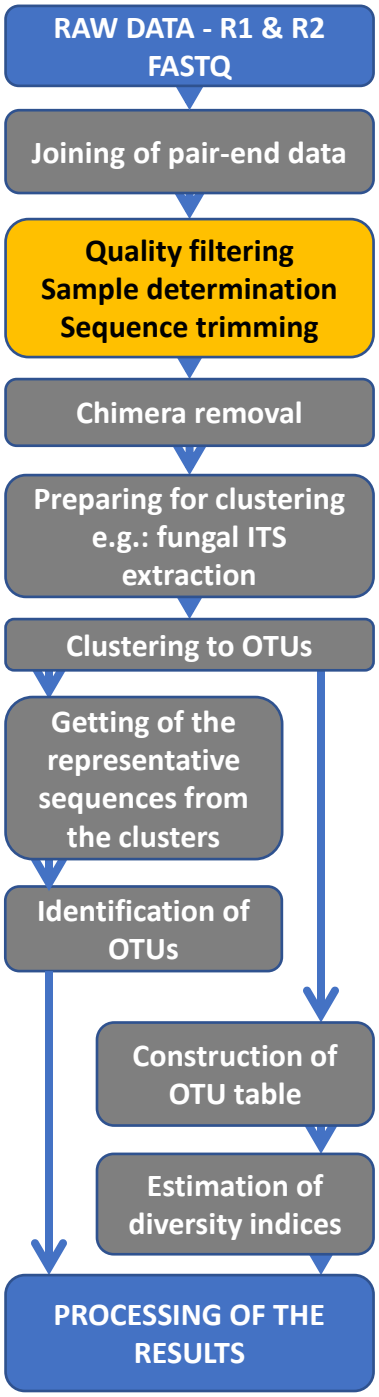
reads quality is dropping at the ends





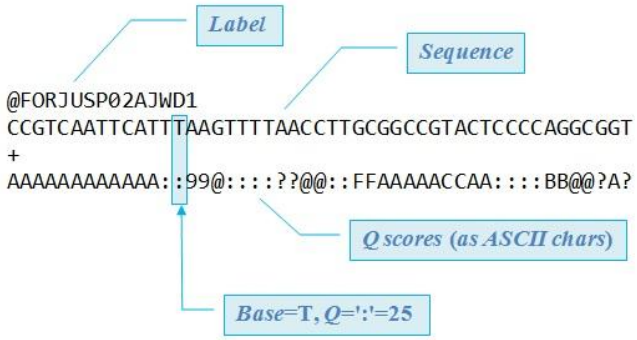
reconstructed sequence are based on bases with higher quality

PROGRAM: **FastqJoin**
set a minimal overlap length and overlap precision



Sequence quality and quality filtering

FASTQ format



$$Q_{illumina} = -10 \times \log_{10}\left(\frac{P_e}{1-P_e}\right),$$

where P_e is the probability of identifying a base incorrectly. For Sanger and other platforms, the formula is as follows [8]:

$$Q_{PHRED} = -10 \times \log_{10}(P_e).$$

$$Q_{illumina} = 10 \times \log_{10}\left(10^{\left\{\frac{Q_{PHRED}}{10}\right\}} + 1\right)$$

Table 2. Phred quality scores are logarithmically linked to error probabilities (http://en.wikipedia.org/wiki/Phred_quality_score)

Phred quality score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.90%
40	1 in 10 000	99.99%
50	1 in 100 000	99.999%
60	1 in 1 000 000	99.9999%

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Multiple samples in one library (Multiplexing)

name	FWDprimer	REVprimer
SAMPLE001	515F_T103	806R_T007
SAMPLE002	515F_T002	806R_T052

spacer is not presented in native sequences, it is used to prevent overestimation of any taxa

TAG **SPACER** **ORIGINAL PRIMER**

515F_T103 AATATACGTGTGCCAGCMGCCGCGGTAA
515F_T002 ACGAAGTGTGCCAGCMGCCGCGGTAA

806R_T007 AGCCACCGGACTACHVGGGTWTCTAAT
806R_T052 ATCCTCCCGGACTACHVGGGTWTCTAAT



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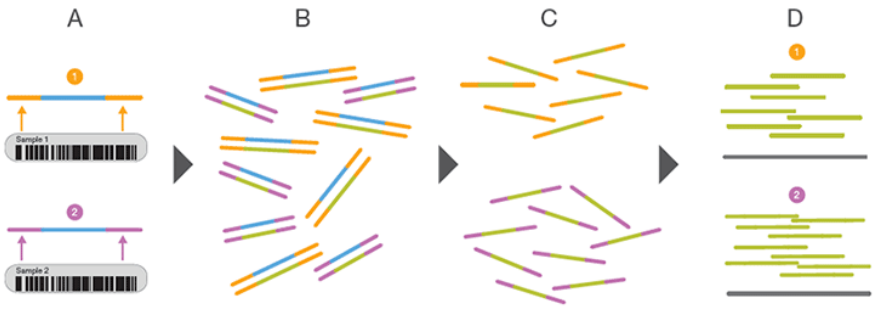
Construction of OTU table

Estimation of diversity indices

PROCESSING OF THE RESULTS

Sample determination (de-multiplexing)

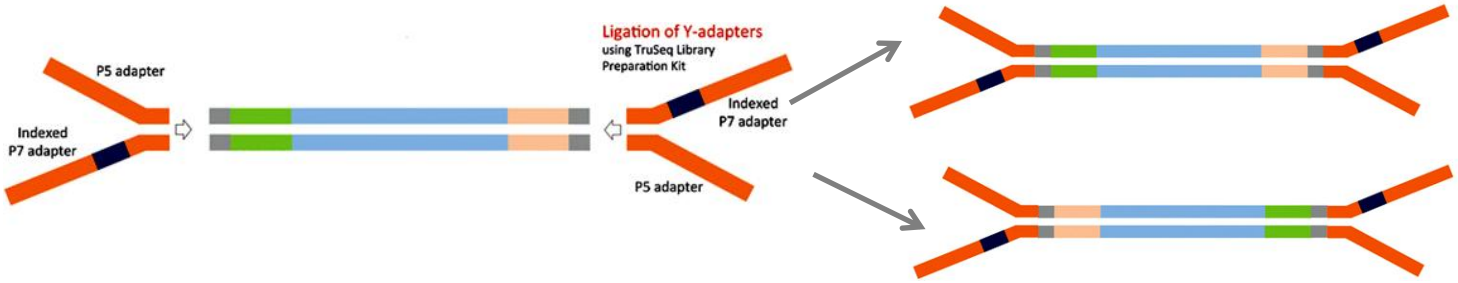
Demultiplex samples



put sample names to sequence titles

>**SAMPLE034**|M03794:8:000000000-AJCUU:1:2114:9990:17907
CCTGTTTGCTCCCCACGCTTTCGTGCCT
CAGTGTCAATGACCGTGTAGCAAGCTG
CA...

Orient sequences



cca 50 % of the reads are reverse complement oriented due to ligation of library adapters

Remove primers

since primer sequences are not native to the sample, they need to be removed before clustering to OTUs

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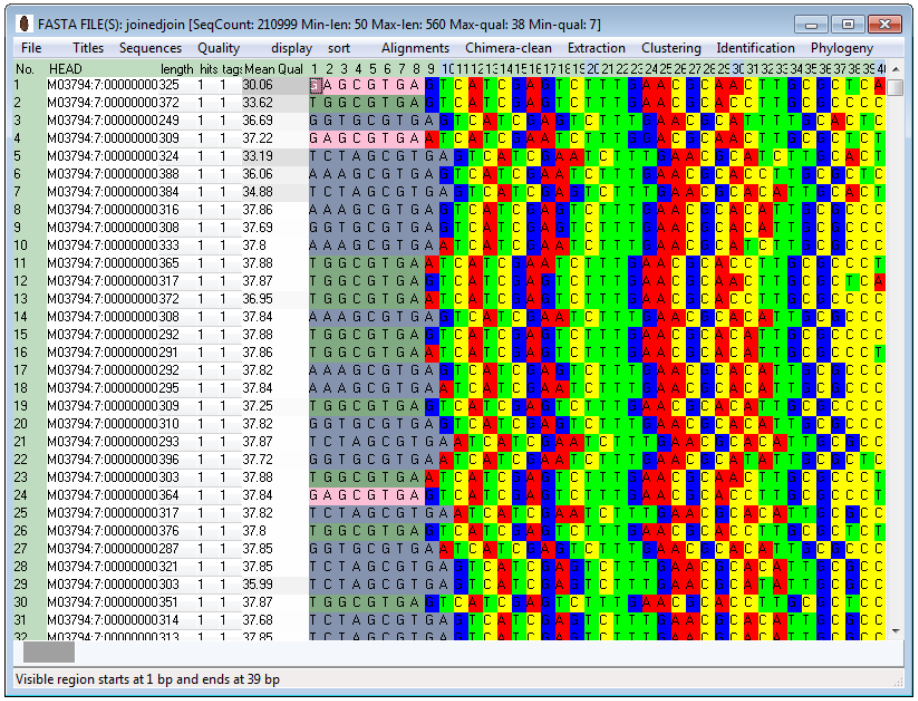
Sample determination (de-multiplexing) and removing barcodes and primers

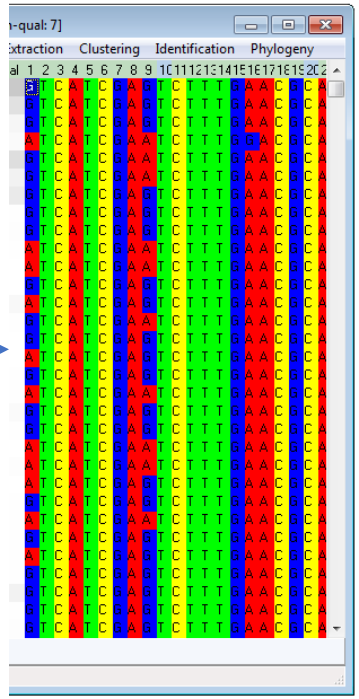
GAGCGTGA	gITS7_T02
TGGCGTGA	gITS7_T03
GGTGCGTGA	gITS7_T06
AAAGCGTGA	gITS7_T08
TCTAGCGTGA	gITS7_T10

search for the sample barcodes at the beginning of reads

Sequence	Query	RESULT
GAGCGTGA	gITS7_T02	14687
TGGCGTGA	gITS7_T03	22568
GGTGCGTGA	gITS7_T06	16835
AAAGCGTGA	gITS7_T08	19258
TCTAGCGTGA	gITS7_T10	20585

remove barcode after putting the name of sample to sequences header…





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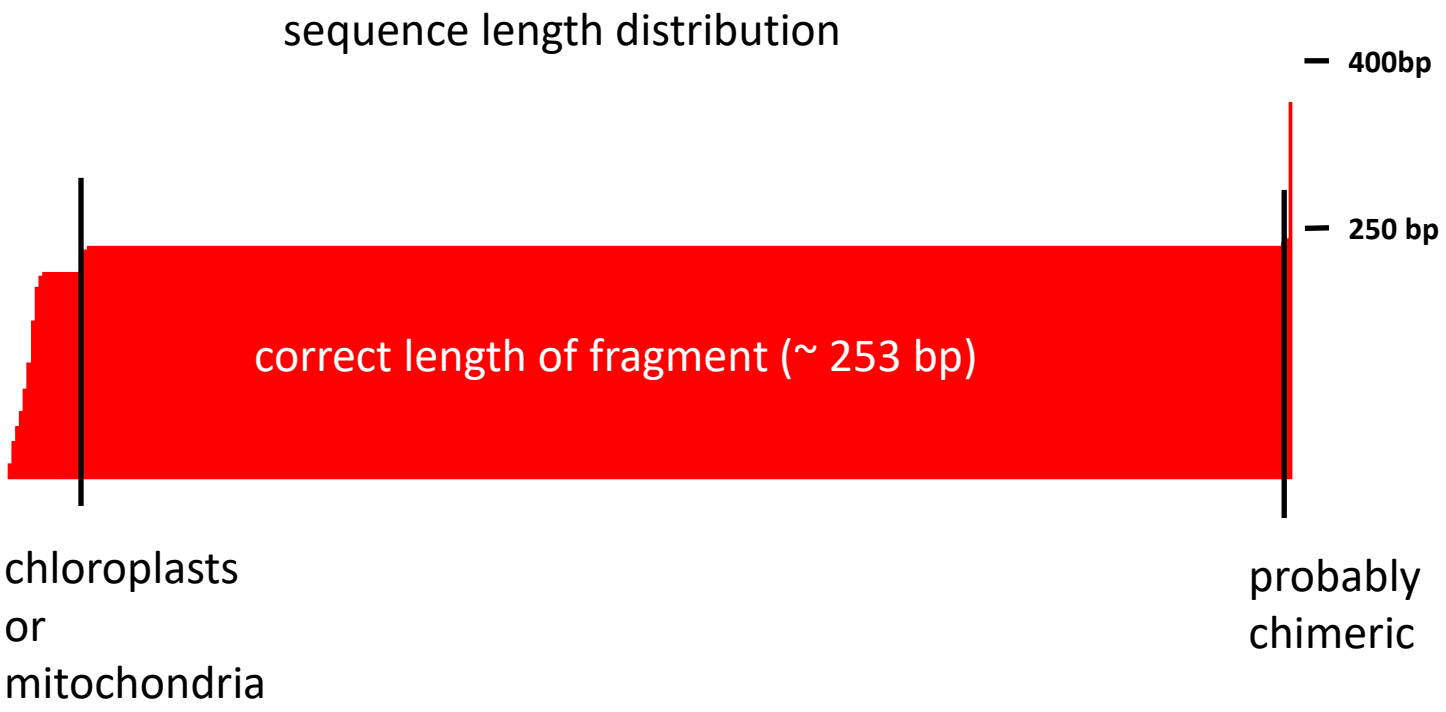
Estimation of diversity indices

PROCESSING OF THE RESULTS

Sequence trimming

removing sequences with aberrated length - depends on the marker gene

Too short – nonspecific PCR products/erroneus sequences
Too long - nonspecific PCR products/chimeric sequences



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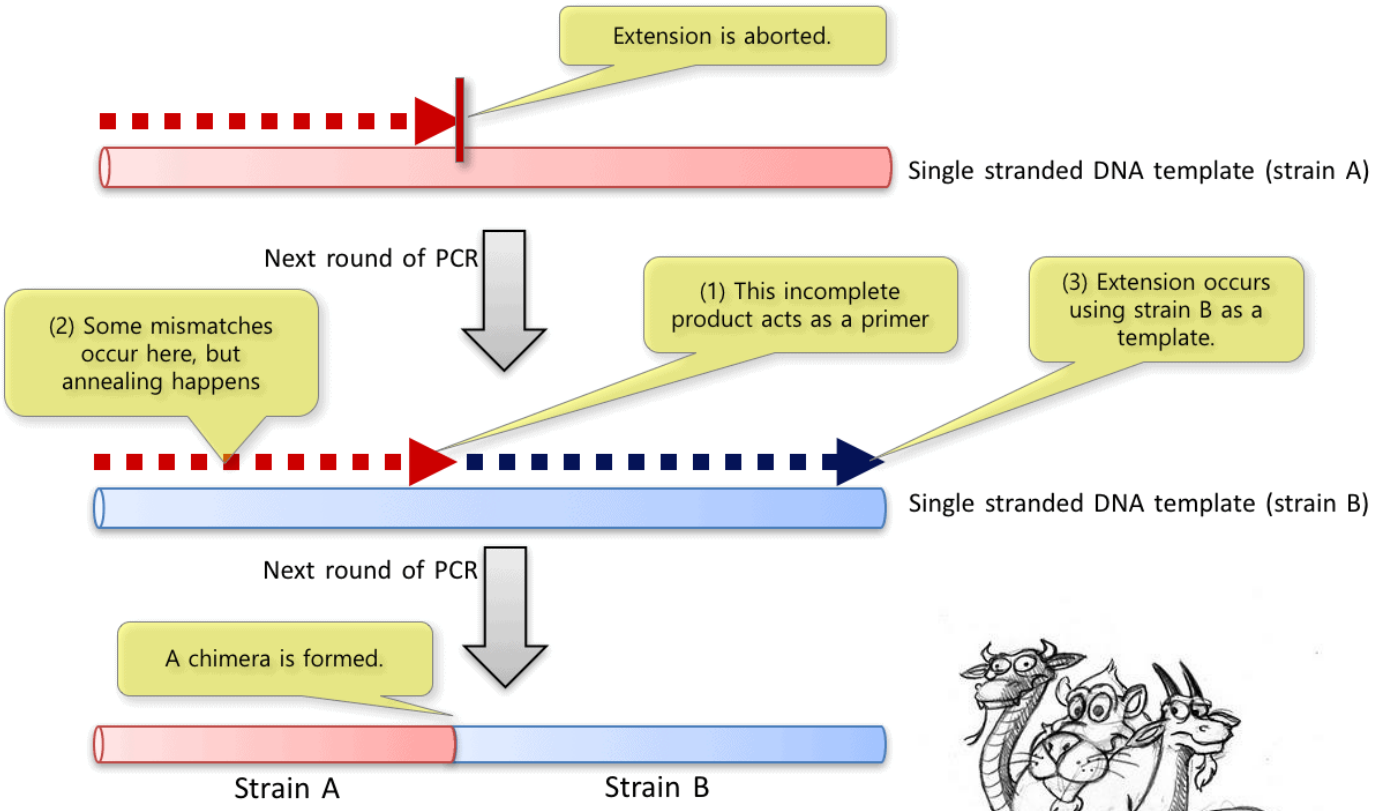
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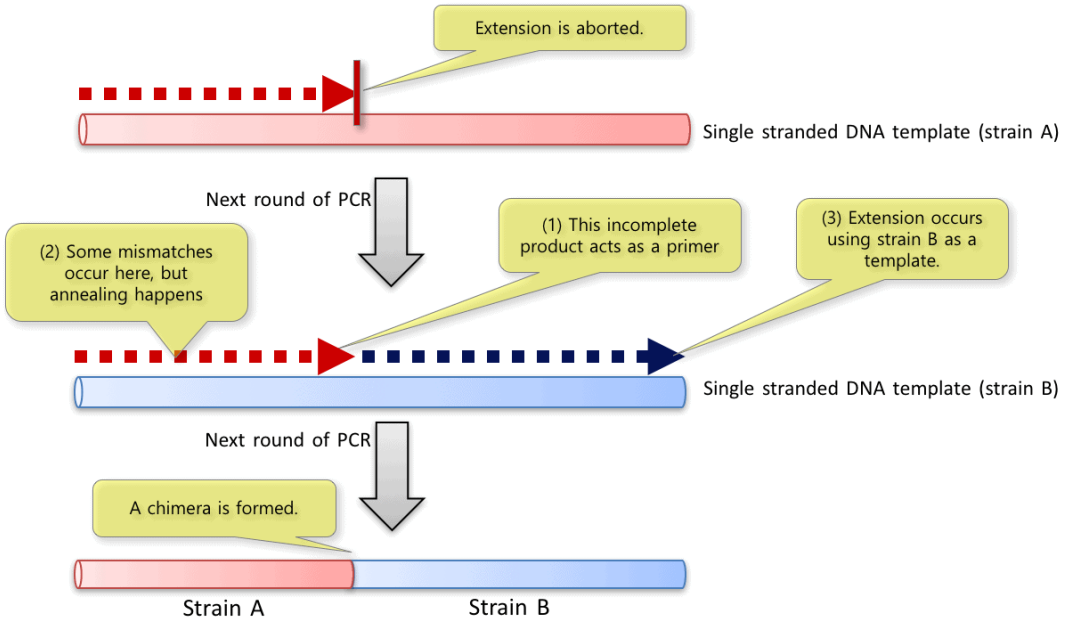
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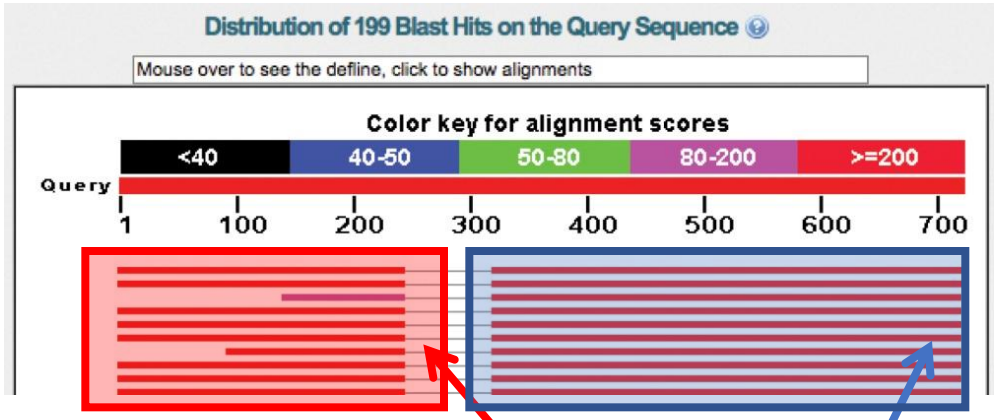
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RAW DATA - R1 & R2 FASTQ

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Chimera removal

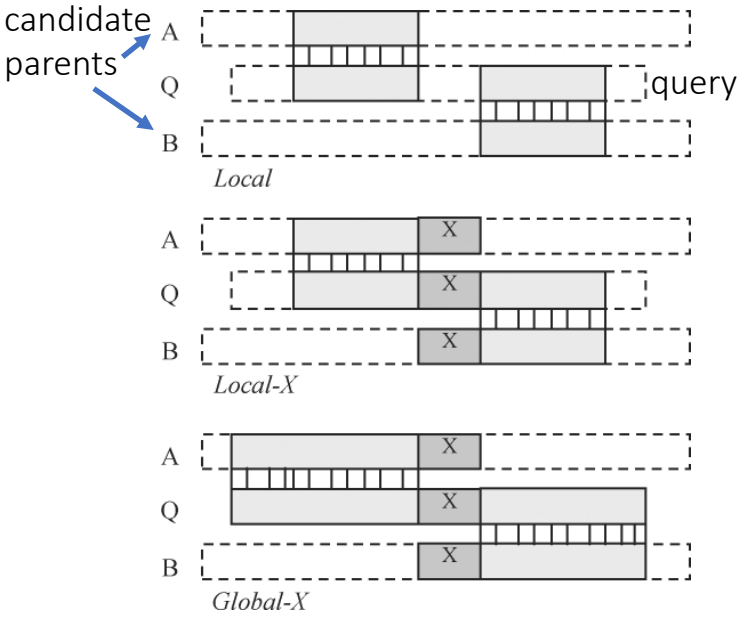
chimeric sequences are derived from parental sequences
-> program search for parents.

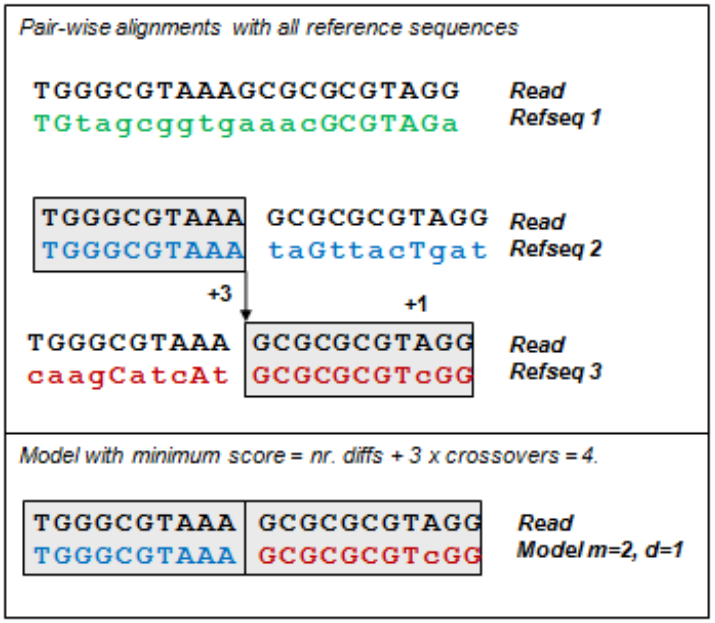
de novo approach:

- no reference database
- usually based on measuring of several sequence parts abundances – parents should be more abundant then its offspring (chimeras)

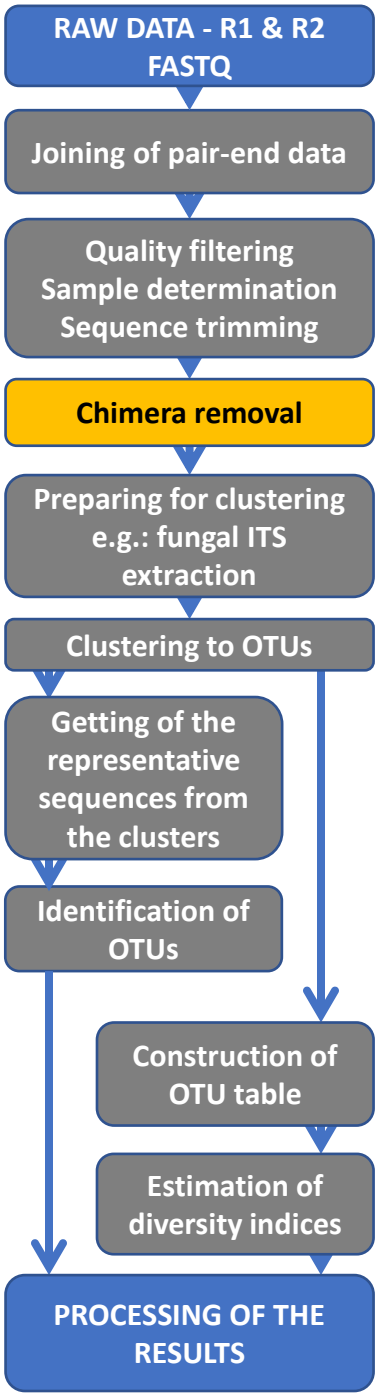
reference based approach:

- pair-wise alignment with reference sequence database





PROGRAMS: **UCHIME, UPARSE**



Chimera removal

chimeric sequences are derived from parental sequences
-> program search for parents.

de novo approach:

- no reference database
- usually based on measuring of several sequence parts abundances – parents should be more abundant then its offspring (chimeras)

Problems

- algorithms are not optimal
- computation cost could be high
- problems with highly similar sequences

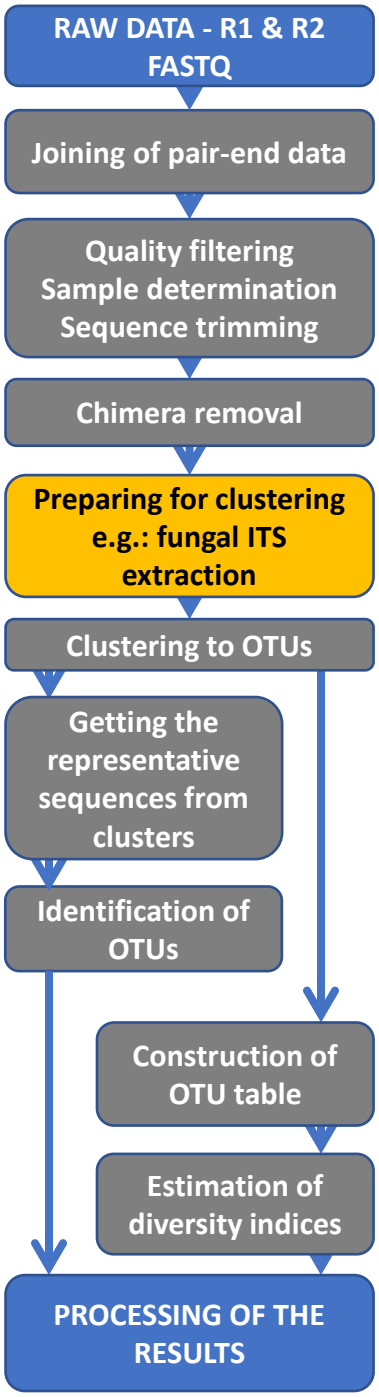
reference based approach:

- pair-wise alignment with reference sequence database

Problems

- no appropriate reference database for environmental samples
- variable quality of reference databases

PROGRAMS: **UCHIME, UPARSE**



Fungal ITS extraction



ITSx

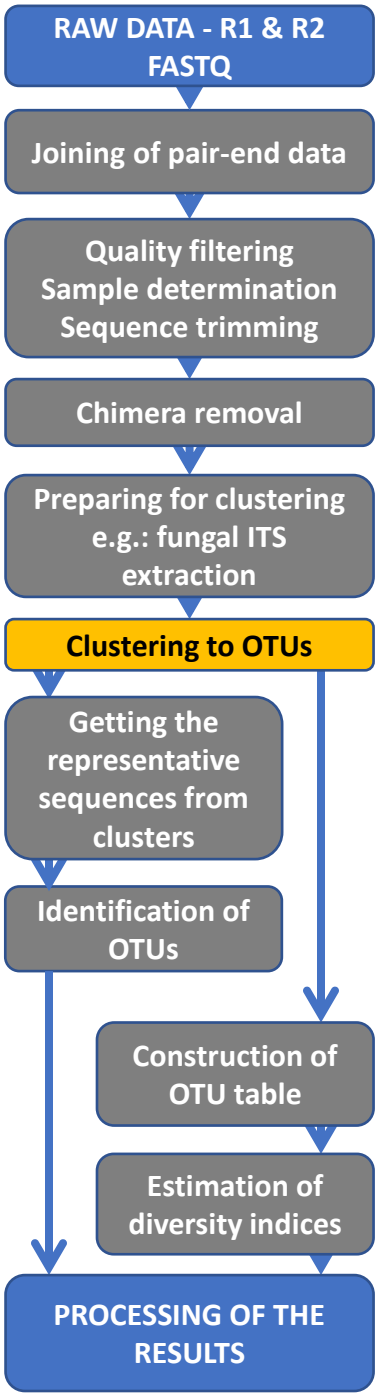
Improved software for detection and extraction of ITS1 and ITS2 from ribosomal ITS sequences of fungi and other eukaryotes for use in environmental sequencing

relies on **HMMER**

searching sequence databases for sequence homologs, and for making sequence alignments. It implements methods using probabilistic models called profile hidden Markov models (profile HMMs).

using the extracted variable ITS improves resolution when the OTUs are created

http://microbiology.se/software/itsx/



Clustering to OTUs

OTU (Operational taxonomic unit) group of similar sequences grouped based on some similarity threshold
usually 97% similarity (16S, ITS) represents Species

Heuristic

- comparison of each sequence with representative sequence („seed“)
- depends on sequence order

FAST

Hierarchical

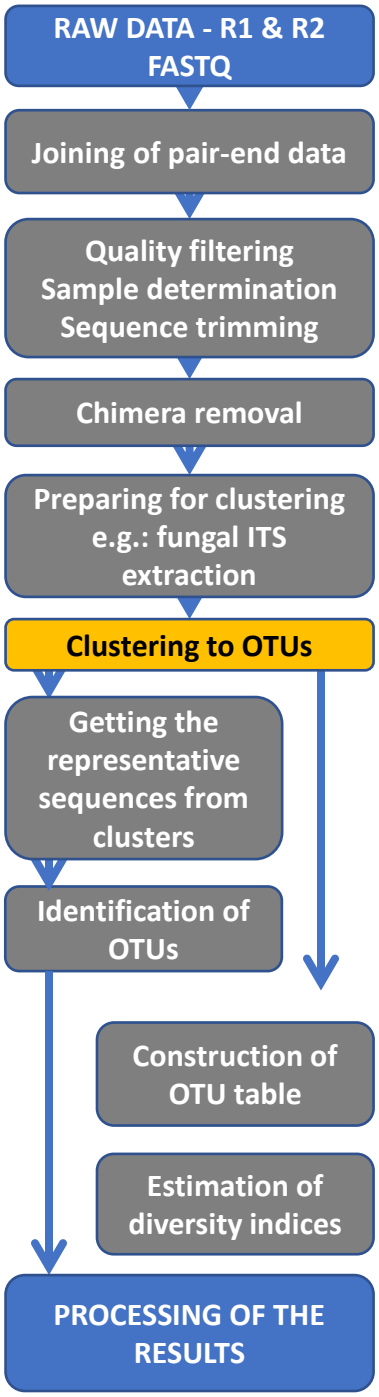
comparison of each sequence with each other (tree contruction)

SLOW

Model based

- probabilistic, iterative
- uses more information than the sequence identity

VERY SLOW



Clustering to OTUs

programs

Tool	Distance calculation	Clustering algorithm	Reference
DOTUR (+ MUSCLE)	MSA_{denovo}	Hierarchical	Schloss et al. 2005
MOTHUR	$MSA_{profile}$	Hierarchical	Schloss et al. 2009
ESPRIT	PSA	Hierarchical	Sun et al. 2009
SLP	PSA	Hierarchical	Huse et al. 2009
ESPRIT-TREE	PSA	Hierarchical	Cai et al. 2011
JMOTU	PSA	Hierarchical	Jones et al. 2011
CD-HIT	PSA	Heuristic	Li et al. 2006
USEARCH/UPARSE	PSA	Heuristic	Edgar et al. 2010/2013
GRAMCLUSTER	PSA	Heuristic	Russell et al. 2010
DNACLUST	PSA	Heuristic	Ghodsi et al. 2011
CRUNCHCLUST	PSA	Heuristic	Hartmann et al. 2012
DYSC	PSA	Heuristic	Zheng et al. 2012
MS-CLUST	PSA	Heuristic	Chen et al. 2013
TBC	PSA	Heuristic	Lee et al. 2012
TSC	PSA	H&H combination	Jiang et al. 2012
CROP	PSA	Model-based (BC)	Hao et al. 2011
BEBAC	PSA	Model-based (BC)	Cheng et al. 2012
DBC454	composition	Model-based	Pagni 2013
DBC	PSA	Model-based	Preheim et al. 2013
M-PICK	graphical	Model-based	Wang et al. 2013

PSA - Pairwise Sequence Alignment
MSA - Multiple Sequence Alignment

RAW DATA - R1 & R2 FASTQ

Joining of pair-end data

Quality filtering
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Preparing for clustering e.g.: fungal ITS extraction

Clustering to OTUs

Getting the representative sequences from clusters

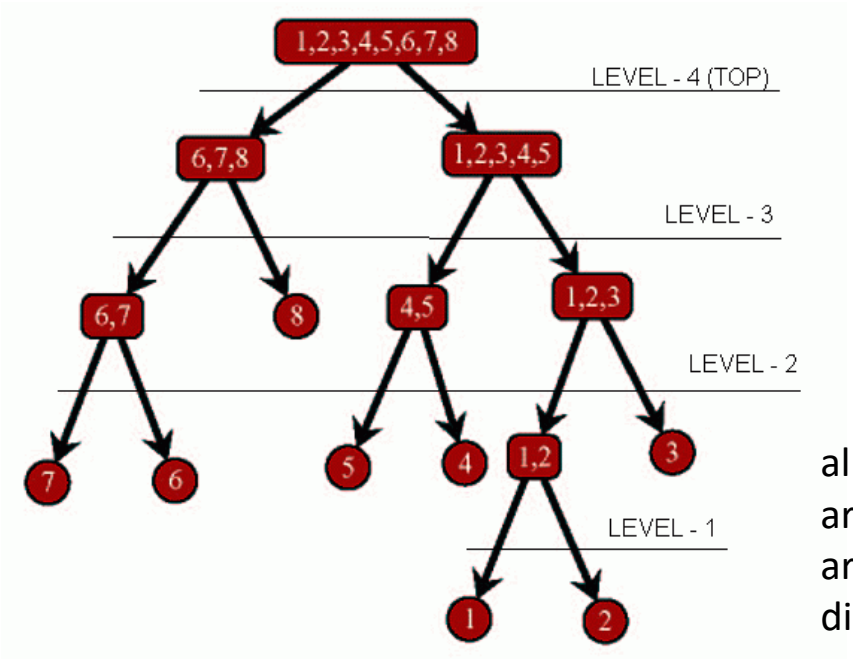
Identification of OTUs

Construction of OTU table

Estimation of diversity indices

PROCESSING OF THE RESULTS

Clustering to OTUs (hierarchical)



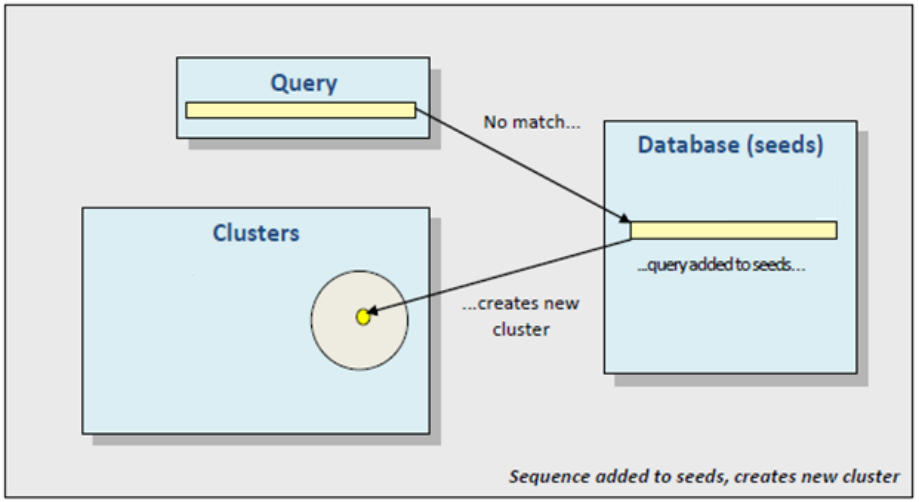
all pairwise comparisons are performed and OTUs are delineated at fixed distance level

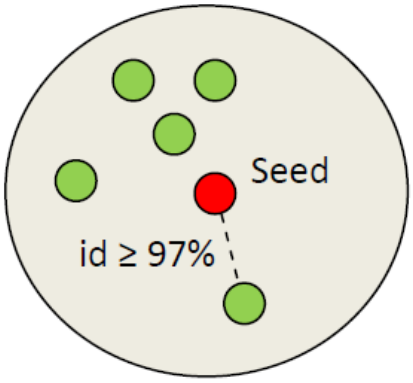
linking method is an important driver of the outcome:

- **single-linkage clustering** (SL) clusters may be merged together due to single sequences being close to each other, even though many of the sequences in each cluster may be very distant to each other
- **complete-linkage clustering** (CL) tends to find compact clusters of approximately equal diameters. With CL, all objects in a cluster are similar to each other
- **average-linkage clustering** (AL) can be seen as an intermediate between single and complete linkage clustering, resulting in more homogeneous clusters than those obtained by the single-linkage method

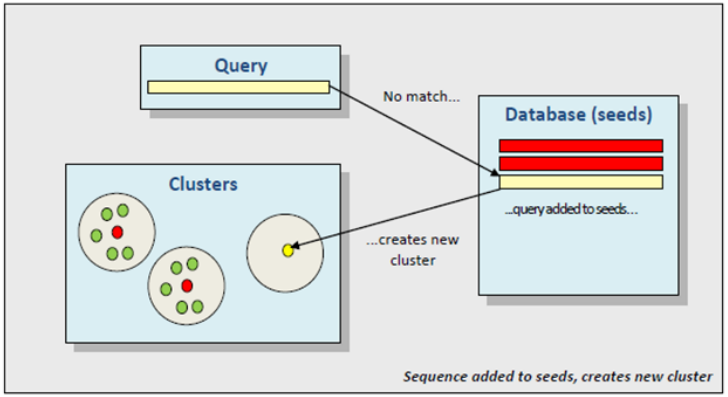
Clustering to OTUs (heuristic - USEARCH)

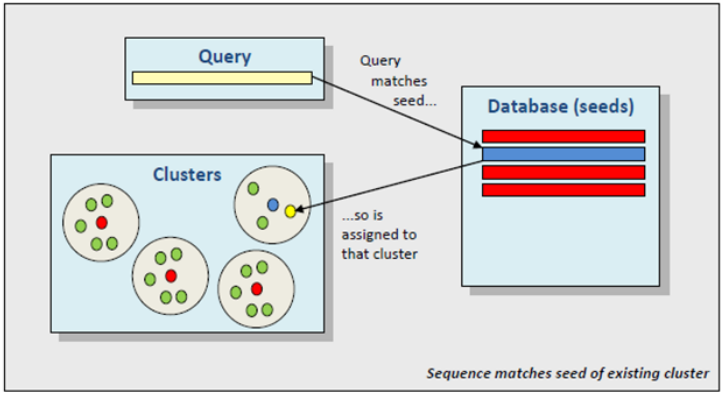
- RAW DATA - R1 & R2 FASTQ
- Joining of pair-end data
- Quality filtering
  Sample determination
  Sequence trimming
- Chimera removal
- Preparing for clustering
  e.g.: fungal ITS extraction
- **Clustering to OTUs**
- Getting the representative sequences from clusters
- Identification of OTUs
- Construction of OTU table
- Estimation of diversity indices
- PROCESSING OF THE RESULTS





cluster definition





RAW DATA - R1 & R2 FASTQ

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Clustering to OTUs

Getting the representative sequences from clusters

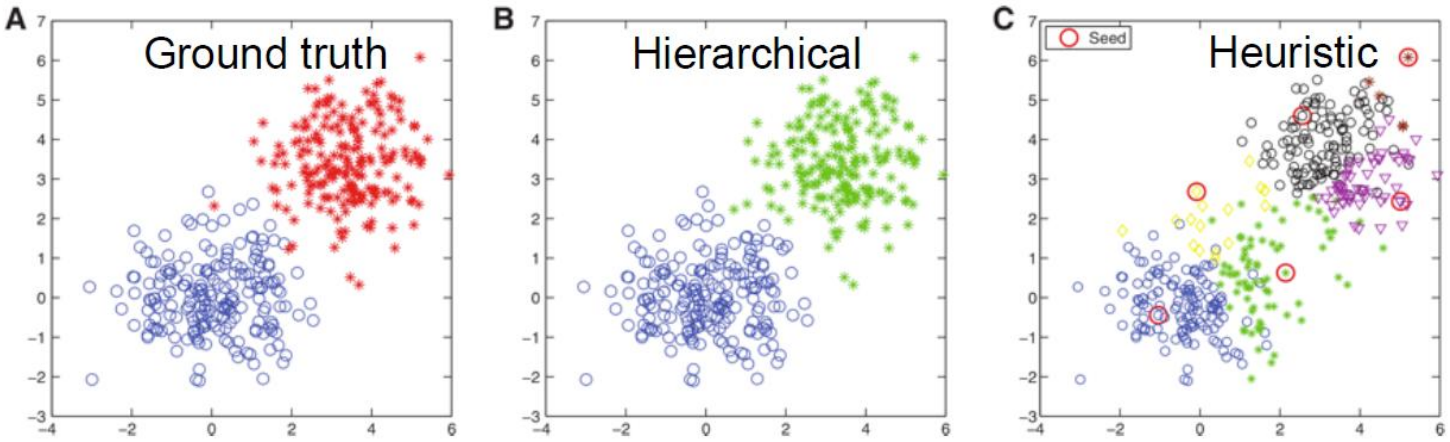
Identification of OTUs

Construction of OTU table

Estimation of diversity indices

PROCESSING OF THE RESULTS

Clustering to OTUs (hierarchical vs. heuristic)



Hierarchical clustering

- is able to identify the real clusters (ideally)
- computationally expensive

x

Heuristic clustering

- computationally cheap
- often generates artificial clusters (overestimated diversity)

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Clustering to OTUs

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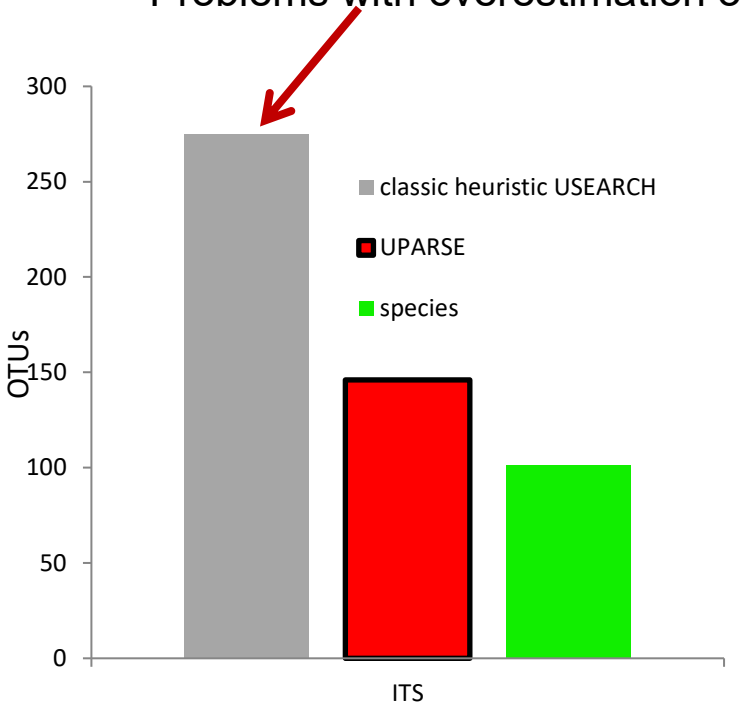
Construction of OTU table

Estimation of diversity indices

PROCESSING OF THE RESULTS

Clustering to OTUs (comparison)

Problems with overestimation of obtained OTUs…

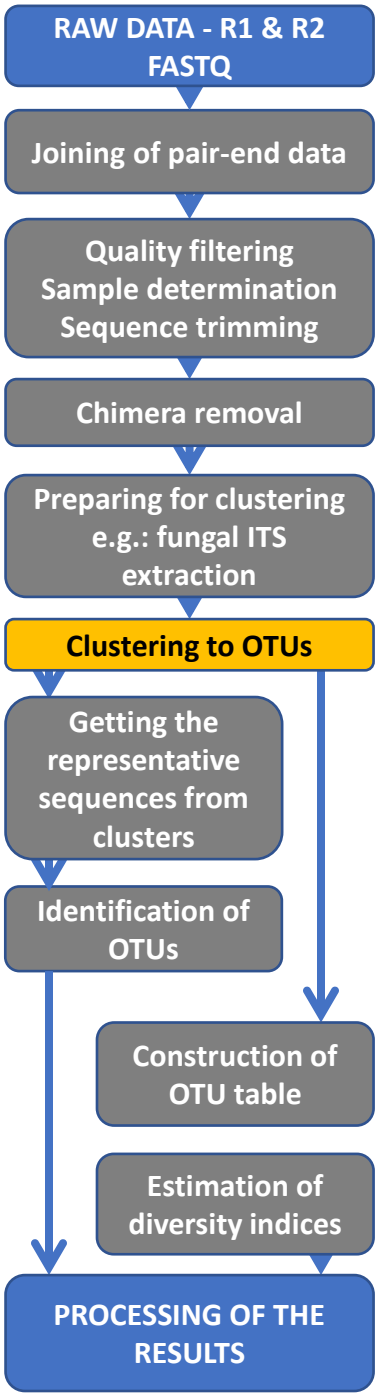


mock fungal community
about 100 fungal species

solution
UPARSE (USEARCH) – improved heuristic algorithm which is able to recognize chimeric sequences

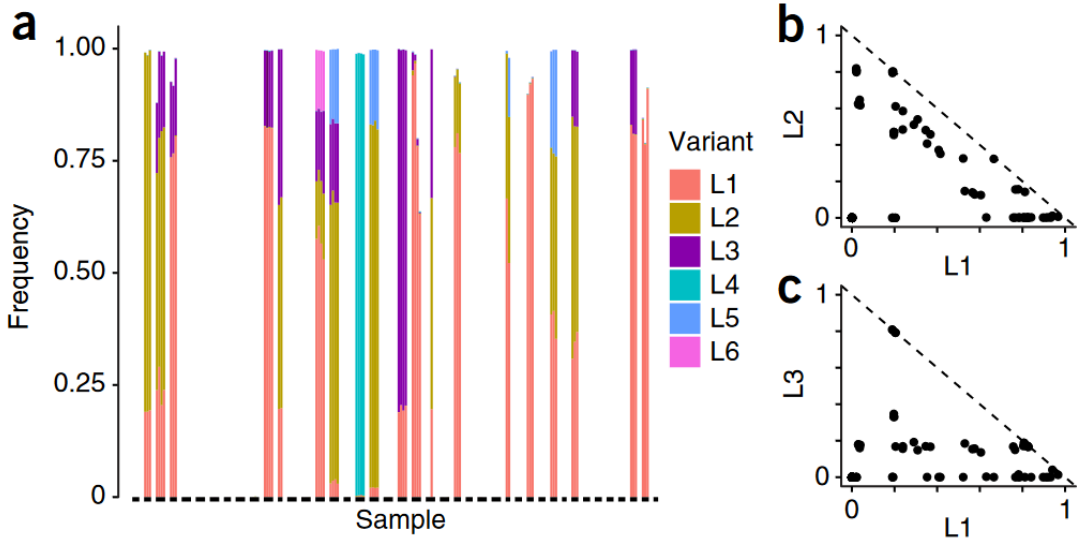
http://drive5.com/uparse/

Edgar, R.C. (2013) UPARSE: Highly accurate OTU sequences from microbial amplicon reads, *Nature Methods* [Pubmed:23955772, dx.doi.org/10.1038/nmeth.2604].



Clustering-independent methods

Callahan, Benjamin J., et al. "**DADA2**: high-resolution sample inference from Illumina amplicon data." *Nature methods* (2016).



L. crispatus sequence variants in the human vaginal community during pregnancy. DADA2 identified six L. crispatus 16S rRNA sequence variants present in multiple samples and a significant fraction of all reads.

Amir, Amnon, et al. "**Deblur** Rapidly Resolves Single-Nucleotide Community Sequence Patterns." *mSystems* 2.2 (2017).

RAW DATA - R1 & R2 FASTQ

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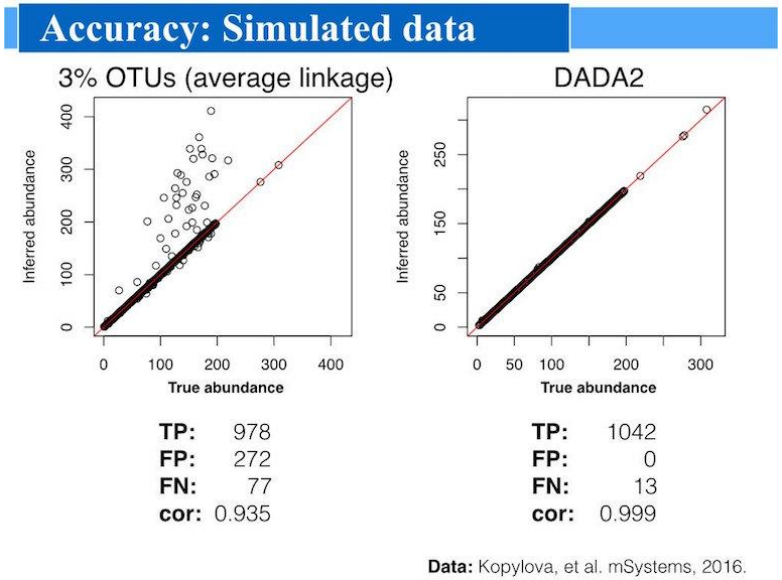
Construction of OTU table

Estimation of diversity indices

PROCESSING OF THE RESULTS

DADA2

Amplicon Sequencing. **Exactly.** *Version 1.18*



Data: Kopylova, et al. mSystems, 2016.

Advantages

Resolution: DADA2 infers exact amplicon sequence variants (ASVs) from amplicon data, resolving biological differences of even 1 or 2 nucleotides.

Accuracy: DADA2 reports fewer false positive sequence variants than other methods report false OTUs.

Comparability: The ASVs output by DADA2 can be directly compared between studies, without the need to reprocess the pooled data.

Computational Scaling: The compute time of DADA2 scales linearly sample number, and memory requirements are essentially flat.

Disadvantage

Sequence variants are not representing the real sequences (they are estimated based on the errors modeling)!!!

Too strict when data are diverse (e.g. different lengths in ITS amplicons) - removing some rare taxa!!!

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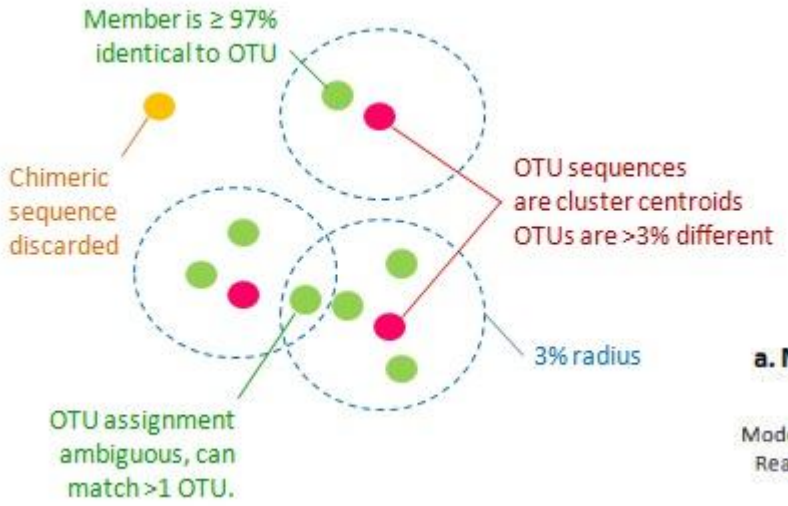
Identification of OTUs

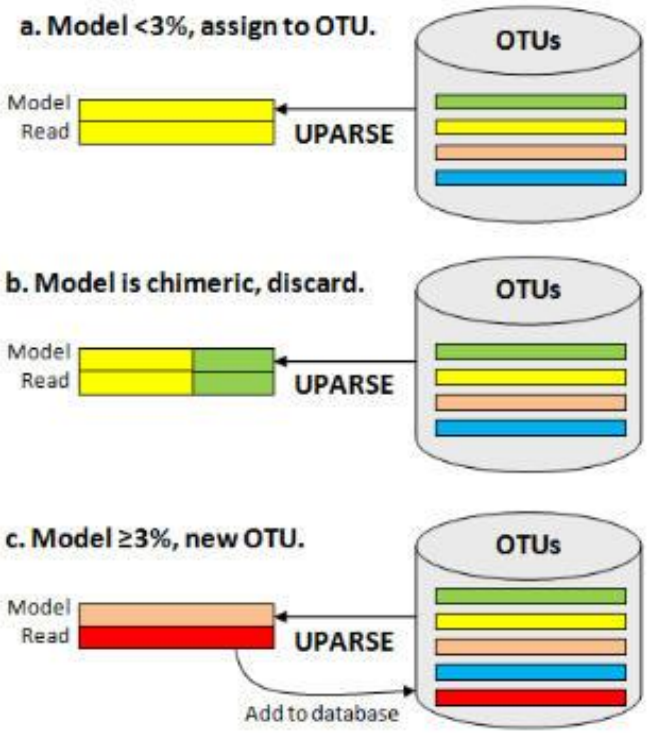
Construction of OTU table

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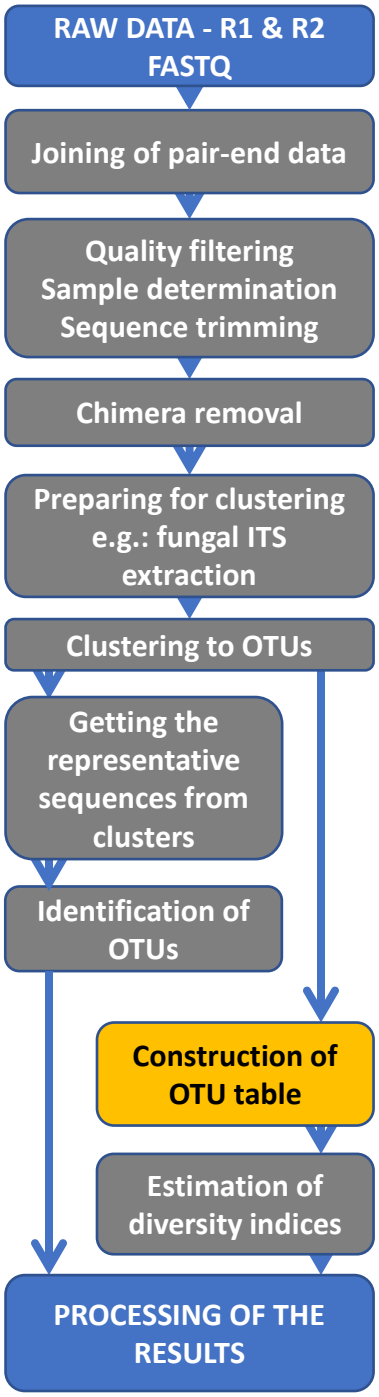
PROCESSING OF THE RESULTS

UPARSE: Clustering and chimera removal in the same time





Edgar, R.C. (2013) UPARSE: Highly accurate OTU sequences from microbial amplicon reads, *Nature Methods*



Construction of OTU table

OTU table - matrix that gives the number of reads per sample per OTU

OTU_ID	SAMPLE_1	SAMPLE_2	SAMPLE_3	SAMPLE_4	SAMPLE_5	SAMPLE_6	SAMPLE_7	SAMPLE_8
CL00001	249	189	220	311	1	16	68	2
CL00002	201	19	169	438	1	8	12	0
CL00003	190	39	176	210	0	21	20	1
CL00004	183	36	195	177	1	16	16	0
CL00005	0	26	2	35	20	164	4	116
CL00006	0	0	0	0	1	0	0	0
CL00007	133	71	125	89	0	3	26	0
CL00008	106	42	96	158	0	10	14	0
CL00009	95	46	108	134	2	7	24	0
CL00010	0	0	0	0	0	0	0	3
CL00011	0	1	0	0	0	0	0	0

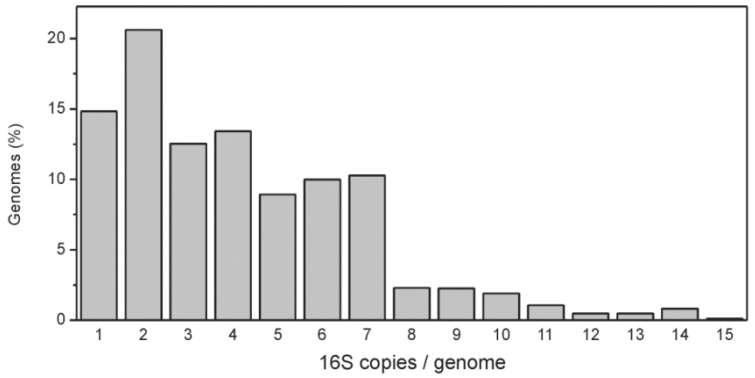
OTU frequency does not correlate with species frequency

This means, for example, that the most abundant OTU does not have to be the most abundant species – especially because of multi-copy nature of target genes as 16S and ITS

Singleton counts are especially suspect

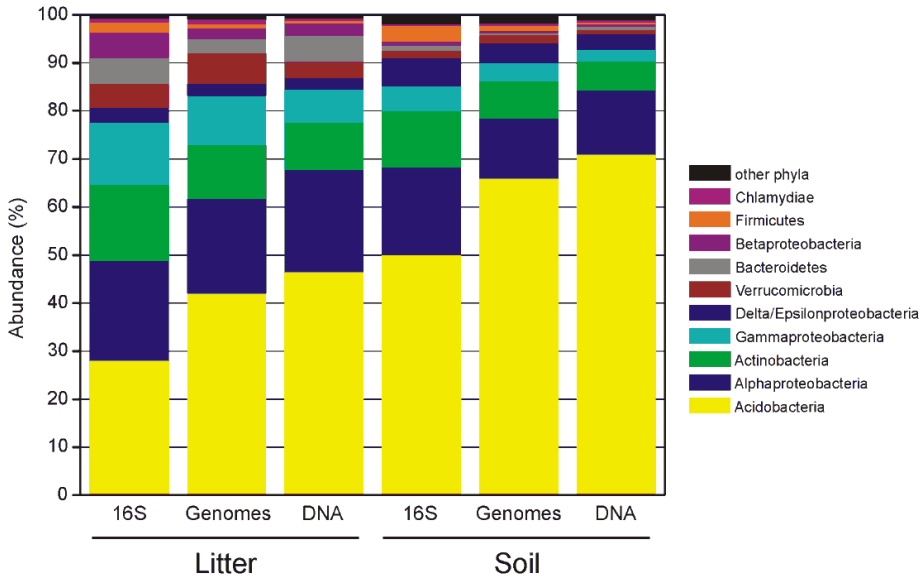
- many OTU table entries are often singletons (have value 1) for smaller OTUs because the total count is distributed over several samples
- Small counts are more likely to be spurious, especially singletons, either because the OTU itself is spurious (e.g., an undetected chimera), or because of cross-talk

Normalize OTU table by 16S copy number



rrnDB

A searchable database documenting variation in ribosomal RNA operons (rrn) in Bacteria and Archaea. Find information such as the 16S gene copy number of an organism by looking up its name under the NCBI or RDP taxonomy or by full-text search of rrnDB's records.



Abundance of bacterial 16S rRNA sequences, genomes and DNA in forest litter and soil.

Relative abundance of bacterial 16S rRNA sequences in the amplicon pool from Picea abies litter and soil (Baldrian et al., 2012), and estimates of the relative abundance of bacterial genomes and DNA. The estimates were calculated using the values of 16S rRNA copy numbers and genome sizes of the closest hits to each bacterial OTU.

T. Větrovský & P. Baldrian - PloS one, 2013 & Stoddard et al. (2015) https://rrndb.umms.med.umich.edu/

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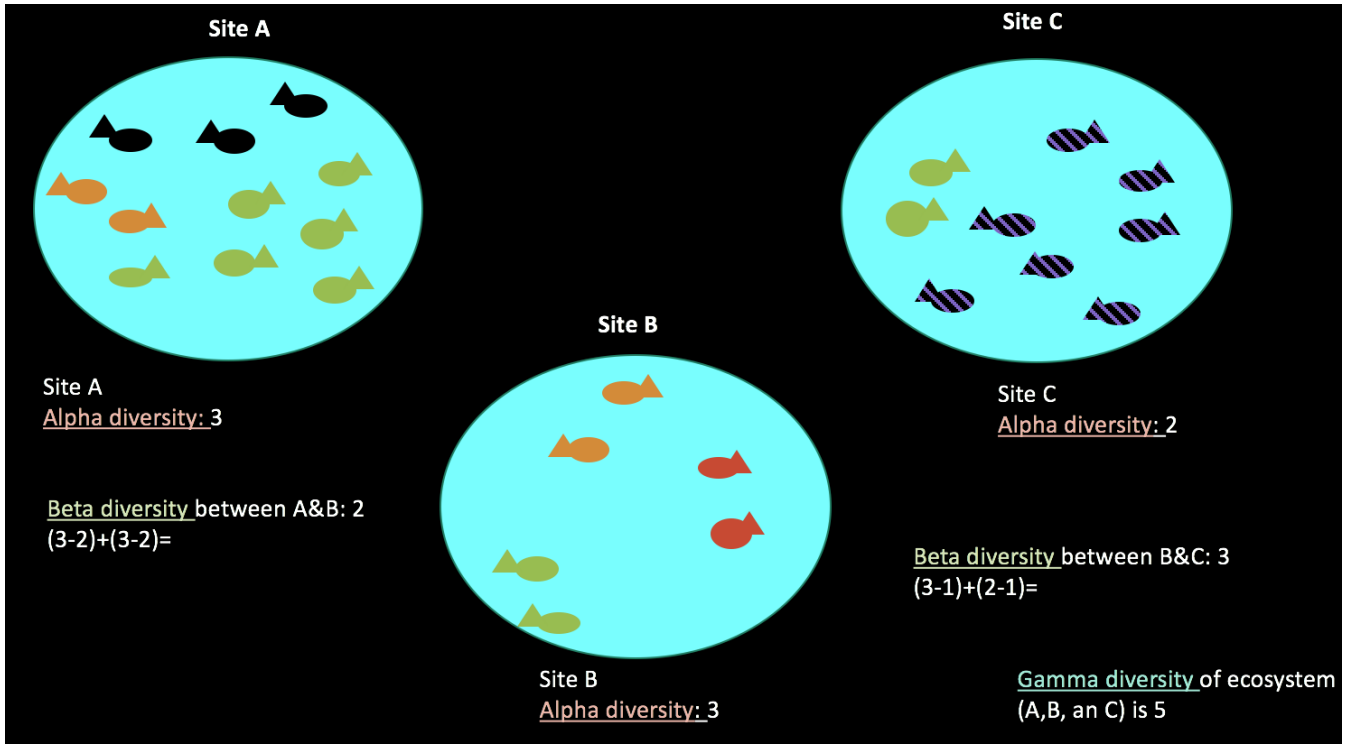
Identification of OTUs

Construction of OTU table

Estimation of diversity indices

PROCESSING OF THE RESULTS

Estimation of diversity indices



- Alpha-diversity: diversity of organisms in one sample / environment
  - Shannon index
  - Chao1
  - Observed OTUs (Richness)

- **Beta-diversity**: differences in diversities across samples or environments
  - **UniFrac** (Lozupone et al, AEM, 2005) (phylogenetic)
  - **Bray-Curtis** dissimilarity measure (OTU abundance)
  - **Jaccard** similarity coefficient (OTU presence/absence)

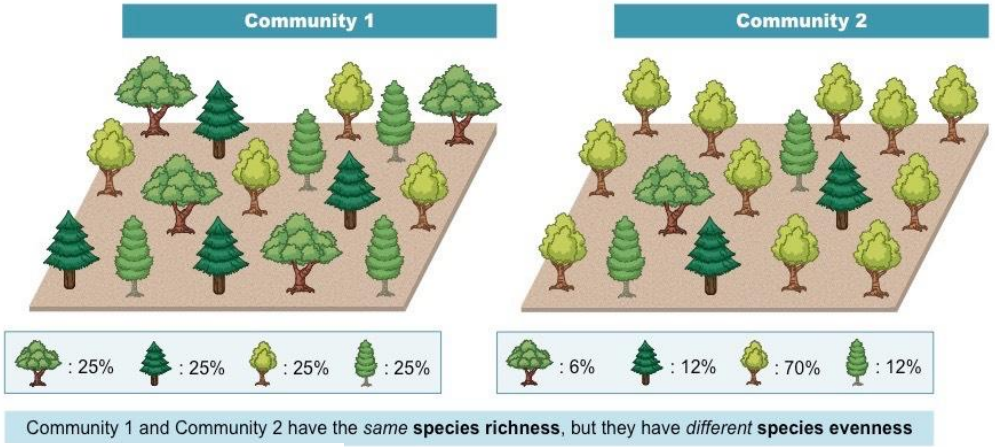
Alpha diversity

$$H' = -\sum_{i=1}^{S} p_i \ln p_i$$

Shannon index (Shannon entropy)

Then the Shannon entropy quantifies the uncertainty in predicting the species identity of an individual that is taken at random from the dataset.

p_i – proportion of the population made up of species i
S – number of species in sample



Species evenness

Species evenness refers to how close in numbers each species in an environment is.
Mathematically it is defined as a diversity index, a measure of biodiversity which quantifies how equal the community is numerically.

$$J' = \frac{H'}{H'_{\max}} \qquad H'_{\max} = -\sum_{i=1}^{S} \frac{1}{S} \ln \frac{1}{S} = \ln S$$

Chao1 index

Estimate diversity from abundance data (importance of rare OTUs)

$$S_{est} = S_{obs} + \left(\frac{{f_1}^2}{2f_2}\right)$$

where S_{obs} is the number of species in the sample, f_1 is the number of singletons and f_2 is the number of doubletons.

Shannon, C. E. (1948) A mathematical theory of communication. The Bell System Technical Journal, 27, 379–423 and 623–656.
Chao, A.; Shen, T-J. (2003)
https://palaeo-electronica.org/2011_1/238/estimate.htm

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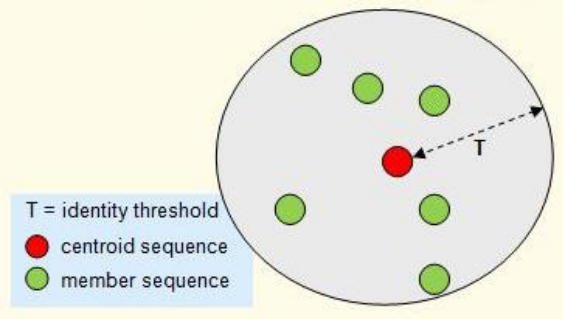
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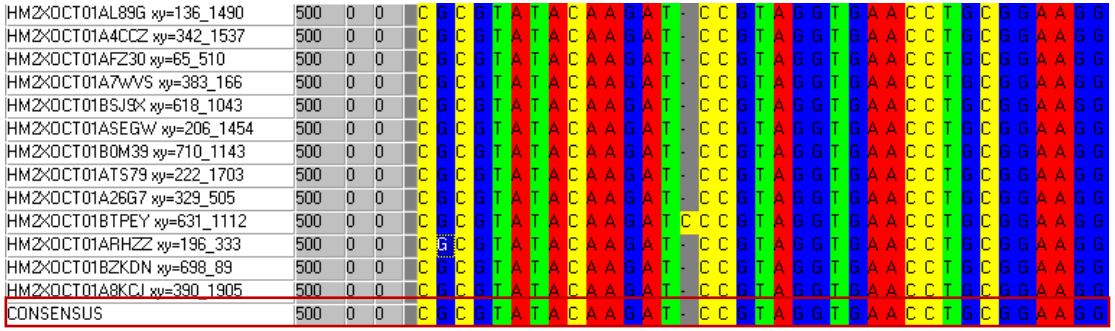
PROCESSING OF THE RESULTS

Getting of the representative sequences from the clusters

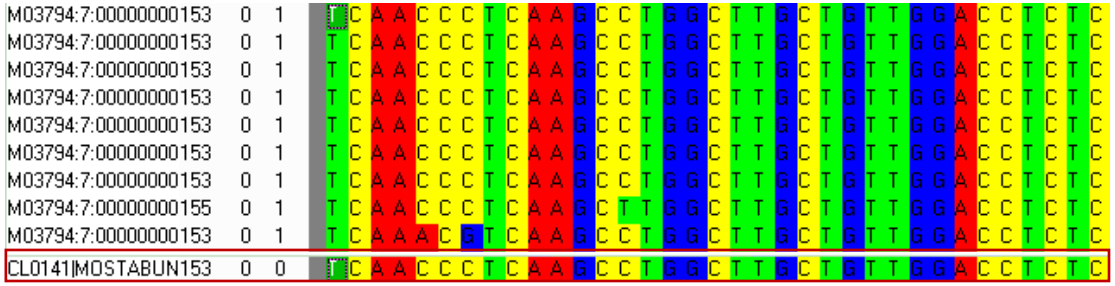
centroid

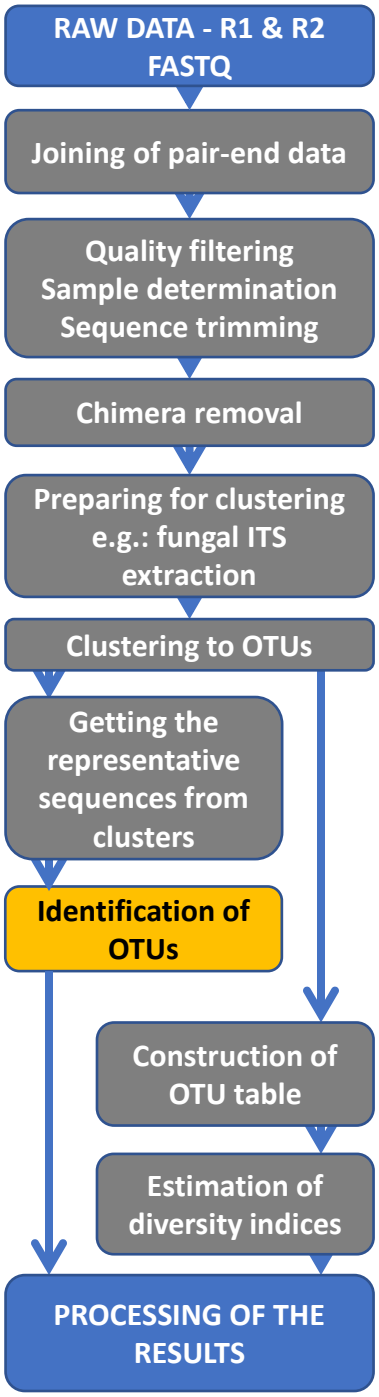


consensus



most abundant





Taxonomic classification of OTUs

Similarity-based

Find homology or minimum alignment distance

Tools:
- local alignments (e.g. BLAST, MEGAN, METAXA2, RTAX)
- global alignments (e.g. GAST)
- overlap alignments (e.g. SINA)

Pro/Con:
- good accuracy for similar sequences
- performs less well on distant lineages
- can be slow on large reference databases

Composition-based

Detect specific features

Tools:
- kmer searches (e.g. NBC/RDP, UTAX, SINTAX)
- hidden Markov models (e.g. PHYMMBL, C16S)

Pro/Con:
- computationally efficient and fast
- performs well on distant lineages
- training required
- limited resolution for shorter sequences

Phylogeny-based

Evolutionary model to determine best placement

Tool:
- ML, NJ, Bayesian (e.g. PPLACER, EPA)

Pro/Con:
- great accuracy for similar sequences
- classification in its evolutionary context
- computationally complex
- requires accurate reference tree
- difficult for non-coding regions

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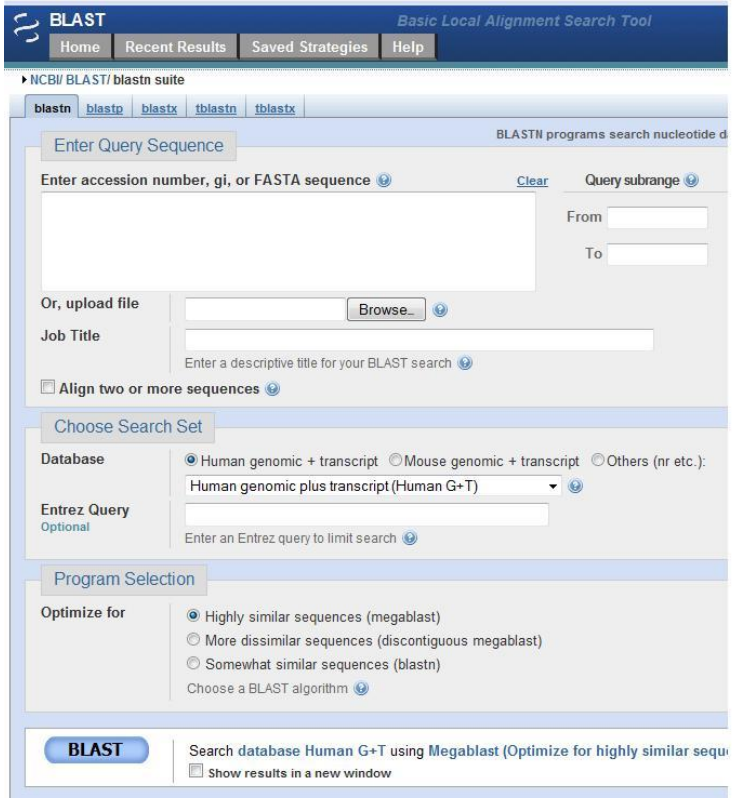
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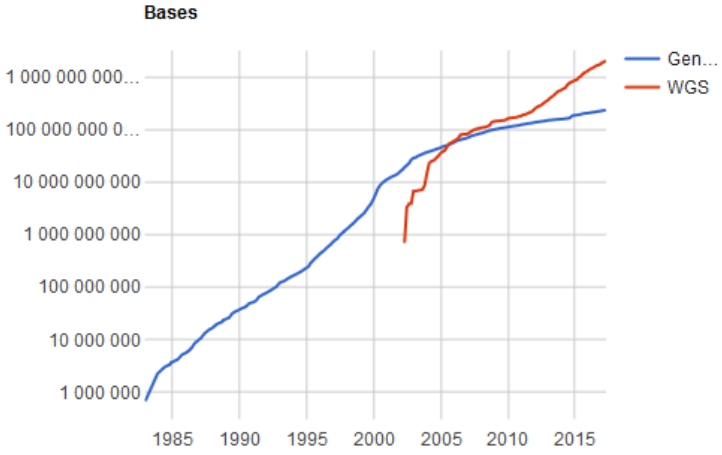
Identification of OTUs

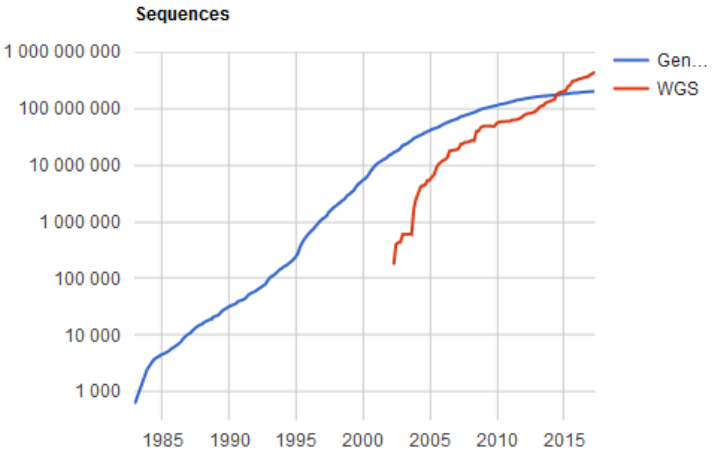
All genes

GenBank - *genetic sequence database, an annotated collection of all publicly available DNA sequence*

- largest ☺
- many errors ☹







https://www.ncbi.nlm.nih.gov/genbank/

Gen – GenBank
WGS – whole genome sequences

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Identification of OTUs

Identification of bacteria

RDP – *Ribosomal Database Project*

provides quality-controlled, aligned and annotated Bacterial and Archaeal 16S rRNA sequences, and Fungal 28S rRNA sequences, and a suite of analysis tools to the scientific community



https://rdp.cme.msu.edu/

silva
high quality ribosomal RNA databases

https://www.arb-silva.de/

SILVA - provides comprehensive, quality checked and regularly updated datasets of aligned small (16S/18S, SSU) and large subunit (23S/28S, LSU) ribosomal RNA (rRNA) sequences for all three domains of life (Bacteria, Archaea and Eukarya).

GREENGENES
The 16S rRNA Gene Database and Tools

http://greengenes.secondgenome.com/

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Clustering to OTUs

Getting the representative sequences from clusters

Identification of OTUs

Construction of OTU table

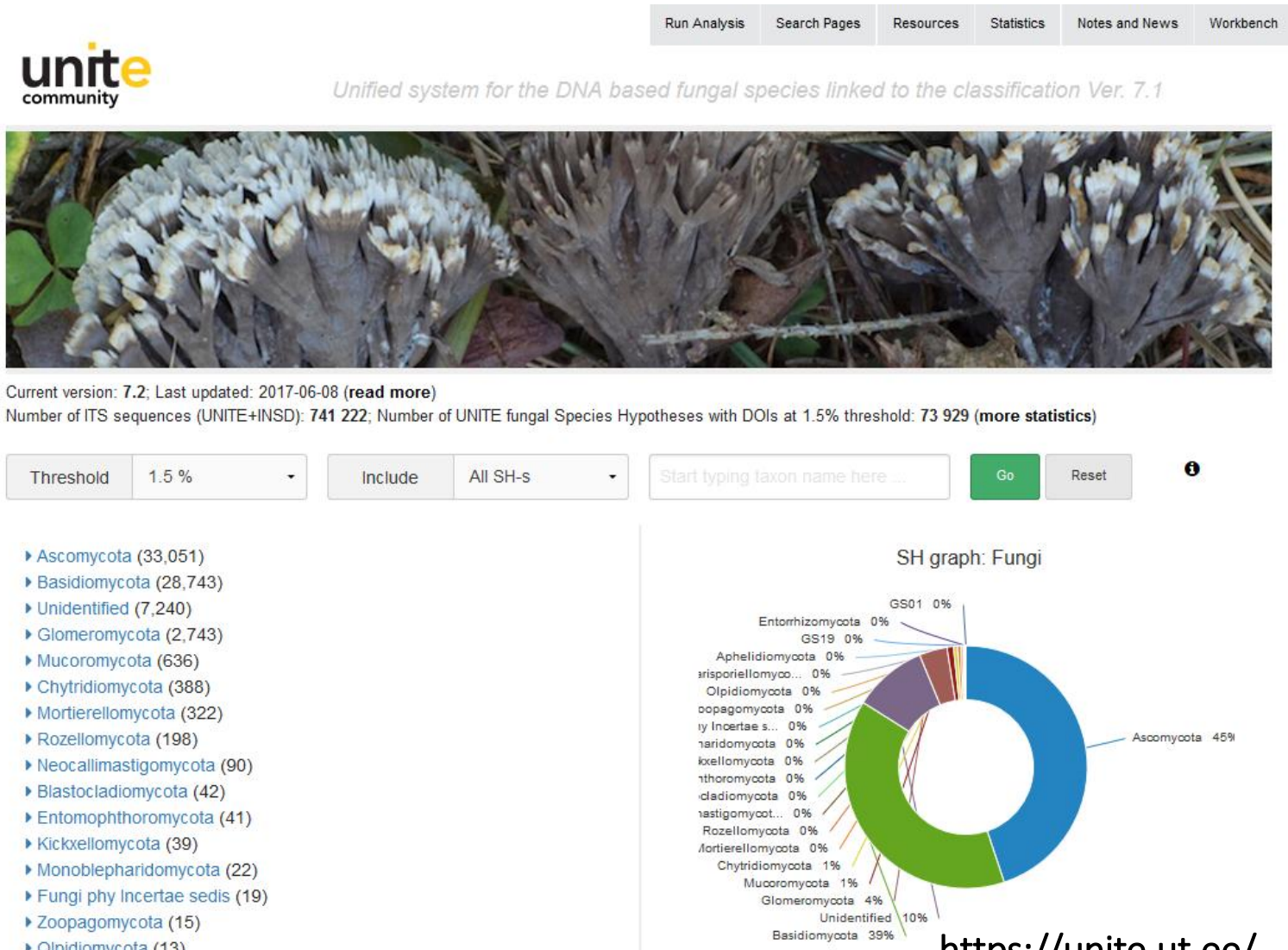
Estimation of diversity indices

PROCESSING OF THE RESULTS

Identification of OTUs

Identification of fungi

UNITE - *Unified system for the DNA based fungal species linked to the classification*



https://unite.ut.ee/

PROCESSING OF THE RESULTS

Introduction to multivariate data analysis